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Units in Integral Group Rings:The Zassenhaus Conjecture and the Prime Graph Question with the use of the HeLP-method

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Brecht Verbeken

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Dankwoord

"Qui dedit beneficium, taceat; narret, qui accepit."

— Lucius Annaeus Seneca, De Beneficiis

De thesis is af. Of toch bijna. Maar het moeilijkste moet misschien nog gebeuren: het dankwoord schrijven. Waar zal ik beginnen? Chronologisch misschien?

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Veel leesplezier!

Brecht Verbeken

21 mei 2018

Samenvatting

> "Als ik zou willen dat je het begrijpt, had ik het wel beter uitgelegd."
>
> — J. Cruijff

Het belangrijkste object doorheen deze thesis is de groep van genormaliseerde torsie-eenheden $V(\mathbb{Z}G)$ van de gehele groepring $\mathbb{Z}G$ van een eindige groep G. Vooraleer we dit van naderbij zullen bestuderen, zullen we de lezer enige achtergrondinformatie verstrekken. Een van de meest fundamentele vragen die iemand kan stellen wat betreft gehele groepringen is de volgende:

(ISO) $\mathbb{Z}G \cong \mathbb{Z}H \Longrightarrow G \cong H$.

Hoewel bekend is dat (ISO) waar is voor verschillende soorten groepen, waaronder meta-abelse [64], nilpotente [58] en enkelvoudige groepen [46], bleef dit probleem open tot in 1998, toen Hertweck met een tegenvoorbeeld kwam, een nilpotent-bij-nilpotente groep van orde $2^{21} \cdot 97^{28}$, zie [35]. Vanzelfsprekend konden we diezelfde vraag stellen voor een willekeurige ring R in plaats van enkel voor $\mathbb{Z}$, maar aangezien er voor elke ring R geldt dat $RG \cong R \otimes_{\mathbb{Z}} \mathbb{Z}G$, zou een negatief antwoord in het geval dat $\mathbb{Z} = R$ een negatief antwoord impliceren voor elke ring R.

Een andere zeer natuurlijke vraag die men zich kan stellen is om de automorfismengroep van $\mathbb{Z}G$ te beschrijven en de relatie ervan met $Aut(G)$. Het volgende is bekend als het Automorfisme Probleem:

(AUT) Onderstel dat θ een willekeurig element in $Aut(\mathbb{Z}G)$ is. Bestaan er dan β in $Aut(G)$ en α in $\mathbb{Q}G$ zodat $\theta(g) = \alpha^{-1} g^{\beta} \alpha$ voor alle g in G?

Een tegenvoorbeeld voor (AUT), een meta-abelse groep werd ontdekt door Roggenkamp en Scott, zie [58].

Men zou zich natuurlijk ook vragen kunnen stellen over eenheden in $\mathbb{Z}G$, meer bepaald zou men de eenheden van eindige orde, die torsie eenheden worden genoemd, in deze ring kunnen bestuderen. De studie van dit onderwerp ving aan in de jaren '40, met de thesis van de Britse wiskundige Graham Higman(1917-2008), zie[41] en heeft in de loop der jaren de aandacht getrokken van verscheidene auteurs. Het meest eenvoudige voorbeeld van torsie eenheden zijn de elementen van G zelf. Men kan zich natuurlijk de vraag stellen of er andere torsie eenheden bestaan dan de elementen van G zelf, of men zou zich kunnen afvragen voor welke groepen alle torsie eenheden net de elementen van G zelf zijn. Higman bewees reeds in zijn thesis dat dit laatste geldt indien G een eindige abelse groep is. Hij bewees zelfs een iets sterker resultaat. Vooraleer we deze stelling kunnen geven hebben we echter nood aan volgende definitie.

Definitie 0.0.1. Een eindige groep G wordt **Hamiltoniaans** genoemd indien al haar deelgroepen normaal zijn.

Stelling 0.1. *(Higman) [41] Alle eenheden van $\mathbb{Z}G$ zijn triviaal als en slechts als G ofwel een abelse groep met exponent $1, 2, 3, 4$ is of indien G een Hamiltoniaanse 2-groep is.*

Maar wat indien G een eindige niet-abelse groep is? Men kan tonen dat de exponenten van G en $V(\mathbb{Z}G)$ gelijk zijn, zie ook Corollary 4.12.1. Wij zullen ons vooral richten op de eenheden van eindige orde in $\mathbb{Z}G$. Tijdens de jaren '70 van de vorige eeuw kwam de Duitse wiskundige Hans Zassenhaus (1912-1991) met drie vermoedens wat betreft de structuur van torsie eenheden in gehele groepringen, die nu bekend zijn als de Vermoedens van Zassenhaus.

Vermoedens 0.0.1 (Vermoedens van Zassenhaus [65])**.**

(ZC1) $u \in V(\mathbb{Z}G)$, $o(u) < \infty \implies u \underset{\mathbb{Q}G}{\sim} g$ *voor een* $g \in G$.

(ZC2) $H \subset V(\mathbb{Z}G)$ *met* $|H| = |G| \implies \alpha^{-1}H\alpha = G$ *voor een* $\alpha \in \mathbb{Q}G$.

(ZC3) $H \subset V(\mathbb{Z}G)$, $|H| < \infty \implies \alpha^{-1}H\alpha \subset G$ *voor een* $\alpha \in \mathbb{Q}G$.

Het is belangrijk om het verband tussen deze drie vermoedens, het Isomorfismeprobleem en het Automorfismeprobleem te benadrukken. Het is eenvoudig in te zien dat $(ZC3)$ ook $(ZC2)$ impliceert, daarnaast hebben we dat:

Stelling 0.2. *Volgende implicaties gelden*

$$\begin{array}{ccccc} (ZC3) & \implies & (ZC2) & \iff & (Aut) \text{ en } (ISO) \\ \Downarrow & & & & \\ (ZC1) & & & & \end{array}$$

Voor een bewijs van deze stelling refereren we aan Sectie 37 in [60], deze referentie bevat overigens de meeste belangrijke achtergrondinformatie wat betereft eenheden in gehele groepringen.

Roggenkamp en Scott [58] vonden een tegenvoorbeeld voor $(ZC2)$, het is niet moeilijk in te zien dat dit ook een tegenvoorbeeld voor $(ZC3)$ moet zijn. Doch $(ZC1)$, nu bekend als (ZC), was een open probleem bij aanvang van deze thesis. Totdat Margolis en Eisele een tegenvoorbeeld vonden in oktober 2017, [31], waarin zij een serie meta-abelse groepen gaven waarvoor (ZC) niet geldt. Hun kleinste tegenvoorbeeld was een groep van orde $2^7 \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 19^2$ waarvan de gehele groepring een element van orde $7 \cdot 19$ bevat dat niet rationaal geconjugeerd is met een groepselement. We zullen dit tegenvoorbeeld echter niet behandelen in deze thesis doch wij zullen focussen op positieve resultaten wat betreft het Vermoeden van Zassenhaus.

Een van de meest gevierde resultaten hieromtrent is het feit dat $(ZC3)$ geldig is voor nilpotente groepen, wegens een resultaat van Weiss [63]. Wat betreft (ZC) werd er in [4] bewezen dat deze vraag affirmatief kan worden beantwoord voor alle groepen wiens orde onder 145 blijft. Daarnaast bewees Hertweck dat (ZC) geldig is voor groepen die een normale Sylow deelgroep met abels complement bezitten [36] en Caicedo-Margolis-del Río bewezen dit voor cyclische-bij-abelse groepen in [23]. Het is ook bekend dat dit vermoeden geldig is voor enkele concrete groepen zoals A_5 [50], S_5 [51], $SL(2,5)$, [30], centrale uitbreidingen van S_5 [10], $PSL(2,7), PSL(2,9) \cong A_6, PSL(2,11)$ en $PSL(2,13)$ [39].

Een van de eerste vragen die iemand zichzelf kan stellen wanneer men voor het eerst geconfronteerd wordt met (ZC) is de volgende: waarom hebben we nood aan rationale conjugatie?

Het blijkt dat reeds voor de symmetrische groep op drie elementen gehele conjugatie niet voltaat, zoals hieronder wordt aangetoond:

Voorbeeld 0.1. *Men kan tonen dat, zie[34]*

$$\rho(1,2) = \left(1, -1, \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}\right)$$

$$\rho(1,2,3) = \left(1, 1, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}\right)$$

aanleiding geeft tot een isomorfisme

$$\rho : \mathbb{Q}S_3 \to \mathbb{Q} \oplus \mathbb{Q} \oplus M_2(\mathbb{Q}).$$

Het is eenvoudig in te zien dat

$$M = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix} = SMS^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}.$$

Maar dit impliceert dat $a = \left(1, -1, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\right) \underset{\mathbb{Q}G}{\sim} (1,2)$ *maar* $a \underset{\mathbb{Z}G}{\not\sim} (1,2)$.

Aangezien het erg moeilijk is om (ZC) te bewijzen, worden zwakkere vormen van deze vraag soms bestudeerd. Omdat er nu een tegenvoorbeeld voor (ZC) gekend is hebben deze aanverwante problemen aan belang gewonnen. Herinner u dat het spectrum van een groep de verzameling van alle ordes van haar torsieelementen is en dat wij een graf kunnen associëren met een groep, de priemgraf, waarbij de knopen gelabeld zijn door de priemgetallen die de orde van de groep delen, en waarbij twee knopen, p en q met elkaar verbonden worden indien er een groepselement van orde $p \cdot q$ bestaat. Voor een natuurlijk getal n zullen wij $G[n]$ schrijven voor de verzameling van elementen van orde n in G.

Vermoedens 0.1.1 (Aanverwante problemen [54])**.**

Kimmerle Probleem (KP)*:Gegeven een eindige groep G en een element u van eindige orde in $V(\mathbb{Z}G)$, bestaat er een eindige groep H die G omvat zodat $u \underset{\mathbb{Q}H}{\sim} g$?*

Spectrum Probleem (SP)*:Gegeven een eindige groep G, bezitten G en $V(\mathbb{Z}G)$ hetzelfde spectrum?*

Priemgraf Vraag (PQ)*: Gegeven een eindige groep G, bezittem G en $V(\mathbb{Z}G)$ eenzelfde priemgraf?*

Bovdi Probleem (BP)*: Gegeven een torsie element van $V(\mathbb{Z}G)$ met als orde een preimmacht p^m. Geldt het volgende dan?*

$$\varepsilon_{G[p^k]}(u) = \begin{cases} 0 \text{ if } k \neq m \\ 1 \text{ if } k = m \end{cases}$$

Veralgemeend Bovdi Probleem (GBP)*: Gegeven een torsie element van $V(\mathbb{Z}G)$ met orde n. Is dan $\varepsilon_{G[m]}(u) = 0$ voor elke $m \neq n$?*

Stelling 0.3. *Volgende implicaties gelden voor een eindige groep G*

$$(ZC) \implies (GBP) \implies (SP) \implies (PQ)$$
$$\Downarrow$$
$$(BP)$$

Bewijs. Zie [54]. ■

Van deze aanverwante problemen zullen wij vooral aandacht besteden aan de Priemgraf Vraag, die geïntroduceerd werd door Kimmerle in [47] en waarvan bekend is dat ze klopt voor oplosbare groepen [38], voor vele sporadische enkelvoudige groepen [49] en voor enkele oneindige series van bijna enkelvoudige groepen [7].

Deze thesis zelf is gewijd aan twee belangrijke onderwerpen, het Vermoeden van Zassenhaus aan de ene kant en de Priemgraf Vraag aan de andere kant. We zullen in Chapter 2 aanvangen met een overzicht van nuttige voorkennis, alvorens de nodige resultaten uit de representatietheorie te bespreken in Chapter 3. In Chapter 4 zullen we naar een equivalente voorwaarde van het Vermoeden van Zassenhaus toewerken wat ons zal aanzetten tot de studie van partiële augmentaties en de HeLP-methode. Nadien zullen wij in Chapter 5 de reeds opgedane kennis illustreren met twee concrete voorbeelden; we zullen bewijzen dat (ZC) geldig is voor A_5 en S_5. In het laatste hoofdstuk, Chapter 6, zullen we onze aandacht vestigen op de Piremgraf Vraag.

De lezer vraagt zich misschien af wat mijn eigen inbreng in deze thesis juist inhoudt. Vooreerst zijn enkele resultaten, zoals het bewijs van (ZC) voor S_5 behandeld op mijn eigen manier en niet zoals dit oorspronkelijk in de literatuur werd behandeld. Daarenboven geef ik in deze thesis de minimale benodigdheden die de lezer toestaat om (PQ) te bewijzen voor verscheidene enkelvoudige groepen. De enige volledig nieuwe resultaten zijn het bewijs voor de Priemgraf Vraag voor de groepen Fi_{22} en $Aut(Fi_{22})$, waarbij ik gebruikmaakte van een methode die ontwikkeld werd door Margolis in [53].

1 Introduction

> "In science one tries to tell people, in such a way as to be understood by everyone, something that no one ever knew before. But in poetry, it's the exact opposite."
>
> — P.A.M. Dirac

The main object of study in this dissertation is the group of normalized torsion units $V(\mathbb{Z}G)$ of the integral group ring $\mathbb{Z}G$ of a finite group G. But before delving deeper into this, we will provide the reader with some background information. One of the most fundamental questions one could ask regarding the integral group ring is the following:

(ISO) $\mathbb{Z}G \cong \mathbb{Z}H \Longrightarrow G \cong H$.

Although it is known that this question holds true for a variety of groups such as metabelian [64], nilpotent [58] and simple groups [46], this problem remained open until 1998, when Hertweck found a counterexample, which is a nilpotent-by-nilpotent group of order $2^{21} \cdot 97^{28}$, see [35]. Of course we could have asked the same question for an arbitrary ring R instead of using $\mathbb{Z}$, but as we have for every ring R that $RG \cong R \otimes_{\mathbb{Z}} \mathbb{Z}G$, a negative solution in the case of the integral group ring implies a negative solution for all rings R.

Another natural question one could ask is to describe the automorphism group of $\mathbb{Z}G$ and its relationship with G. The following is known as the Automorphism Problem:

(AUT) Let θ be any element in $Aut(\mathbb{Z}G)$. Does there exist a β in $Aut(G)$ and α in $\mathbb{Q}G$ such that $\theta(g) = \alpha^{-1} g^{\beta} \alpha$ for all g in G?

A metabelian counterexample to (AUT) was discovered by Roggenkamp and Scott, see [58].

One could also start asking questions about the units of $\mathbb{Z}G$, in particular one could take an interest in the units of finite order, which are called torsion units. The study of this topic started in the 1940s with the thesis of British mathematician Graham Higman (1917-2008) [41] and has attracted the attention of many authors since. The easiest example of torsion units in the integral group ring are the elements of G itself. But one could raise the question if there are any other units in the integral group ring, or if there do exist groups for which the trivial units are the only units in $V(\mathbb{Z}G)$. Higman proved already in his thesis that if G is a finite abelian group, then the torsion units of $V(\mathbb{Z}G)$ are trivial. He even proved a little more. Before stating the theorem of Higman we need the following definition.

Definition 1.0.1. A finite group G is said to be **Hamiltonian** if all of its subgroups are normal.

Theorem 1.1. *(Higman) [41] All units of $\mathbb{Z}G$ are trivial if and only if G is either an abelian group of exponent $1, 2, 3, 4$ or 6 or G is a Hamiltonian 2-group.*

But what if G is a finite non-abelian group? It can be shown that the exponent of G, equals the exponent of $V(\mathbb{Z}G)$, see Corollary 4.12.1. The main objects of interest in this thesis are the units of finite order in $\mathbb{Z}G$. During the 1970s the German mathematician Zassenhaus (1912-1991) came up with three conjectures regarding the structure of the torsion units of the integral group ring, which are now known as the Zassenhaus Conjectures.

Conjecture 1.1.1 (Zassenhaus Conjectures [65])**.**

(ZC1) $u \in V(\mathbb{Z}G)$, $o(u) < \infty \implies u \underset{\mathbb{Q}G}{\sim} g$ *for some* $g \in G$.

(ZC2) $H \subset V(\mathbb{Z}G)$ *with* $|H| = |G| \implies \alpha^{-1}H\alpha = G$ *for some* $\alpha \in \mathbb{Q}G$.

(ZC3) $H \subset V(\mathbb{Z}G)$, $|H| < \infty \implies \alpha^{-1}H\alpha \subset G$ *for some* $\alpha \in \mathbb{Q}G$.

It is important to note the connections between the three Zassenhaus Conjectures, the Isomorphism Problem and the Automorphism problem. It is easy to see that $(ZC3)$ implies $(ZC2)$ moreover we have that:

Theorem 1.2. *The following implications hold*

$$\begin{array}{ccccc} (ZC3) & \implies & (ZC2) & \iff & (Aut) \text{ and } (ISO) \\ \Downarrow & & & & \\ (ZC1) & & & & \end{array}$$

For a proof of this theorem we refer the reader to Section 37 in [60], which contains almost all background information about units in integral group rings.

Roggenkamp and Scott [58] came with a counterexample to $(ZC2)$ and it is not so difficult to see that this has to be a counterexample to $(ZC3)$ as well. But (ZC1), now known as (ZC), was open at the start of this dissertation until Margolis and Eisele found a counterexample in October 2017, [31] in which they gave a series of metabelian groups for which (ZC) does not hold. Their smallest counterexample was a metabelian group of order $2^7 \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 19^2$ whose integral group ring contains a unit of order $7 \cdot 19$ which is not rationally conjugate to a group element. We will not deal with this counterexample, yet we will focus on positive results regarding the Zassenhaus conjecture.

One of the most celebrated results is the fact that $(ZC3)$ holds for nilpotent groups, by a result of Weiss [63]. For (ZC) it was shown in [4] that this question holds for all groups of order up to 144. Moreover Hertweck proved that (ZC) holds for groups which have a normal Sylow subgroup with an abelian complement [36] and Caicedo-Margolis-del Río proved this for cyclic-by-abelian groups in [23]. It is also known to hold for some concrete groups like A_5 [50], S_5 [51], $SL(2,5)$, [30], central extensions of S_5 [10], $PSL(2,7), PSL(2,9) \cong A_6, PSL(2,11)$ and $PSL(2,13)$ [39].

One of the first questions one might ask oneself if one is confronted with (ZC) is why do we need rational conjugation? It turns out that the symmetric group on 3 elements is already a counterexample if one should only allows integral conjugation, as is shown below:

Example 1.3. *One can show that, see [34]*

$$\rho(1,2) = \left(1, -1, \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}\right)$$

$$\rho(1,2,3) = \left(1, 1, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}\right)$$

gives rise to an isomorphism

$$\rho : \mathbb{Q}S_3 \to \mathbb{Q} \oplus \mathbb{Q} \oplus M_2(\mathbb{Q}).$$

One can easily see that

$$M = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix} = SMS^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}.$$

This implies that $a = \left(1, -1, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\right) \underset{\mathbb{Q}G}{\sim} (1,2)$ *but* $a \underset{\mathbb{Z}G}{\not\sim} (1,2)$.

As (ZC) is very hard to proof, weaker versions are studied sometimes. With the existence of a counterexample to (ZC), the study of those other, related problems about the structure of the unit group became more important. Recall that the spectrum of a group is the set of all orders of its torsion elements and that we can associate a graph with a group, the prime graph, a graph for which the vertices are labeled by the primes dividing the order of the torsion elements of the group and where two vertices p and q are connected if and only if there exists units of order $p \cdot q$. For a positive integer n we will write $G[n]$ for the set of elements in G of order n.

Conjecture 1.3.1 (Related problems [54])**.**

> ***The Kimmerle Problem (KP)****: Given a finite group G and an element u of finite order in $V(\mathbb{Z}G)$, does there exists a finite group H containing G as a subgroup such that $u \underset{\mathbb{Q}H}{\sim} g$?*
>
> ***The Spectrum Problem (SP)****: Given a finite group G, do G and $V(\mathbb{Z}G)$ have the same spectrum?*
>
> ***The Prime Graph Question (PQ)****: Given a finite group G, do G and $V(\mathbb{Z}G)$ have the same prime graph?*
>
> ***Bovdi Problem (BP)****: Given a torsion element of $V(\mathbb{Z}G)$ of prime power order p^m. Does the following hold?*
>
> $$\varepsilon_{G[p^k]}(u) = \begin{cases} 0 \text{ if } k \neq m \\ 1 \text{ if } k = m \end{cases}$$
>
> ***General Bovdi Problem (GBP)****: Given a torsion element of $V(\mathbb{Z}G)$ of order n. Is $\varepsilon_{G[m]}(u) = 0$ for every $m \neq n$?*

Theorem 1.4. *The following implications hold for a finite group G*

$$(ZC) \implies (GBP) \implies (SP) \implies (PQ)$$
$$\Downarrow$$
$$(BP)$$

Proof. See [54]. ■

We will focus on the Prime Graph Question, which was introduced by Kimmerle in [47] and which is known to hold for solvable groups [38], for many sporadic simple groups [49] and for some infinite series of almost simple groups [7].

This thesis itself is devoted to two main subjects, the Zassenhaus Conjecture on the one hand and the Prime Graph Question on the other hand. We will start in Chapter 2 with a summary of some useful preliminaries before diving into both ordinary and Brauer representation theory, which will be treated in Chapter 3. In Chapter 4 we will work towards an equivalent criterion of the Zassenhaus Conjecture which will pave the way to partial augmentations and the HeLP-method. Thereafter, in Chapter 5 we will work us through two concrete cases of (ZC) proving that this question holds for A_5 and S_5. In the last chapter, Chapter 6, we will turn our attention to the Prime Graph Question.

The reader may ask what my own contribution to the thesis is. First of all some known results like the proof of (ZC) for S_5 was written down in my own way. But more importantly I gave minimal tables in order to prove the Prime Graph Question for various sporadic simple groups, which was not done before. The only truly new result is the proof of the Prime Graph Question for Fi_{22} and $Aut(Fi_{22})$, using a method developed by Margolis in [53].

Notation

$[x, y]$	$xy - yx$.
$[R, R]$	Additive group generated by Lie products of the form $[x, y]$.
$Cl(G)$	Conjugacy classes of G.
$Cl(G)_p^{\text{reg}}$	p-regular conjugacy classes of G.
1_G	Neutral element of the group G.
1_R	Neutral element of the ring R.
$\Delta_R(G)$	Kernel of the augmentation map of RG.
$\Gamma(R)$	Prime graph of the normalized unit group $V(R)$.
$\hat{x}$	Sum of all powers of x.
Λ	$[\mathbb{Z}G, \mathbb{Z}G]$.
$\mathbb{F}_p$	Field with p elements.
$\mathbb{Q}_p$	p-adic numbers.
$\mathbb{N}$	Non-negative integers.
$\otimes_R$	Tensor product over F.
$\overline{z}$	Complex conjugate of $z \in \mathbb{C}$.
$\prod(G)$	Prime graph of G.
$\sim$	Conjugation.
$\varepsilon_X(g)$	Partial augmentation with respect to X.
$\mathbb{Z}$	Integers.
$\mathbb{Z}_{\geq 1}$	$\{1, 2, 3, \cdots\}$.
A_n	Alternating group on n elements.
$Aut(X)$	Automorphism group of X.
G	Group.
$G[n]$	Elements of order n in G
G_{reg}^p	p-regular elements of G.
$GL(V)$	General linear group of V.
$GL_n(A)$	General linear group over A.

$IBr(G)$	Irreducible Brauer characters of G.
$Irr(G)$	Irreducible characters of G.
$J_k(a)$	Elementary Jordan matrix with a underneath the diagonal.
K	Field.
$M_n(D)$	$n \times n$ matrix ring over D.
$o(g)$	Order of g.
$Out(G)$	Outer automorphism group of G.
R	Ring.
S_n	Symmetric group on n elements.
$soc(G)$	Socle of a group G.
Tr	Matrix trace.
$Tr_{L/K}$	Field trace with respect to the field extension L/K.
$U(\mathbb{Z}G)$	Unit group of integral group ring.
$V(\mathbb{Z}G)$	Normalized torsion units of integral group ring.
$V(\mathbb{Z}G)^p_{reg}$	Normalized p-regular torsion units in $\mathbb{Z}G$.

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2 Preliminaries

2.1 Groups

> "Begin at the beginning," the King said gravely, "and go on till you come to the end: then stop."
>
> — Lewis Carroll, Alice in Wonderland

We will start this thesis by recalling some background information and providing the reader with references for the main results we will use later on. Most of the material in this chapter is taken from [60] and [62].

Definition 2.0.1. A group G is said to be **metabelian** if and only if there exists an abelian normal subgroup N such that the quotient group G/N is abelian.

Definition 2.0.2. A group G is said to be a **Hamiltonian group** if and only if all of its subgroups are normal.

Definition 2.0.3. A group G is said to be an **almost simple group** if it satisfies the following equivalent conditions:

1. G contains a non-abelian simple group and is contained within the automorphism group of that simple group, i.e.

   $$S \leq G \leq Aut(S) \text{ for a simple group } S.$$

   In this case G is sometimes called S-simple.

2. G contains a unique non-abelian minimal normal subgroup S which is simple.

Theorem 2.1. *(Classification of Finite Simple Groups) [3], [25]. Every finite simple group is isomorphic to one of the following groups:*

1. *A cyclic group of prime order.*
2. *An alternating group of degree at least* 5.
3. *A group of Lie type.*
4. *One of* 26 *groups, the sporadic simple groups.*
5. *The Tits group.*

The following theorem is now known as a corollary of the Classification of Finite Simple Groups but is still known as Schreiers Conjecture:

Theorem 2.2. *(Schreiers Conjecture) For a finite simple group S, the group $Out(S)$ is solvable. In particular, it follows that an almost simple group is the extension of a solvable group by a simple group.*

Definition 2.2.1. Given two groups. Q and N, a group G is said to be an extension of Q by N if there exists a short exact sequence

$$1 \to N \to G \to Q \to 1.$$

In this case N is normal in G and the quotient group G/N is isomorphic to Q.

Definition 2.2.2. Let G be a group, a **composition series** of G is a finite collection of subgroups H_i such that

$$1 = H_0 \triangleleft H_1 \triangleleft \cdots \triangleleft H_n = G,$$

where each H_i is a maximal normal subgroup of H_{i+1}. The quotients of the form H_i/H_{i+1} are called **composition factors**.

Definition 2.2.3. A finite group G is called solvable if and only if it admits a composition series all of whose factors are cyclic groups of prime order.

Theorem 2.3. *(Burnside) Let p and q be primes and let a and b be strictly positive integers. If G is a group for which $|G| = p^a q^b$ then G is solvable.*

Proof. See Theorem 3.10 in [43]. ■

Definition 2.3.1. Let G be a group, a **chief series** of G is a finite collection of subgroups $N_i \triangleleft G$ with

$$1 = N_0 \subset N_1 \subset N_2 \subset \cdots \subset N_n = G$$

and such that there does not exists any $H \triangleleft G$ with $N_i \subset H \subset N_{i+1}$ for any i.

Definition 2.3.2. The **socle** of a nontrivial group G, $soc(G)$, is the subgroup generated by the set of all minimal normal subgroups of G. If G has no minimal normal subgroups we set $soc(G) = 1$.

Remark. *As any automorphism of G has to permute the minimal normal subgroups of G, $soc(G)$ is invariant under automorphisms of G and hence is a characteristic subgroup.*

The following theorem will be useful in our study of almost simple groups:

Theorem 2.4. *Let G be a nontrivial finite group*

1. *If M is a minimal normal subgroup of G and N is an arbitrary normal subgroup of G, then either $M \subset N$ or $\langle M, N\rangle = M \times N$.*
2. *There exist minimal normal subgroups $M_1, \cdots M_n$ of G such that $soc(G) = M_1 \times M_2 \times \cdots \times M_n$.*
3. *Every minimal normal subgroup M of G is a direct product $M = S_1 \times \cdots \times S_k$ where all S_i are simple normal subgroups of M which are conjugate in G.*
4. *If all M_i above are non-abelian, then the set of all M_i contains all minimal normal subgroups of G.*

Proof. Since $M \cap N \triangleleft G$ the minimality of M implies that either $M \subset N$ or $M \cap N = \{1\}$, the latter case, as M and N are both normal, implies that $\langle M, N\rangle = MN = M \times N$, which proves 1.

Thanks to the finiteness of G we can find a set $X = \{M_1, \cdots, M_v\}$ of minimal normal subgroups which is maximal with respect to the property that $H = \langle X\rangle = M_1 \times M_2 \times \cdots \times M_v$. We have to show that H contains all minimal subgroups of G, but this follows by 1.

Let now S be a minimal normal subgroup of M, then all conjugates of S are still minimal normal subgroups of M. Choose a set of these conjugates $X = \{S_1, S_2, \cdots, S_v\}$ which is maximal with respect to the property that $H = \langle X\rangle = S_1 \times S_2 \times \cdots \times S_v$. As H contains all conjugates of S under G it follows that $H \triangleleft G$. But since $H \subset M$ and M is minimal normal this implies that $M = H = S_1 \times S_2 \times \cdots \times S_v$. Every S_i has to be simple because S_i has to be minimal normal in $S_1 \times S_2 \times \cdots \times S_v$, which settles 3.

Let M be a minimal normal subgroup which is distinct from all M_i, applying 1. with $L = M_i$ we obtain that $\langle M, M_i\rangle = M \times M_i$, in particular M centralizes each of the M_i. Thus $M \subset Z(soc(G))$ but as each M_i is non-abelian, 3. implies that $Z(soc(G)) = \{1\}$, a contradiction, so 4. holds. ■

2.2 Group Rings

Let us start with recalling the definition of the most important object throughout this dissertation and giving some properties:

Definition 2.4.1. Let G be a group, and R be a ring with identity. The **group ring** RG is the free R-module with basis G and where the multiplication is defined by

$$\Big(\sum_{g\in G} r_g g\Big)\Big(\sum_{g\in G} r'_g g\Big) = \sum_{g\in G}\Big(\sum_{st=g} r_s r'_t\Big)g.$$

The multiplicative identity of RG is the product of the identity of R with the identity of G in the group ring, we will often denote this by 1. In the case where R is a field, RG is sometimes called the **group algebra** .

The map $\varepsilon : RG \to R$ given by

$$\varepsilon\Big(\sum_{g\in G} r_g g\Big) = \sum_{g\in G} r_g$$

is a ring homomorphism which is called the **augmentation map** of RG. Its kernel

$$\Delta_R(G) = \Big\{\sum_{g\in G} r_g g \in RG : \sum_{g\in G} r_g = 0\Big\}$$

is called the **augmentation ideal** of RG.

Definition 2.4.2. Let G be a group and let X be a subset of G and suppose that $u \in RG$. The **partial augmentation** of u with respect to X is denoted by $\varepsilon_X(u)$ and given by

$$\varepsilon_X(u) = \varepsilon_X\Big(\sum_{g\in G} r_g g\Big) = \sum_{x\in X} r_x.$$

An element x of a ring R is called a **unit** if and only if there exists an element y in R such that $xy = 1 = yx$. The set of all units of R form a multiplicative group which is called the unit group of R and which is denoted by $U(R)$. A unit is called **torsion** or **periodic** if and only if it has finite order.

As the augmentation map is a homomorphism, it follows that, for every unit u of an integral group ring $\mathbb{Z}G$, we have that $\varepsilon(u) = \pm 1$. It will be sufficient for our purposes to study the torsion units of augmentation 1, the so-called normalized torsion units which we denote by $V(\mathbb{Z}G)$, as $U(\mathbb{Z}G) = \pm V(\mathbb{Z}G)$. Those normalized torsion units will be the main object of study in this thesis. We provide some examples of units in $\mathbb{Z}G$.

Examples 2.5. (1) *Elements of the form $\pm g$, $g \in G$, which are called* ***trivial units****.*

(2) *For $x \in G$, we write $\hat{x}$ for the sum of all the powers of x:*

$$\hat{x} = \sum_{i=1}^{o(x)} x^i.$$

As $\hat{x}(1-x) = 0$ it follows that $((1-x)y\hat{x})^2 = 0$ for every $y \in G$. Hence $u_{x,y} = 1 + (1-x)y\hat{x}$ is a unit with inverse $1 - (1-x)y\hat{x}$. In the same spirit one can define units of the form $1 + \hat{x}y(1-x)$. Units of these forms are called ***bicyclic units****.*

The following theorem will be useful when we will talk about Brauer trees:

Theorem 2.6. *Let G be a group and let R be a commutative ring. Then, the set of all class sums, i.e. the set consisting of sums of the form $\sum_{x \in C} x$ where C is a conjugacy class of G, forms a basis of $\mathcal{Z}(RG)$ which is the centre of RG.*

Proof. See Theorem 3.6.2 in [56]. ■

Note that RG is commutative if and only if R and G are commutative. In this thesis we will mainly deal with the case that $R = \mathbb{Z}$ or $R = \mathbb{Q}$.

2.3 Modules and Rings

In this section we will give a brief account of some results concerning modules and rings. We will only consider left modules unless stated otherwise. The material inhere is mainly taken from [43].

Definition 2.6.1. An R-module M is **semisimple** if and only if every submodule of M is a direct summand of M. A ring R is called **semisimple** if and only if it is semisimple as a (left) module over itself.

Theorem 2.7. *(Wedderburn - Artin) A ring R is semisimple if and only if it is isomorphic to a direct sum of matrix algebras over division rings:*

$$R \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

Furthermore the division rings D_i and the numbers k and n_i are uniquely determined up to isomorphism.

Proof. See Theorem 2.1.3 in [62]. ■

This decomposition of R into matrix rings over division rings is called the Wedderburn Decomposition of R.

Now we want to know when a group ring is in fact a semisimple ring, this result is due to Maschke.

Theorem 2.8. *(Maschke) Let G be a group and R be a ring. The group ring RG is semisimple if and only if the following conditions hold:*

(1) *R is a semisimple ring.*

(2) *G is a finite group.*

(3) *$|G|$ is invertible in R.*

Proof. See Theorem 1.2.1 in [62]. ■

Definition 2.8.1. A **Dedekind domain** is an integral domain in which every nonzero proper ideal can be written as the product of maximal ideals in a unique way.

Definition 2.8.2. A module P is called **projective** if and only if for every surjective module homomorphism $a : M \to P$ there exists a module homomorphism $b : P \to M$ such that $ab = 1_P$.

Theorem 2.9. *For an R-module P, the following properties are equivalent:*

1. *P is a projective module.*

2. *P is a direct summand of a free module.*

Proof. See Proposition 7.1.3 in [62]. ■

Remark that every free module is projective.

3 Representation Theory

3.1 A brief recall of ordinary representation theory

> "Representation theory is like a George R.R. Martin novel. There are lots and lots of characters; most of them are complex, and many of them are unfaithful. But everything gets a lot tensor: there are a lot of duals, and some of the characters end up decomposing!"
>
> –Unknown

We will recall the basics of ordinary representation theory that we will need in the remainder of the text. Moreover we provide the reader with some important theorems about modular representation theory. For more information and proofs we refer the reader to [61] and [43].

Definition 3.0.1. A (linear) **representation** of a finite group G to a K-vector space V is a group homomorphism

$$\rho : G \to GL(V) : g \to \rho(g).$$

We will call the space V a representation of G and $n = \dim_K(V)$ is called the degree of ρ.

Definition 3.0.2. A subspace W of V is called G**-invariant** if and only if we have for all $w \in W$ that

$$\rho(g)(w) \in W.$$

Definition 3.0.3. A representation $\rho : G \to GL(V)$ is said to be **irreducible** if the only G-invariant subspaces of V are $\{0\}$ and V.

Definition 3.0.4. The (left) **regular representation** of G with respect to RG is the homomorphism

$$\rho_g : G \to GL(RG)$$
$$\text{given by } \rho_g(h) = gh, \text{ for all } h \in H.$$

Definition 3.0.5. Let G be a group and let $\rho : G \to GL(V)$ be a representation of G, the map

$$\chi_\rho : G \to K : g \to \mathrm{Tr}(\rho_g)$$

is called the **character** of ρ.

It is easily seen that characters are constant on conjugacy classes. Moreover, characters which correspond to irreducible representations are called **irreducible characters**, these irreducible characters and their values are often presented in a table, the so-called **charactertable**.

Definition 3.0.6. Let ρ and τ be representations $G \to GL(V)$. Then ρ and τ are called isomorphic if and only if there exists a K-vector space isomorphism $\phi : V \to V$ for which, for all $g \in G$:

$$\phi \circ \rho(g) \circ \phi^{-1} = \tau(g).$$

Theorem 3.1. *Two complex representations are isomorphic if and only if they have the same character.*

Proof. See Lemma 2.3 in [43]. ■

Lemma 3.2. *Let G be a finite group and let R be an irreducible complex representation of degree n. If M is an $n \times n$ complex matrix which commutes with $R(g)$ for all $g \in G$ then $R = zI$ for a $z \in C$.*

Proof. See Lemma 2.25 in [43]. ■

Definition 3.2.1. A matrix M is called a **Vandermonde matrix** if and only if it is of the following form:

$$M = \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^n \end{bmatrix}$$

In particular

$$\det(M) = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Definition 3.2.2. Let K be a field and let L be a finite field extension of K, take $\alpha \in L$ and let $\sigma_1(\alpha), \cdots, \sigma_n(\alpha)$ be the roots of the minimal polynomial of α over K. The **field trace** of α with respect to K is given by

$$\mathrm{Tr}_{L/K}(\alpha) = [L : K(\alpha)] \sum_{j=1}^{n} \sigma_j(\alpha).$$

Moreover, if L/K is a Galois extension, then the trace of α is the sum of all the Galois conjugates of α:

$$\mathrm{Tr}_{L/K}(\alpha) = \sum_{g \in \mathrm{Gal}(L/K)} g(\alpha).$$

Definition 3.2.3. Let F be a field and let A be a finite dimensional F-algebra. Then the norm of $x \in F$ is given by

$$\mathrm{Nr}_{A/F}(x) = \det(\rho(x)).$$

Where $\rho : A \to End_F(A)$ is the regular representation of A.

3.1.1 Some Representation Theory of Symmetric Groups

Let S_n denote the symmetric group of degree n. We can use the partitions of n to label the conjugacy classes of S_n. For a prime r and a positive integer j such that $rj \leq n$ we will denote by $\mathsf{r.j}$ the conjugacy class of S_n which corresponds to the partition $(r^j, 1^{n-rj})$, the permutation consisting of j disjoint r-cycles. Remark that there are exactly $\lfloor \frac{n}{r} \rfloor$ conjugacy classes of elements of order r in S_n and that, for two different primes p and q there is an element of order pq in S_n if and only if $p + q \leq n$.

Also the ordinary irreducible representations are in natural bijection with the partitions of n. For a partition λ of n we will denote by χ_λ the ordinary irreducible character afforded by the Specht module S^λ, see [32]. In particular $\chi_{(n)} = 1$ is the principal character of S_n and $\chi_{(1^n)} = sgn$ is the alternating character. Set $\pi = \chi_{(n-1,1)} = 1$. Remark that $\pi(1) = n - 1$ and that for a prime r and $j \in \{1, \cdots, \lfloor \frac{n}{r} \rfloor\}$ we have that $\pi(\mathsf{r.j}) = \pi(1) - rj$. We will start be establishing a way to calculate characters associated with certain partitions of S_n. Let $\lambda = (\lambda_1, \cdots, \lambda_k)$ be a partition of n and let C_i denote any conjugacy class of S_n and let $i \in \{1, \cdots, n\}$. Now write an element in C_i as the product of disjoint cycles and let i_j be the number of cycles of length j in this product. Introduce k independent variables: $X_1, \cdots X_k$ and define their jth power sum to be $P_j(X) = X_1^j + \cdots X_k^j$. Define the discriminant of the variables $X_1, \cdots X_k$ as

$$\Delta(X) = \begin{vmatrix} 1 & X_k & \cdots & X_k^{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_1 & \cdots & X_1^{k-1} \end{vmatrix}$$

Let $l_s = \lambda_s + k - s$ for $s \in \{1, \cdots k\}$ and last but not least let $[f(x)]_{(l_1, \cdots l_k)}$ denote the coefficient of f at $X_1^{l_1} \cdots X_k^{l_k}$, where f is a polynomial function in the X_i. We then have:

Theorem 3.3. *(Frobenius Formula) The character of the irreducible representation of S_n associated with λ is given by*

$$\chi_\lambda(C_i) = \Big[\Delta(X) \prod_j P_j(X)^{i_j}\Big]_{(l_1, \cdots l_k)}.$$

Proof. See Section 4.15 in [32]. ■

We will illustrate this theorem by two examples:

Examples 3.4.

1. *Consider the group S_3 and let us compute $\chi_{2,1}$ for the conjugacy class of order 2. So $k = 2$ and we introduce the variables X_1 and X_2. Each element is a product of a 2- and a 1-cycle so $i_1 = i_2 = 1$ and $i_3 = 0$. An easy calculation shows that this implies that the polynomial inside the brackets is $X_2^4 - X_1^4$. Thus the coefficient of $X_1^2 X_2$ equals zero.*
2. *If we consider S_n and the partition $\lambda = (n-1, 1)$ we obtain $(l_1, l_2) = (n-1, 1)$ and using our formula yields*

$$\chi_\lambda(C_i) = \Big[(X_1 - X_2) \prod_j (X_1^j + X_2^j)^(i_j)\Big]_{(n-1,1)} = i_1 - 1.$$

3.2 Modular Representation Theory

In this section we will give a brief overview of modular representation theory, with special emphasis on the results and ideas we will need later on in this thesis. The material is based on [28], for a more elaborate treatment of the topic, please see [62].

In contrast with ordinary representation theory, modular representation theory concerns itself with cases where the characteristic of the base field divides the order of the group G. This will make a substantial difference, as Maschke's theorem, a cornerstone of ordinary representation theory no longer holds in the modular case. In this way it will be much more complicated to describe how a representation breaks down into irreducible representations. Let us first start with an example to illustrate this.

Definition 3.4.1. A (modular) representation V of G is called **indecomposable** if and only if it is not isomorphic to a direct sum of representations of G, $W \oplus W^{'}$.

Example 3.5. *Let V be the two-dimensional representation of C_p over $\mathbb{F}_p$ where*

$$x^n \to \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}.$$

It is easily verified that this is indeed a representation. It is obvious to see that the space

$$W = \left\{ \begin{pmatrix} 0 \\ z \end{pmatrix} : z \in \mathbb{F}_p \right\}$$

is invariant under this action. This means that our representation is reducible. Suppose now that there exists a G-invariant $W^{'}$ such that $V = W \oplus W^{'}$. But if we take a look at

$$\begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} y \\ ny + z \end{pmatrix} = \begin{pmatrix} \lambda y \\ \lambda z \end{pmatrix} \quad \textit{with } \lambda \in \mathbb{F}_p$$

we see that $\lambda = 1$ and that W is the only invariant subspace. So contrary to the ordinary case, there do exist representations which are reducible but indecomposable.

In the ordinary case we know, thanks to Maschke, that all representations in characteristic zero are semisimple modules. Our example above shows that this will not be the case if we work with modular representations. On the other hand, suppose V is a finite dimensional representation and $\{0\} \neq W \subsetneq V$ is G-variant, then V/W will be a representation of G and both spaces will have a dimension strictly less than the dimension of V, as one-dimensional representations are irreducible, repeating the process will eventually terminate and will provide a finite set of irreducible representations which are called **composition factors**. As it can be shown that those composition factors are unique up to isomorphism, the following definition is justified

Definition 3.5.1. Let K be a field, the **semisimplification** of a KG-module is the direct sum of its composition factors.

3.2.1 Reduction of Representations

Definition 3.5.2. Let F be a field, a multiplicative **valuation** on F is a mapping $\phi : F \to \mathbb{R}_{\geq 0}$ such that

- $\phi(x) = 0$ if and only if $x = 0$.
- $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in F$.
- $\phi(x + y) \leq \phi(x) + \phi(y)$ for all $x, y \in F$.

Definition 3.5.3. A valuation ϕ is called **discrete** if and only if $\{\phi(x) : x \in F, x \neq 0\}$ is an infinite cyclic group under multiplication.

The set $R_\phi = \{x \in F : \phi(x) \leq 1\}$ is a ring, which is called the **valuation ring** of ϕ. If ϕ is a discrete valuation it is called a **discrete valuation ring**.

Definition 3.5.4. Let F be a field and let A be a finite dimensional F-algebra. A simple A-module S is said to be **absolutely simple** if and only if $E \otimes_F S$ is a simple $E \otimes_F A$ module for all extension fields E of F.

Definition 3.5.5. An extension field E of F is called a **splitting field for a finite dimensional algebra** A if and only if every simple $E \otimes_F A$ module is absolutely simple.

If A is a group algebra, one says that E is a splitting field of the corresponding group G. In particular one can prove that:

Theorem 3.6. *(Brauer) Let G be a finite group of exponent n and let F be a field that contains a primitive n-th root of unity, then F is a splitting field of G.*

Proof. See Theorem 9.2.7 in [62]. ■

Our main goal in this section is to reduce modules from characteristic zero to characteristic p for some prime p. In order to do that we will need three things, which we will list as a triple (F, R, k) where F is a field of characteristic zero equipped with a discrete valuation, R is the valuation ring in F with maximal ideal (π) and $k = R/(\pi)$ is the residue field of R. Such a triple is called a **p-modular system** . In the remainder of the text we will implicitly assume that both F and k are splitting fields for all subgroups of a fixed group G. In most cases we will deal with the p-modular system of the form $(\overline{\mathbb{Q}_p}, R, \mathbb{F}_p)$, where $\overline{\mathbb{Q}_p}$ is an algebraic closure of the p-adic numbers.

Definition 3.6.1. Let K be a field, a **lattice** in a K-vector space V is the R_ϕ span of a K-basis for V.

Lemma 3.7. *There exists a G-invariant lattice over $\mathbb{Q}_p$, in particular, if $\rho : G \to GL(V)$ is a representation over K there exists a lattice $\Gamma \subset V$ such that $\rho(g)\Gamma \subset \Gamma$.*

Proof. See Proposition 1.2.2 in [28]. ■

If we take now a representation ρ and a G-invariant lattice Γ, it follows that we can view ρ as a map into $GL_n(R_\phi)$, as the action of ρ on Γ fixes Γ. As the entries of the matrix of ρ will live in R_ϕ we can reduce them modulo (π), which yields matrices with coefficients in $k = \mathbb{R}_\phi/(\pi)$, in this way we have obtained a representation $\overline{\rho} : G \to GL_n(k)$ which is called the **reduction of a representation** modulo p.

One can show that the reductions of irreducible characters of a group G are still linearly independent over k.

3.2.2 Modular Characters

Definition 3.7.1. An element g of G is **p-regular** if $p \nmid \mathrm{o}(g)$. A conjugacy class is p-regular if its elements are p-regular. The set of p-regular elements will be denoted by G^p_{reg}.

Remark. *One can easily extend this definition to torsion units as it will later turn out to be very useful to not just consider p-regular elements of the group G but to consider p-regular torsion units as well. We will denote the set of normalized p-regular torsion units by* $V(\mathbb{Z}G)^p_{reg}$.

Fix now a p-modular system (F, R, k) where both F and k are splitting fields for all subgroups of G. In particular, this implies, by Theorem 3.6 that both F and k contain a primitive n-th root of unity where n is the l.c.m. of the orders of p-regular elements in G. Let us write

$$\mu_F = \{n^{\text{th}} \text{ roots of unity in } F\},$$

$$\mu_k = \{n^{\text{th}} \text{ roots of unity in } k\}.$$

Lemma 3.8. *With the above notation one has*

1. $\mu_F \subset R$.
2. *Reduction mod p gives an isomorphism from* μ_F *to* μ_k.

Proof. The first claim follows because roots of unity have value 1 under the valuation. The polynomial $X^n - 1$ is separable in both $k[X]$ and $F[X]$ as its formal derivative is not zero and has no factors in common with $X^n - 1$. As the polynomial $X^n - 1$ over F reduces to a polynomial with the same coefficients over k, the linear factors over F will reduce to a complete set of linear factors over k. As the set of linear factors are in bijection with each other, $X - \hat{\lambda}$ will reduce to $X - \lambda$. ■

Definition 3.8.1. Let $g \in G$ be a p-regular element and let $\rho : G \to GL(V)$ be a representation over k. Then $\rho(g)$ is diagonalizable and all eigenvalues of $\rho(g)$, say $\lambda_1, \cdots \lambda_w$ will lie in k as they are roots of unity. The **Brauer character** of g with respect to V is defined as

$$\phi_V(g) := \hat{\lambda_1} + \cdots + \hat{\lambda_n}.$$

Remark. *This definition can be linearly extended to p-regular torsion units.*

Theorem 3.9. *(Properties of Brauer characters) For a p-modular system (F, R, k), finite dimensional kG-modules U, V and S and a finite dimensional FG module M we have that:*

1. $\phi_V(1) = dim_k(V)$.
2. ϕ_V *is a class function on* G^p_{reg}.
3. $\phi_{U\otimes V} = \phi_U \cdot \phi_V$.
4. *If* $0 \to U \to V \to S \to 0$ *is a short exact sequence of* kG *modules then* $\phi_V = \phi_U + \phi_S$. *In particular, let* $S_1, \cdots S_n$ *be the composition factors of* S, *then* $\phi_S = \phi_{S_1} + \cdots + \phi_{S_n}$.
5. *If* M *has character* χ_M *then* $\phi_{\overline{M}} = \chi_M|_{G^p_{reg}}$, *where* $\overline{M}$ *denotes the reduction of* M *modulo* p.

Proof.

1. As 1 acts as the identity map on V, all of its eigenvalues are equal to 1 and there are precisely $\dim_k(V)$ eigenvalues.
2. This follows as conjugate matrices have the same eigenvalues.
3. If u is a p-regular torsion unit then U and V have bases $a_1, \cdots a_p$ and $b_1, \cdots b_q$ resepectively, consisting of eigenvectors of u with eigenvalues $\lambda_1, \cdots \lambda_p$ and $\mu_1, \cdots \mu_q$ respectively. But the tensors $a_i \otimes b_j$ form a basis of eigenvectors of $U \otimes V$ with eigenvalues $\lambda_i \mu_j$. So $\phi_{U\otimes V}(u) = \phi_U(u) \cdot \phi_V(u)$.
4. If u is a p-regular torsion unit then $k\langle u\rangle$ is semisimple by Maschke's theorem, so $V \cong U \oplus W$ as $k\langle u\rangle$ modules. It now follows that the eigenvalues induced by u on V are the union of the eigenvalues on U and S. The result for composition factors follows inductively.
5. For a p-regular torsion unit u, we have that $\chi_M(u) = \lambda_1 + \cdots \lambda_n$ for $\lambda_i \in \mu_F$. So $\phi_{\overline{M}}(u)$ is nothing else as the reduction of $\chi_M(u)$ modulo p.

■

3.2.3 Some results concerning characters and the CDE-triangle

We will need the following properties, all concerned with the CDE-triangle, which lies at the heart of modular representation theory. The name CDE triangle refers to the matrices C, D and E below. A detailed explanation of this material would lead us to far. But the interested reader may consult [26], [27] or [62] for a comprehensive treatment.

- $D = (d_{ij})$, the **decomposition** matrix, is the matrix one gets if one writes the restriction of an irreducible ordinary character χ_i to the p-regular elements as a linear combination of the irreducible Brauer characters ϕ_j. In particular, suppose that the ordinary irreducible characters are given by $\chi_1, \cdots \chi_h$ and let $\phi_1, \cdots \phi_r$ be the irreducible Brauer characters of G. Then D is the matrix (d_{ij}) such that, for any u in the p-regular torsion units:

$$\chi_i(u) = \sum_{j=1}^{r} d_{ij}\phi_j(u).$$

- The matrix $C = D^t D$ is called the **Cartan matrix**, which is an invertible matrix, while $E = D^t$.

Furthermore, analogous to the ordinary case, one has:

Theorem 3.10. *(Brauer) The irreducible modular characters constitute a basis for the vector space of class functions of G^p_{reg}.*

Proof. See Theorem 1.3.4 in [28]. ■

Corollary 3.10.1. *The number of irreducible characters is equal to the number of p-regular conjugacy classes of G.*

Theorem 3.11. *(Brauer-Nesbitt) Let n be the largest power of p dividing $|G|$ and let χ be an irreducible character of degree m such that $n|m$, then is $\phi_{\overline{M}}$, the reduction of χ_M modulo p an irreducible Brauer character.*

Proof. See Theorem 9.4.7 in [62]. ■

3.3 Brauer Graphs

Fix a prime p. We will denote the irreducible characters of a group G by $Irr(G)$ and we will write $IBr(G)$ for the set of irreducible Brauer characters relative to p.

Let G be a group and let χ be an ordinary character of G associated with the representation R. We will define a map from $\mathcal{Z}(\mathbb{C}G)$ to $\mathbb{C}$. Let x be an element of $\mathcal{Z}(\mathbb{C}G)$. Using Lemma 3.2 yields that $R(x) = zI$ for some $z \in \mathbb{C}$, where z is independent of the choice of the representation because $R(x)$ is a diagonal matrix. We now define the map $\omega : \mathcal{Z}(\mathbb{C}G) \to \mathbb{C}$ by setting

$$R(x) = \omega(x)I,$$

for all $x \in \mathcal{Z}(\mathbb{C}G)$. We will often denote this map by ω_χ as it depends on χ. Those maps are determined by their values on the class sums of G, which we will denote by C_i. For two ordinary irreducible characters χ_1 and χ_2 of G we write $\chi_1 \sim \chi_2$ if and only if $\widehat{\omega_{\chi_1}(K_i))}^{-1} = \widehat{\omega_{\chi_2}(K_i))}^{-1}$ for all class sums of G, Where $\widehat{\ }^{-1}$ denotes the inverse image under the isomorphism between the roots of unity in $\mathbb{C}$ and those in F. One can easily check that this defines an equivalence relation on $Irr(G)$.

Definition 3.11.1. Let G be a finite group, a **p-block** of G is a subset $X \subset Irr(G) \cup IBr(G)$ such that:

1. The intersection $X \cap Irr(G)$ is an equivalence class under the relation defined above.
2. The intersection $X \cap IBr(G)$ equals the set $\{\phi_j \in IBr(G) \mid \exists \chi_i \in B \cap Irr(G)$ such that $d_{ij} \neq 0\}$ where d_{ij} is an entry of the decomposition matrix D.

Definition 3.11.2. A **Brauer graph** of a finite group G is a graph where the vertex set is given by $Irr(G)$ and where two vertices χ_i and χ_j are linked if and only if there exists a Brauer character ϕ_k in $IBr(G)$ such that d_{ik} and d_{jk} are both nonzero.

Brauer graphs can be calculated in GAP if the Brauer tables are available, see [19]. We will need the Brauer graphs of Fi_{22} and $Aut(Fi_{22})$, which can also be found in [42].

3.3.1 Example of a Brauer character table, S_5 modulo 5

This section serves two purposes. In the first place it is an ideal opportunity to demonstrate how to construct a modular character table and how to use the machinery of modular representation theory. On the other hand, we will use this Brauer table later on, when we will give a proof of (ZC) for S_5 in Theorem 5.2.

Example 3.12. *S_5, the symmetric group on 5 elements, has 60 elements, so its modular representation breaks down into the cases $p = 2$, $p = 3$ and $p = 5$. We will focus on the case $p = 5$. As conjugacy depends only on cycle type in a symmetric group, there are seven conjugacy classes, namely: $1, (a,b), (a,b)(c,d), (a,b,c), (a,b,c,d), (a,b,c,d,e)$ and $(a,b)(c,d,e)$. The character table of S_5 in characteristic zero is given by:*

	1	(a,b)	$(a,b)(c,d)$	(a,b,c)	(a,b,c,d)	(a,b,c,d,e)	$(a,b)(c,d,e)$
χ_1	1	1	1	1	1	1	1
$\chi_{1'}$	1	−1	1	1	−1	1	−1
χ_4	4	2	0	1	0	−1	−1
$\chi_{4'}$	4	−2	0	1	0	−1	1
χ_5	5	1	1	−1	−1	0	1
$\chi_{5'}$	5	−1	1	−1	1	0	−1
χ_6	6	0	−2	0	0	1	0

Remark. *χ_4 comes from the representation ρ_4 which acts on five-tuples summing to zero by the permutation representation and $\chi_{4'} = \chi_4 \cdot \chi_{1'}$.*

The only conjugacy class with order dividing 5 is (a,b,c,d,e) so the 5-regular conjugacy classes are $1, (a,b), (a,b)(c,d), (a,b,c), (a,b,c,d)$ and $(a,b)(c,d,e)$. As χ_1 and $\chi_{1'}$ are distinct on these conjugacy classes, they correspond to distinct irreducible Brauer characters which we will call ϕ_1 and $\phi_{1'}$.

As $5 \mid |S_5| = 60$ but $25 \nmid |S_5|$ it follows by Theorem 3.11 that the five-dimensional representations remain irreducible in characteristic five, therefore χ_5 and $\chi_{5'}$ correspond to the Brauer characters ϕ_5 and $\phi_{5'}$.

Lemma 3.13. *χ_4 and therefore $\chi_{4'}$ are reducible in characteristic 5.*

Proof. *If they were irreducible, then we would have six irreducible characters, and by Theorem 3.10 these would be all of the modular characters. In particular, as we know from the decomposition matrix D, this would imply that we can construct any ordinary character χ as a linear combination of our six irreducible characters on the 5-regular classes. But $(1,2)(3,4)$ is a 5-regular conjugacy class, and $\phi_1, \phi_{1'}, \phi_5$ and $\phi_{5'}$ are all non-negative on this class, but $\chi_6((a,b)(c,d)) = -2$, a contradiction.* ■

Lemma 3.14. *Let U denote the subspace of the five dimensional vector space over $\mathbb{F}_5$ which stays invariant under the permutation representation of S_5 and let $V = \{(a,a,a,a,a) : a \in \mathbb{F}_5\}$. If $W \subset U/V$ denotes the subspace of tuples that sum to zero, then W is an irreducible S_5-invariant subspace.*

Proof. *It is easy to see that W is indeed a subspace, and we can represent each element in W in a unique way by a four-tuple whose entries sum to zero by subtracting off $(1,1,1,1,1)$ times the last entry and then taking only the first four entries into account. For example $(4,4,4,1,2)$ becomes $(2,2,2,4,0)$ which we will represent by $(2,2,2,4)$. This space is three dimensional as $e_1 = (1,-1,0,0)$, $e_2 = (0,1,-1,0,0)$ and $e_3 = (0,0,1,-1)$ forms a basis. To show that W is irreducible we will consider all types of those four tuples, i.e. equivalence classes under permutation of elements by S_4 and scalar multiplication. In particular any nonzero tuple has a nonzero element, which we can permute into the first index, and we can make this element 1 by an appropriate scalar multiplication. A little calculation shows that we have the following five types of tuples:*

$$\begin{aligned} t_1 &= (1,4,0,0),\\ t_2 &= (1,1,3,0),\\ t_3 &= (1,1,1,2),\\ t_4 &= (1,4,1,4),\\ t_5 &= (1,4,2,3). \end{aligned}$$

To prove that W is irreducible it suffices to show that each tuple generates W, i.e. that we can get from any type to any other type.

$t_1 \to t_2$ *: As* $2t_1 = (2,3,0,0)$ *we see that* $(1,4,0,0) + (0,2,3,0) = (1,1,3,0) = t_2$.

$t_2 \to t_3$ *:* $(1,1,3,0) + (1,0,3,1) = (2,1,1,1) = t_3$.

$t_3 \to t_4$ *:* $(1,1,1,2) + (1,1,2,1) = (2,2,3,3)$ *and* $3(2,2,3,3) = (1,1,4,4)$.

$t_4 \to t_5$ *:* $2(1,4,4,1) = (2,3,3,2)$ *and* $(2,3,3,2) + (1,4,1,4) = (3,2,4,1)$.

$t_5 \to t_1$ *:* $(1,4,2,3) + (1,4,3,2) = (2,3,0,0) = 2t_1$.

We conclude that W has to be irreducible.

■

If we now take the quotient of the space of five-tuples which stays invariant under χ_4 and W we will find a one dimensional space which corresponds to a one dimensional representation, but which one is it? Using the basis (e_1, e_2, e_3) from above (we can write any tuple $(a,b,c,-a-b-c)$ as $ae_1 + (a+b)e_2 + (a+b+c)e_3$) we can calculate the character of W:

- 1 *acts as the identity matrix so it will have trace* 3.
- $(1,2)$ *sends* $e_1 \to -e_1$, $e_2 \to e_1 + e_2$ *and* $e_3 \to e_3$. *So the trace is* 1.
- $(1,2)(3,4)$ *sends* $e_1 \to -e_1$, $e_2 \to e_1 + e_2 + e_3$ *and* $e_3 \to -e_3$ *thus, its trace is* -1.
- $(1,2,3)$ *sends* $e_1 \to e_2$, $e_2 \to -e_1 - e_2$ *and* $e_3 \to e_1 + e_2 + e_3$, *which implies the trace is* 0.
- $(1,2,3,4)$ *sends* $e_1 \to e_2$, $e_2 \to e_3$ *and* $e_3 \to -e_1 - e_2 - e_3$. *So the trace is* -1.

- $(1,2)(3,4,5)$ *sends* $e_1 \to -e_1$, $e_2 \to e_1 + e_2 + e_3$. *But* e_3 *is sent to* $(0,0,0,1,-1)$ *which is equivalent to* $(1,1,1,2) = e_1 + 2e_2 - 2e_3$ *in our notation, so the trace equals* -2 *in this case.*

On the five-regular elements $\chi_4 - \chi_W$ *must be one of the one-dimensional representations we already know. Moreover*

	1	(a,b)	$(a,b)(c,d)$	(a,b,c)	(a,b,c,d)	$(a,b)(c,d,e)$
χ_4	4	2	0	1	0	-1
$-\chi_W$	3	1	-1	0	-1	-2
ϕ_1	1	1	1	1	1	1

But since χ_W *is irreducible, we will call this character* ϕ_3*. As this modular character is not invariant under tensoring with* $\phi_{1'}$*, the resulting representation will be another irreducible representation, which we will denote by* $\phi_{3'}$*.We have now found six irreducible characters ans so we are done. The charactertable of* S_5 *in characteristic* 5 *now reads:*

	1	(a,b)	$(a,b)(c,d)$	(a,b,c)	(a,b,c,d)	$(a,b)(c,d,e)$
ϕ_1	1	1	1	1	1	1
$\phi_{1'}$	1	-1	1	1	-1	-1
ϕ_3	3	1	-1	0	-1	-2
$\phi_{3'}$	3	-1	-1	0	1	2
ϕ_5	5	1	1	-1	-1	1
$\phi_{5'}$	5	-1	1	-1	1	-1

We can now also find the decomposition matrix and the Cartan matrix:

$$D = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$$C = DD^t = \begin{pmatrix} 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

4 The HeLP-method

4.1 Some properties regarding torsion units

> "When we talk mathematics, we may be discussing a secondary language built on the primary language of the nervous system."
>
> — John von Neumann

In this chapter we will start by studying some basic properties of group rings, afterwards we will turn our attention to the Zassenhaus conjecture and prove its equivalence to a criterion discovered by Marciniak-Ritter-Sehgal and Weiss. This criterion will translate the Zassenhaus Conjecture into a question about partial augmentations. We will also provide the reader with some results about the orders of torsion units in the integral group ring, in order to have enough machinery to prove the Zassenhaus Conjecture for two concrete cases in the next chapter. G will always be a finite group in this chapter. Most of the material here is taken from [60].

Lemma 4.1. *Let G be a finite group and let K be a field. Let ρ be the regular representation of KG and let $\alpha = \sum_{g\in G} \alpha_g g \in KG$. Then the trace of $\rho(g)$ is given by*

$$\mathrm{Tr}(\rho(\alpha)) = |G|\alpha_1.$$

Proof. Let $|G| = n$. As the trace of a matrix is base-independent, we can choose the set $G = \{g_1, \cdots, g_n\}$, where $g_1 = 1_G$, as a K-basis for KG. Then we see that

$$\rho(\alpha) = \rho\Big(\sum_{g\in G} \alpha_g g\Big) = \sum_{g\in G} \alpha_g \rho(g).$$

But for a non-identity element $g \in G$ we have that $gg_i \neq g_i$ for all i. This implies, by the definition of the regular representation, that all the diagonal entries of the matrix $\rho(g)$ are equal to zero. So it follows that $\mathrm{Tr}(\rho(g)) = 0$ for all such g. On the other hand, as $\rho(1_G)$ is the identity matrix, we have that $\mathrm{Tr}(\rho(1_G)) = n$. So we arrive at

$$\mathrm{Tr}(\rho(\alpha)) = \sum_{g\in G} \alpha_g \,\mathrm{Tr}(\rho(g)) = \alpha_1 \,\mathrm{Tr}(\rho(1_G)) = \alpha_1|G|.$$

■

This seemingly easy results has some very useful applications. We will start by a theorem of Berman and Higman which is of vital importance to the study of units in integral group rings.

Theorem 4.2. *(Berman-Higman) Let $\alpha = \sum_{g\in G} \alpha_g g$ be a torsion unit of $\mathbb{Z}G$, if $\alpha_1 \neq 0$ then $\alpha = \alpha_1 = \pm 1$.*

Proof. Let $|G| = n$ and assume that $\alpha^k = 1$ for some $k \in \mathbb{N}$. Let ρ be the regular representation of $\mathbb{C}G$ and consider $\mathbb{Z}G$ as a subring of $\mathbb{C}G$. By Lemma 4.1 it follows that $\mathrm{Tr}(\rho(\alpha)) = n\alpha_1$. As $\alpha^k = 1$ we have that $\rho(\alpha)^k = \rho(\alpha^k)$ equals the identity matrix. So $\rho(\alpha)$ is a root of the polynomial $x^k - 1$ which possesses distinct roots. This implies that there exists a basis, of $\mathbb{C}G$ for which the matrix of $\rho(\alpha)$ is diagonal and of the following form:

$$M = \begin{bmatrix} \varepsilon_1 & & \\ & \ddots & \\ & & \varepsilon_n \end{bmatrix} \quad \text{with } \varepsilon_i^k = 1.$$

Then $\mathrm{Tr}(\rho(\alpha)) = \sum_{i=1}^n \varepsilon_i$ and thus

$$n\alpha_1 = \sum_{i=1}^n \varepsilon_i.$$

Taking absolute values yields that

$$|n\alpha_1| = \Big|\sum_{i=1}^n \varepsilon_i\Big| \leq \sum_{i=1}^n |\varepsilon_i| = n.$$

As $|n\alpha_1| \leq n$ we must have $|\alpha_1| = 1$ and also $|\sum_{i=1}^n \varepsilon_i| = \sum_{i=1}^n |\varepsilon_i|$. This can only happen if $\varepsilon_1 = \varepsilon_2 = \cdots = \varepsilon_n$. So $n\alpha_1 = n\varepsilon_1$ which implies that $\alpha_1 = \varepsilon_1 = \pm 1$. We conclude that $\rho(\alpha) = \pm I$ which gives us that $\alpha = \pm 1$. ■

Corollary 4.2.1. *Let A be an abelian group, then all torsion units of $\mathbb{Z}A$ are trivial.*

Proof. Let α be a torsion unit of $\mathbb{Z}A$ of order n. Suppose that $\alpha_{g_0} \neq 0$ and that k is the order of g_0. Due to the commutativity of A it follows that $(\alpha g_0^{-1})^{nk} = 1$. Moreover $(\alpha g_0)_{1_G} = \alpha_{g_0} \neq 0$ so Theorem 4.2 implies that $\alpha g_0^{-1} = \pm 1$ and thus $\alpha = \pm g_0$. ■

The complex group algebra $\mathbb{C}G$ admits an involution. For $\alpha = \sum_{g\in G} \alpha_g g$ we can define $\alpha^* := \sum_{g\in G} \overline{\alpha_g} g^{-1}$. The following properties are easily verified:

- $(\alpha + \beta)^* = \alpha^* + \beta^*$
- $(\alpha\beta)^* = \beta^*\alpha^*$
- $(\alpha^*)^* = \alpha$

Lemma 4.3. *Let G be a finite group and let α be an element of $\mathbb{Z}G$, then $\alpha\alpha^* = 1$ if and only if $\alpha = \pm g$ for some $g \in G$.*

Proof. If $\alpha = \pm g$ then $\alpha^* = \pm g^{-1}$ so $\alpha\alpha^* = 1$. Suppose now that $\alpha = \sum_{g\in G} \alpha_g g$, $\alpha^* = \sum_{g\in G} \alpha_g g^{-1}$ with $\alpha\alpha^* = 1$, then:

$$1 = \alpha\alpha^* = \sum_{g\in G} \alpha_g^2 1_G + \sum_{g\in G, g\neq 1_G} \beta_g g.$$

It follows that $\sum_{g\in G} \alpha_g^2 = 1$ and thus $\alpha_h = \pm 1$ for a unique $h \in G$ and $\alpha_g = 0$ for all $g \neq h$. Hence $\alpha = h$ as claimed. ■

Lemma 4.4. *Let K be a field of characteristic 0 and let G be a group. Suppose that $e = \sum_{g\in G} e_g g$ is a non-trivial idempotent of KG, then $e_{1_G} \in \mathbb{Q}$, $|G|e_{1_G} \in \mathbb{Z}$ and $0 < e_{1_G} < 1$.*

Proof. Let ρ be the regular representation of $\mathbb{C}G$ and consider $\mathbb{Z}G$ as a subring of $\mathbb{C}G$. As e is an non-trivial idempotent, $\rho(e)$ will be an idempotent too. So the eigenvalues of $\rho(e)$ are zeros and ones. In particular, by Lemma 4.1 we see that:

$$\operatorname{Tr}(\rho(e)) = \sum_{g\in G} e_g \operatorname{Tr}(\rho(g)) = e_{1_G}|G| = m.$$

Where m is the number of eigenvalues of $\rho(e)$ equal to one. As e is a non-trivial idempotent, it follows that $0 < m < |G|$, which implies that $0 < e_{1_G} < 1$. Moreover $e_{1_G} = \frac{m}{|G|} \in \mathbb{Q}$. ■

Theorem 4.5. *Let $\alpha \in \mathbb{Z}G$ be a normalized torsion unit of order n. Then n is a divisor of $|G|$.*

Proof. Consider $e = \frac{1+\alpha+\cdots+\alpha^{n-1}}{n}$, an easy calculation shows that $e^2 = e$. Besides $e_{1_G} = \frac{1}{n}\sum_{i=0}^{n-1} \alpha_1^i$. Since the order of α is n, $\alpha^i \neq \pm 1$ for $1 \leq i < n$. By Theorem 4.2, $\alpha_1^i = 0$, from which we deduce that $e_{1_G} = \frac{1}{n}$. But from Lemma 4.4 we know that $e_{1_G} = \frac{k}{|G|}$, thus $kn = |G|$ as claimed. ■

This result shall later be strengthened in Corollary 4.12.1.

4.2 An equivalent criterion to (ZC)

This section is devoted to the Zassenhaus Conjecture which is one of the most important topics in this thesis, we will start by stating this conjecture:

Zassenhaus Conjecture 4.5.1.

(ZC) Let u be a normalized torsion unit in $V(\mathbb{Z}G)$, does there exists a $g \in G$ for which $u \underset{\mathbb{Q}G}{\sim} g$.

If one sees this conjecture for the first time, there is no obvious way to tackle this problem, even in a specific case it is not clear how to start proving this. It shall be shown in this section that there exists an interesting connection between (ZC) and a property regarding partial augmentations. In this way, results about partial augmentations can be used to study this conjecture.

We start by doing some preliminary work in order to prove an equivalent criterion for this conjecture. For any ring R, define $[R,R]$ to be the additive group generated by all the Lie products of the form $[x,y] = xy - yx$ with $x, y \in R$. It follows immediately from this definition that we have for a group ring RG that $[RG,RG] = \sum_{g,h\in G} R[g,h]$.

Theorem 4.6. *(Cliff's Formula) Let R be a ring of prime power characteristic p^n. For a positive integer $k \geq n$ and $a_1, \cdots a_m \in R$ we have that*

$$\Big(\sum_{i=1}^{m} a_i\Big)^{p^k} \equiv \sum \big(a_{i_1} \cdots a_{i_s}\big)^{p^{k-n+1}} \mod [R,R].$$

Where $s = p^{n-1}$, $\delta \in [R,R]$ and where the sum on the right-hand side runs over all s-tuples $(i_1, \cdots, i_s)$ with $1 \leq i_j \leq m$.

Proof. Set $t = p^k$ for convenience. Expanding the right-hand side using distributivity we find that

$$\Big(\sum_{i=1}^{m} a_i\Big)^{p^k} = \sum \big(a_{i_1} \cdots a_{i_t}\big)$$

where the sum on the right-hand side runs over all t-tuples $(i_1, \cdots, i_t)$ with $1 \leq i_j \leq m$. Let σ denote the cyclic permutation such that $\sigma(i_1, \cdots, i_t) = (i_2, i_3, \cdots, i_t, i_1)$ and let us define $\sigma(a_{i_1} \cdots a_{i_t}) := a_{i_2} a_{i_3} \cdots a_{i_t} a_{i_1}$. Then we see that:

$$\sigma(a_{i_1} \cdots a_{i_t}) - a_{i_1} a_{i_2} \cdots a_{i_t} = \big[a_{i_2} a_{i_3} \cdots a_{i_t}, a_{i_1}\big] \in \big[R,R\big]$$

Since $\sigma^t = I$ and by the orbit-stabilizer theorem, we know that the number of elements in any σ-orbit is a power of p. If $x = (a_{j_1} \cdots a_{j_t})$ is not of the form $(a_{i_1} \cdots a_{i_s})^{p^{k-n+1}}$ then $\sigma^{p^{n-1}}(j_1, j_2, \cdots, j_t) \neq (j_1, j_2, \cdots j_t)$. So the number of elements in the σ-orbit of x is a multiple of p^n and each of its elements is congruent to $x \mod [R,R]$. As the characteristic of R is exactly p^n, the proof is complete. ∎

In particular we observe that the following holds:

Corollary 4.6.1. *Let R be a ring with identity of prime characteristic p, for $x, y \in R$ and for every positive integer n we have that*

$$(x+y)^{p^n} \equiv x^{p^n} + y^{p^n} \mod [R,R].$$

Corollary 4.6.2. *Let R be a ring with identity of prime characteristic p, then the following holds:*

$$\alpha \in [R,R] \text{ implies that } \alpha^p \in [R,R].$$

Proof. Using the statement above it suffices to show that $(xy-yx)^p \in [R,R]$ for all $x, y \in R$. But

$$\begin{aligned}(xy-yx)^p &\equiv (xy)^p - (yx)^p \mod [R,R] \\ &\equiv [x,(yx)^{p-1}y] \mod [R,R].\end{aligned}$$

■

We will now turn to a more specific example, let us first denote $[\mathbb{Z}G, \mathbb{Z}G]$ by Λ.

Remark. *Observe that we have the following equality for intersection of $\mathbb{Q}G$ with Λ:*

$$\Lambda \cap \mathbb{Q}G = [\mathbb{Q}G, \mathbb{Q}G].$$

Lemma 4.7. *Let R be a ring, than we have the following characterization of Λ:*

$$\Lambda = [RG, RG] = \{\alpha = \sum_{g\in G} \alpha_g g \in RG : \varepsilon_C(\alpha) = 0 \text{ for every } C \in Cl(G)\}.$$

Proof.

Let us start by proving the inclusion from left to right. Suppose $\alpha, \beta \in RG$ then

$$[\alpha,\beta] = \Big[\sum_{g\in G} \alpha_g g, \sum_{h\in G} \beta_h h\Big] = \sum_{g\in G}\sum_{h\in G} \alpha_g\beta_h[g,h] = \sum_{g\in G}\sum_{h\in G} \alpha_g\beta_h(gh-hg).$$

So we conclude that Λ is the linear span of $(gh-hg)$, but since gh and hg are conjugate elements, we see that $\varepsilon_C(\alpha) = 0$ for every $\alpha \in \Lambda$ and every $C \in Cl(G)$.

For the other direction, suppose that $\alpha = \sum_{h\in G} \alpha_h h$ such that $\varepsilon_g(\alpha) = 0$, $\forall g \in G$. Let T be a transversal to the conjugacy classes. Take $g \in T$, for every $h \in g^G/\{g\}$ we can find an element $k_h \neq g$ such that $h = k_h^{-1} g k_h$. Remark that $h - g = [k_h^{-1}g, k_h]$, thus $\sum_{h\in G} \alpha_h(h-g) \in [RG, RG]$. But now we see that

$$\begin{aligned}\sum_{h\in G} \alpha_h(h-g) &= \sum_{g\in T}\sum_{h\in g^G} \alpha_h(h-g) \\ &= \sum_{g\in T}\sum_{h\in g^G} \alpha_h h - \sum_{g\in T}\sum_{h\in g^G} \alpha_h g \\ &= \alpha - \sum_{g\in T} g \sum_{h\in g^G} \alpha_h = \alpha\end{aligned}$$

which completes the proof.

■

Lemma 4.8. *[29]. Let L/K be an extension of infinite fields, let A be a finite dimensional K-algebra and let $B = L \otimes_K A$. Let X and Y be finite subsets of A which are conjugate in B. Then they are conjugate in A too.*

Proof. Let us start by fixing a K-basis $\{v_1, \cdots v_k\}$ of A and let u be a unit of B such that $u^{-1}Xu = Y$. Let y_x denote $u^{-1}xu$ for every $x \in X$. Our assumption now implies that the system of equations $Zy_x = xZ$ has a solution on the units of B. We can express this in terms of our F-basis of A to obtain a system of homogeneous linear equations in k unknowns, with coefficients in K. This system has a solution $(s_1, \cdots s_k)$ in L such that $s_1v_1 + \cdots s_kv_k$ is a unit of B. Let $c_1, \cdots c_l$ be a K-basis of the set of solutions of this equations and take look at the polynomial :

$$P(Z_1, \cdots Z_l) = \mathrm{Nr}_{A/K}(Z_1c_1 + \cdots + Z_lc_l) = \mathrm{Nr}_{B/L}(Z_1c_1 + \cdots + Z_lc_l).$$

It follows by linear algebra that $c_1, \cdots c_l$ is also an L-basis of the sets of solutions in L. So we have

$$s_1v_1 + \cdots s_kv_l = z_1c_1 + \cdots z_kc_l,$$

for some $z_1, \cdots z_l \in L$ which implies that $f(z_1, \cdots z_l) \neq 0$. In particular, P can not be the zero polynomial so $P(u_1, \cdots, u_l) \neq 0$ for some $u_1 \cdots u_l \in K$ as K is an infinite field. Therefore $c = u_1c_1 + \cdots + u_lc_l$ is an element of A for which $\mathrm{Nr}_{A/K} \neq 0$ and we have that

$$cy_x = xc \text{ for all } x \in X.$$

This implies that c is a unit of A for which $c^{-1}Xc = Y$. ■

The following is an immediate consequence of this lemma:

Corollary 4.8.1. *Two elements x and y in $U(\mathbb{C}G)$ are conjugate in $U(\mathbb{Q}G)$ if and only if x and y are conjugate elements in $U(\mathbb{C}G)$.*

Lemma 4.9. *Let M and N be subgroups of $V(\mathbb{Z}G)$, then M and N are conjugate if and only if there exists an isomorphism $f : M \to N$ such that $\chi(m) = \chi(f(n))$ for all $m \in M$ and for all irreducible characters χ of G.*

Proof. Necessity is obvious as characters remain constant under conjugation. Now start by fixing a representation ρ_χ affording χ for all irreducible representations χ of G. Then the map

$$T = (\rho_\chi)_{\chi \in Irr(C)} : \mathbb{C}G \to \prod_{\chi \in Irr(G)} M_{\chi(1)}(\mathbb{C}),$$

which corresponds to the Wedderburn Decomposition, is an isomorphism. Moreover the representations $\rho_\chi \mid_M$ and $\rho_\chi \mid_N \circ f$ of M afford the same characters, so they are similar representations, which means that there is a matrix $U_\chi \in M_{\chi(1)}(\mathbb{C})$ such that

$$\rho_\chi f(m) = U_\chi^{-1}\rho_\chi(m)U_\chi,$$

for every $m \in M$. Therefore the image of $(U_\chi)_{\chi \in Irr(G)}$ is a unit in $\mathbb{C}G$ such that

$$u^{-1}mu = f(m), \text{ for all } m \in M.$$

In particular $u^{-1}Mu = f(M) = N$, so M and N are conjugate in $\mathbb{C}G$ and thus rationally conjugate by Lemma 4.8. ■

We can now finally state the principal result of this section.

Theorem 4.10. *Marciniak-Ritter-Sehgal-Weiss [52]*
Let U be a subgroup of torsion units in $V(\mathbb{Z}G)$. Then the following are equivalent:

1. *For every $u \in U$ there exists a group element $g \in G$ such that $u \underset{\mathbb{Q}G}{\sim} g$.*
2. *For every $u = \sum_{g\in G} u_g g \in U$ there exists a $g_0 \in G$, unique up to conjugacy, such that $\varepsilon_{g_0}(u) \neq 0$*

Proof. Let us first start by showing that 1. implies 2. We know that there exists an $\alpha \in \mathbb{Q}G$ and a $g_0 \in G$ such that $u = \alpha g_0 \alpha^{-1}$. Therefore

$$u - g_0 = \alpha g_0 \alpha^{-1} - g_0 = \left[\alpha g_0, \alpha^{-1}\right] \in \Big[\mathbb{Q}G, \mathbb{Q}G\Big] \cap \mathbb{Z}G.$$

So by lemma 4.2 and lemma 4.7 we see that $u - g_0 \in \Lambda$ so $\varepsilon_h(u) = \varepsilon_h(g_0) = 1$ and $\varepsilon_g(u) = 0$ for $g \not\sim g_0$.

For the converse, assume that for every $u \in U$ there is a unique g_0 up to conjugacy such that $\varepsilon_{g_0}(u) \neq 0$, we shall prove that $u \underset{\mathbb{Q}G}{\sim} g_0$. Using the same argument as in the second part of the proof of lemma 4.7 we can write

$$u = \sum_{g\in G} u_g g = \varepsilon_{g_0}(u) \cdot g_0 + \lambda, \quad \lambda \in \Lambda.$$

But since u has augmentation 1, we get

$$u = g_0 + \lambda, \quad \lambda \in \Lambda.$$

Then we use lemma 4.6.1, such that for every prime p we have

$$u^p \equiv g_0^p \mod (\Lambda + p\mathbb{Z}G).$$

Now applying our assumption to $u^p = \sum_{g\in G} k_g g$ we find an $h \in G$ for which $\varepsilon_h(u^p) \neq 0$. Thus $u^p \equiv h \mod \Lambda$, and combining this with the previous equation we arrive at $h \equiv g_0^p \mod (\Lambda + p\mathbb{Z}G)$.

Now suppose that $h \not\sim g_0^p$ and take $\lambda \in \Lambda, k \in \mathbb{Z}G$ such that $h - g_0^p + pk = \lambda$. Then

$$0 = \varepsilon_h(\lambda) = \varepsilon_h(h - g_0^p + pk) = 1 - 0 + pn \neq 0 \quad \text{with } n \in \mathbb{Z}.$$

So $h \sim g_0^p$ but this implies that $h - g_0^p \in \Lambda$, so $u^p \equiv g_0^p \mod \Lambda$. Going through all primes p we arrive at

$$u^k \equiv g_0^k \mod \Lambda \quad \forall k \in \mathbb{N}.$$

But then, u and g_0 need to have the same order. Suppose not, then there exists an $n \in \mathbb{N}$ for which $u^n = 1$ and $g_0^n \neq 1$ and thus $1 - g_0^n \in \Lambda$, which is a contradiction as

$$\varepsilon_1(1 - g_0^n) = 1.$$

So we conclude that $o(u) = o(g_0)$.
In addition as $u^k - g_0^k \in \Lambda$ we see that

$$\chi(u^k - g_0^k) = \chi\Big(\sum_{g\in G} \alpha_g g\Big) = \sum_{C\in Cl(G)} \sum_{g\in C} \alpha_g \chi(g) = \sum_{C\in Cl(G)} \chi(C) \sum_{g\in C} \alpha_g = 0.$$

Where we used in the last step the fact that $u^k - g_0^k \in \Lambda$ implies that $\sum_{g\in C} \alpha_g = 0$. So $\chi(u^k) = \chi(g_0^k)$ for each $k \in \mathbb{N}$ and we can use Corollary 4.9 to finish our proof. ■

Corollary 4.10.1. *(ZC) holds for G if and only if for every torsion unit $u = \sum_{g\in G} u_g g \in V(\mathbb{Z}G)$ there exists a $g_0 \in G$ unique up to conjugacy, with $\varepsilon_{g_0}(u) \neq 0$.*

4.3 Partial augmentations of units in $\mathbb{Z}G$

Theorem 4.11. *Let u be a torsion unit in $V(\mathbb{Z}G)$ of order n and $g \in G$. If there exists a prime p such that $p|o(g)$ but $p \nmid n$ then $\varepsilon_g(u) = 0$.*

Proof. Since p is a unit of finite order in $\mathbb{Z} \mod o(u)$ there exist infinitely many k's such that $o(u)|(p^k - 1)$. Fix one such $k > n$. It follows that $u^{p^k} = u$. By Cliff's formula, Theorem 4.6 we have that:

$$\begin{aligned} u &= \sum_{g\in G} u_g g = \Big(\sum_{g\in G} u_g g\Big)^{p^k} \\ &= \sum (u_{i_1} u_{i_2} \cdots u_{i_s})^{p^{k-n+1}} (g_{i_1} g_{i_2} \cdots g_{i_s})^{p^{k-n+1}} + \lambda \end{aligned}$$

with $\lambda \in [\mathbb{Z}G, \mathbb{Z}G] + p^n\mathbb{Z}$ and $s = p^{n-1}$. Now take k so large that all the orders of the $(g_{i_1} g_{i_2} \cdots g_{i_s})^{p^{k-n+1}}$ on the right-hand side are relatively prime to p. As the orders of elements in the same conjugacy class are equal it follows that $\varepsilon_g(u) \equiv 0 \mod p^n$. Increasing n yields that $\varepsilon_g(u) = 0$. ■

Theorem 4.12. *(Zassenhaus) Let $u \in V(\mathbb{Z}G)$ be a torsion unit of prime power order p^n. Then there exists an element $g \in G$ in the support of u such that the order of g is p^n and with $\varepsilon_g(u) \neq 0$.*

Proof. Let $u = \sum_{g\in G} u_g g$, it then follows by Corollary 4.6.1 that

$$1 = u^{p^n} \equiv \sum_{g\in G} u_g^{p^n} g^{p^n} + \lambda \mod p\mathbb{Z}G$$

for some $\lambda \in [\mathbb{Z}G, \mathbb{Z}G]$. Thanks to Lemma 4.7 we know that the coefficient of the identity in λ equals zero. Therefore, taking augmentations, we have that

$$\begin{aligned} 1 &\equiv \sum_{g\in G, g^{p^n}=1} u_g^{p^n} \mod p \\ &\equiv \sum_{g\in G, o(g)=p^n} u_g^{p^n} + \sum_{g\in G, g^{p^{n-1}}=1} u_g^{p^n} \mod p \\ &\equiv \sum_{g\in G, o(g)=p^n} u_g^{p^n} + \Big(\sum_{g\in G, g^{p^{n-1}}=1} u_g^{p^{n-1}}\Big)^p \mod p. \end{aligned}$$

But since the order of u is p^n we know by Theorem 4.2 that

$$\varepsilon_1(u^{p^{n-1}}) \equiv \sum_{g\in G, g^{p^{n-1}}=1} u_g^{p^{n-1}} \equiv 0 \mod p.$$

Thus we can conclude that

$$\sum_{g\in G, o(g)=p^n} u_g^{p^n} \not\equiv 0 \mod p.$$

Moreover, there exists an element $g \in G$ of order p^n such that $\varepsilon_g(u) \neq 0$. ■

Corollary 4.12.1. *The exponent of the normalized torsion unit group $V(\mathbb{Z}G)$ and the exponent of G are equal.*

Proof. For every prime p we know that there exists an element in G of order p^k if and only if there exists an element in $V(\mathbb{Z}G)$ of the same order. In particular the least common multiple of orders of torsion elements in $V(\mathbb{Z}G)$ coincides with the least common multiple of orders of elements in G. ■

4.4 The HeLP-method

The HeLP-method was born in 1989, when the Indian mathematicians Luthar and Passi developed a method to prove the Zassenhaus Conjecture for the alternating group of degree 5 in [50]. Only ordinary characters where used until 2007 when Hertweck [37] extended this method by the use of Brauer characters. This method is named after its creators HErtweck, Luthar and Passi.

We will start by doing some linear algebra. Let M be an $n \times n$ matrix of order k over an algebraic closed field F with $k \geq 1$. It follows by linear algebra that M is a diagonalizable matrix and that the eigenvalues of M are roots of unity. Let Z be a primitive k-th root of unity and denote the multiplicity of Z^l as an eigenvalue of M by μ_l.

Lemma 4.13. *With the same notation as above we have that*

$$\mu_l = \frac{1}{k}\sum_{r=0}^{k-1} \mathrm{Tr}(M^r)Z^{-lr} \quad l \in \{0, 1, \cdots k-1\}.$$

Proof. As M is a diagonalizable matrix we know that

$$\mathrm{Tr}(M) = \mu_0 1 + \mu_1 Z + \cdots + \mu_{k-1}Z^{k-1}$$

from which it follows that

$$\mathrm{Tr}(M^r) = \mu_0 1 + \mu_1 Z^r + \cdots + \mu_{k-1}Z^{(k-1)r} \quad \text{for all } r.$$

Putting those equations in matrix form we arrive at

$$\begin{bmatrix} 1 & Z & Z^2 & \cdots & Z^{k-1} \\ 1 & Z^2 & Z^4 & \cdots & Z^{2(k-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & Z^{(k-1)} & Z^{2(k-1)} & \ldots & Z^{(k-1)^2} \end{bmatrix} \begin{bmatrix} \mu_0 \\ \mu_1 \\ \vdots \\ \mu_{k-1} \end{bmatrix} = \begin{bmatrix} \mathrm{Tr}(M) \\ \mathrm{Tr}(M^2) \\ \vdots \\ \mathrm{Tr}(M^{k-1}) \end{bmatrix}$$

As the $k-1 \times k-1$ matrix on the left is a Vandermonde matrix, it is an invertible matrix. Solving for the μ_i now yields

$$\mu_l = \frac{1}{k}\sum_{r=0}^{k-1} \mathrm{Tr}(M^r)Z^{-lr} \quad l \in \{0, 1, \cdots k-1\}.$$

■

This lemma will be of vital importance in the development of the HeLP-method which will be treated in the remainder of this section but before diving deeper into this we need the following two lemmas:

Lemma 4.14. *Let $\{C_1, C_2, \cdots C_r\}$ be the conjugacy classes of G and let u be a torsion unit in $V(RG)$. Then the following formula holds for every ordinary character of G:*

$$\chi(u) = \sum_{i=1}^{r} \varepsilon_{C_i}(u)\chi(C_i).$$

Proof. Let $u = \sum_{g\in G} u_g g$ be a torsion unit in $V(RG)$ and let χ be an ordinary character of G then we have that:

$$\chi(u) = \chi\Big(\sum_{g\in G} u_g g\Big) = \sum_{i=1}^{r}\sum_{g\in C_i} u_g\chi(C_i) = \sum_{i=1}^{r} \varepsilon_{C_i}(u)\chi(C_i).$$

■

But one could do something similar for Brauer characters:

Lemma 4.15. *[36] Let u be a p-regular torsion unit and let ϕ be a Brauer character of G relative to p. If $\{g_1, g_2, \cdots g_r\}$ are representatives of the p-regular conjugacy classes of G then*

$$\phi(u) = \sum_{i=1}^{r} \varepsilon_{g_i}(u)\phi(g_i)$$

Proof. By Theorem 3.9 it is sufficient to prove this theorem for irreducible Brauer characters relative to p. Let $\chi_1, \cdots \chi_l$ be the ordinary irreducible characters of G and let $\phi_1, \cdots \phi_r$ be the Brauer characters of G relative to p. By the definition of the decomposition matrix D we have:

$$D\begin{pmatrix}\phi_1(u)\\ \phi_2(u)\\ \vdots\\ \phi_r(u)\end{pmatrix} = \begin{pmatrix}\chi_1(u)\\ \chi_2(u)\\ \vdots\\ \chi_l(u)\end{pmatrix} = \sum_{i=1}^{r}\varepsilon_{g_i}(u)\begin{pmatrix}\chi_1(g_i)\\ \chi_2(g_i)\\ \vdots\\ \chi_l(g_i)\end{pmatrix} = \sum_{i=1}^{r}\varepsilon_{g_i}(u)D\begin{pmatrix}\phi_1(g_i)\\ \phi_2(g_i)\\ \vdots\\ \phi_r(g_i)\end{pmatrix} = D\begin{pmatrix}\sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_1(g_i)\\ \sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_2(g_i)\\ \vdots\\ \sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_r(g_i)\end{pmatrix}.$$

Multiplying both sides with D^t we find that

$$C\begin{pmatrix}\phi_1(u)\\ \phi_2(u)\\ \vdots\\ \phi_r(u)\end{pmatrix} = C\begin{pmatrix}\sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_1(g_i)\\ \sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_2(g_i)\\ \vdots\\ \sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_r(g_i)\end{pmatrix},$$

where C is the Cartan matrix, an invertible matrix, so we end up with

$$\begin{pmatrix}\phi_1(u)\\ \phi_2(u)\\ \vdots\\ \phi_r(u)\end{pmatrix} = \begin{pmatrix}\sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_1(g_i)\\ \sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_2(g_i)\\ \vdots\\ \sum_{i=1}^{r}\varepsilon_{g_i}(u)\phi_r(g_i)\end{pmatrix}$$

by which we are done. ■

Let u be a unit in $\mathbb{Z}G$ with $u^k = 1$ and $k \geq 1$ and let χ be an ordinary character of G of degree n which is afforded by the representation ρ. Using Lemma 4.13 and the fact that an ordinary character is just a trace, we see that

$$\mu_l(u;\chi) = \frac{1}{k}\sum_{r=0}^{k-1} \chi(u^r)Z^{-lr} \quad l \in \{0,1,\cdots k-1\}.$$

Remark that we can use the same reasoning as in the proof of Lemma 4.13 to conclude that this formula also holds true if u is a p-regular torsion unit of $\mathbb{Z}G$ and χ a Brauer character of G relative to p.

In each case, if we collect all those r which have the same greatest common divisor we can rewrite this as

$$\mu_l(u;\chi) = \frac{1}{k}\sum_{d|k} \sum_{\substack{r \bmod \frac{k}{d} \\ (r,\frac{k}{d})=1}} \chi(u^{dr})Z^{-dlr} \quad l \in \{0,1,\cdots k-1\}.$$

Let us now consider the field $\mathbb{Q}(Z^d)$, the field automorphisms are given by mappings of the form $\sigma_r : Z^d \to Z^{dr}$. As $(u^d)^{\frac{k}{d}} = 1$ it follows that $\chi(u^d)$ is a sum of n $\frac{k}{d}$-roots of unity, say $\epsilon_1, \cdots, \epsilon_n$. We now see that

$$\chi(u^{dr}) = \epsilon_1^r + \cdots \epsilon_n^r = \sigma_r(\chi(u^d))$$

if we combine this with the fact that $\mathbb{Q}(Z^d)$ is a Galois extension over $\mathbb{Q}$, we finally arrive at

$$\mu_l(u;\chi) = \frac{1}{k}\sum_{d|k} \mathrm{Tr}_{\mathbb{Q}(Z^d)/\mathbb{Q}}\left(\chi(u^d)Z^{-dl}\right) \quad l \in \{0,1,\cdots k-1\}.$$

Rewriting $\chi(u^d)$ using Lemma 4.14 or 4.15 yields:

Theorem 4.16. *Let u be a torsion unit in $\mathbb{Z}G$ of order k with $k > 1$ and let Z be a primitive k-root of unity. Let p be a prime divisor of $|G|$, coprime with k. Let χ be either an ordinary character or a Brauer character relative to p of G. Then the following formula equals a non-negative integer*

$$\frac{1}{k}\sum_{d|k} \mathrm{Tr}_{\mathbb{Q}(Z^d)/\mathbb{Q}}\left(\chi(u^d)Z^{-dl}\right) = \frac{1}{k}\sum_{d|k}\sum_{C^* \in Cl(G)} \varepsilon_C(u^d)\,\mathrm{Tr}_{\mathbb{Q}(Z^d)/\mathbb{Q}}\left(\chi(C)Z^{-dl}\right).$$

Where C^* denotes a p-regular conjugacy class in the case that χ is a Brauer character.

The fact that this formula equals a non-negative integer will be the cornerstone of the HeLP-method. This will be illustrated in Chapter 5, were it shall be used over and over again to prove (ZC) for A_5 and S_5.

The remainder of this section is devoted to a rather deep result of Hertweck which will turn out to be important in the proof of (ZC) for S_5:

Theorem 4.17. *(Hertweck) [37] For u a torsion element of $V(\mathbb{Z}G)$ of order n, $C \in Cl(G)$, for which the order of the elements in C does not divide n, we have that $\varepsilon_C(u) = 0$.*

We will need the following results:

Theorem 4.18. *Let R be a Dedekind domain of characteristic zero and let G be a finite group. Suppose that ${}_{RG}M$ is a projective module, let χ denote the character afforded by ${}_{RG}M$ and suppose that $g \in G$ such that $o(g)$ is not invertible in RG then $\chi(g) = 0$.*

Proof. See (32.15) in [26]. ■

Theorem 4.19. *Let F be a field of characteristic a prime p and let C be a finite cyclic p-group with generator c. If M is a finite dimensional FC-module of degree k then M is indecomposable if and only if $1 \leq k \leq |C|$ and the Jordan form of the matrix which denotes multiplication by c, $\rho(c)$, is an elementary Jordan matrix $J_k(a)$. Moreover M is a projective module if and only if $k = |C|$.*

Proof. See Chapter 2 of [2]. ■

Remark that the order of $\rho(c)$ and thus of the elementary Jordan matrix $J_k(a)$ is a power of p. So a has to be a root of unity of order a power of p. But F has characteristic p which implies that $0 = (a^{p^n} - 1) = (a-1)^{p^n}$ which in turn implies that $a = 1$. So M is an indecomposable module if and only if $\rho(c) \sim J_k(1)$. Moreover FC has a unique indecomposable projective FC-module because such a module has dimension $|C|$, and FC itself is a projective module of dimension $|C|$.

Theorem 4.20. *Let R be a Dedekind domain of characteristic zero and let M be a representation of RG. If H is a subgroup of G then ${}_{RG}M$ is projective if and only if ${}_{RH}M$ is projective and $R/P \otimes_R M$ is projective as a $(R/P)G$-module for every maximal ideal P of R containing $[G.H]$.*

Proof. See Theorem 9.1 in [9]. ■

The following short introduction to the double action formalism is taken from [31].

Let G be a finite group and let $C = \langle c \rangle$ be a cyclic group of order k. Let u be a torsion unit of order k in $V(RG)$, then the left $R(C \times G)$ module, ${}_G(RG)_u$ is defined in the following way: as an R-module it equals RG and the right action of $C \times G$ is given by:

$$(C \times G) \times_G (RG)_u : ((c^i, g), x) \to u \cdot x \cdot g^{-1}.$$

Where the product on the left hand side is taken in the ring RG. This $R(C \times G)$-module is called the **double action module** associated with the unit u.

Lemma 4.21. *Let u be a torsion unit in $V(RG)$ of order k and let*

$$\chi_u : C \times G \to R$$

denote the character of the $R(C \times G)$ module ${}_G(RG)_u$. Then

$$\chi_u((u^i, g)) = |C_G(g)|\varepsilon_g(u^i).$$

Proof. The trace of the linear map $f(g,h) : RG \to RG : x \to g \cdot x \cdot h^{-1}$ for arbitrary $g, h \in G$ equals the number of $x \in G$ for which $g \cdot x \cdot h^{-1} = x$ which is the same as the number of $x \in G$ for which $g = x \cdot h \cdot x^{-1}$. This number equals $|C_G(g)|$ if g and h are conjugate and is zero otherwise. If we now consider the torsion unit $u^i = \sum_{g \in G} a_g g$ we see that the linear endomorphism of RG which is induced by (c^i, g) equals $\sum_{h \in G} a_h f(h,g)$. The character value $\chi_u((u^i, g))$ is nothing more than the trace of this map so we see that

$$\sum_{h \in G} a_h \operatorname{Tr}(f(h,g)) = \sum_{h \sim g} a_h |C_G(g)| = |C_G(g)|\varepsilon_g(u^i).$$

■

Lemma 4.22. *Let M be an RG-module and let H be a subgroup of G such that $[G : H]$ is invertible in R. If ${}_{RH}M$ is projective then so is ${}_{RG}M$.*

Proof. The proof follows the same structure as the famous proof of Maschke's Theorem. Suppose that ${}_{RH}M$ is projective and let $\alpha : N \to M$ be a surjective RG-module homomorphism. We have to show that there exists an RG-module homomorphism $\beta : M \to N$ such that $\alpha\beta = 1_N$.

As ${}_{RH}M$ is projective it follows that there exists a homomorphism $\gamma : M \to N$ of RH-modules such that $\alpha\beta = 1_M$. Now fix a right transversal T of H in G. Then we have that for all $g \in G$ there are unique $t_g \in T$ and h_g in H such that $tg = h_g t_g$. Moreover the map $t \to t_g$ is a permutation of elements of T. Now define

$$\beta(m) = \frac{1}{[G : H]} \sum_{t \in T} t^{-1}\gamma(tm).$$

This is a homomorphism of RG-modules because for $g \in G$ and $m \in M$ we see that

$$\begin{aligned}
\beta(gm) &= \frac{1}{[G : H]} \sum_{t \in T} t^{-1}\gamma(tgm) = \frac{1}{[G : H]} \sum_{t \in T} t^{-1}\gamma(h_g t_g m) \\
&= \frac{1}{[G : H]} \sum_{t \in T} t^{-1} h_g \gamma(t_g m) = g\frac{1}{[G : H]} \sum_{t \in T} t_g^{-1}\gamma(t_g m) \\
&= g\frac{1}{[G : H]} \sum_{t \in T} t^{-1}\gamma(tm) = g\beta(m).
\end{aligned}$$

Moreover $\alpha\beta(m) = \frac{1}{[G:H]} \sum_{t \in T} t^{-1}\alpha\gamma(tm) = m$ as α is a homomorphism of RG-modules. So $\alpha\beta = 1_M$. ■

Lemma 4.23. *(Hertweck) [36] Let p be a prime integer and let F be a field of characteristic p. Let C be a non-trivial cyclic p-group and let P be the subgroup of C of order p. Let M be an FC-module which is finitely generated over F. Then M_{FC} is projective if and only if M_{FP} is projective.*

Proof. If M_{FC} is projective then FC_{FP} is free which implies that M_{FP} is projective. To prove the converse it is sufficient to prove the result for indecomposable modules. So assume that M_{FC} is indecomposable with $|C| > p$, denote the generator of C by c. Thanks to Theorem 4.19 the matrix expression of the multiplication map by c in a suitable basis $v_1, \cdots v_k$ is given by

$$\rho(c) = J_k(1) = \begin{pmatrix} 1 & & & & \\ 1 & 1 & & & \\ & & \ddots & \ddots & \\ & & & 1 & 1 \end{pmatrix}$$

with $1 \leq k \leq |C|$. We also know that M_{FC} is projective if and only if $k = |C|$ so we will show the latter. Suppose that M_{FP} is projective and let $q = \frac{|C|}{p}$. Then $P = \langle g^q \rangle$ and $\rho(g^q) = J_k(1)^q$. Therefore:

$$g^q v_i = \begin{cases} v_i + v_{i+q} & i + q \leq k \\ v_i & \text{otherwise} \end{cases} \tag{1}$$

But as M_{FP} is projective, the action of p on M can not be trivial, in particular $J_k(1)^q \neq 1$, which implies that $k > q$ because we work in characteristic p. Now choose non-negative integers s and t with $t < q$ such that

$$k - q + 1 = sq + t. \tag{2}$$

Let $I = \{t + iq \mid i \in \{0, 1, \cdots, s\}\}$ and $J = \{1, \cdots, k\}\ I$. Define $M_I = \sum_{i \in I} Fv_i$ and $M_J = \sum_{j \in J} Fv_j$. It is easy to see that $M = M_I \oplus M_J$ and (1) implies that both M_I and M_J are submodules of FP. But M_{FP} is a projective module and hence both M_I and M_J are projective as well which implies that the dimension of both has to be a multiple of the dimension of the unique projective indecomposable FP-module, which has dimension p. So we deduce that $p|(s+1)$ and $p|k$. In view of (2), this implies that $t \equiv 1 \mod p$ and hence $|C| \mid (s+1)q = k + 1 - t \geq |C|$. But we know that $1 \leq k \leq |C|$ and that $1 - t \leq 1$, so this implies that $k = |C|$ as desired. ■

Finally we will prove our main result:

Proof of Theorem 4.17. Take a $g \in C$, so we know that $o(g) \nmid o(u)$ which implies that there exists a prime integer p and a non-negative integer n such that $p^n | o(g)$ but $p^n \nmid o(u)$. Let $R = \mathbb{Z}_{(p)}$ be the localization of $\mathbb{Z}$ at (p) and take $F = R/pR \cong \mathbb{Z}/p\mathbb{Z}$. Define $M =_G (RG)_u$. Take $C = \langle (u, g) \rangle = P \times H$ where P is the Sylow p-group of C and let Q be the subgroup of order p in P. By our assumption on the orders of u and g it follows that $Q = \langle (1, k) \rangle$ where $\langle k \rangle$ is the subgroup of order p of $\langle g \rangle$. Then ${}_{FQ}(F \otimes_R M) \cong_{F\langle k \rangle} (F \otimes_R M) \cong_{F\langle k \rangle} FG = F\langle k \rangle^{[G:\langle k \rangle]}$. But this is free and hence projective so we can use Theorem 4.20, with $G = P$ and $H = 1$ to see that ${}_{FP}(F \otimes_R M)$ is projective. Because of the fact that $[C : P]$ is invertible in R we can invoke Lemma 4.22 to deduce that ${}_{RC}M$ is projective . Furthermore $p \mid o((u, g))$ so $o((u, g))$ is not invertible in R and we can use Theorem 4.18 to arrive at $\chi((u, g)) = 0$. Finally, using Lemma 4.21, we obtain that $\varepsilon_g(u) = \frac{\chi((u,g))}{|C_G(g)|} = 0$. ■

5 The HeLP-method at work: proving (ZC)

5.1 Luthar-Passi and A5

> "The source of all great mathematics is the special case, the concrete example. It is frequent in mathematics that every instance of a concept of seemingly great generality is in essence the same as a small and concrete special case."
>
> — P.R. Halmos, I Want to be a Mathematician, 1985.

The main purpose of this chapter is to illustrate all the machinery developed so far in order to tackle (ZC) in two concrete cases, namely for A_5 and S_5. We will start by proving that (ZC) holds for the alternating group on 5 elements. As the use of Brauer characters does not simplify the calculations in this case, we will only use ordinary characters in the first section. We follow [50].

Theorem 5.1. *For every torsion unit $u \in V(\mathbb{Z}A_5)$ there exists a $g \in A_5$ such that $u \underset{\mathbb{Q}G}{\sim} g$, i.e. (ZC) holds for A_5.*

Proof. Let $S = (1,2)(3,4)$, $T = (1,2,3)$, $V = (1,2,3,4,5)$ and $V^2 = (1,3,5,2,4)$. The character table for A_5 is given by

		S	T	V	V^2
	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_3	3	-1	0	$\frac{1+\sqrt{5}}{2}$	$\frac{1-\sqrt{5}}{2}$
$\chi_{3'}$	3	-1	0	$\frac{1-\sqrt{5}}{2}$	$\frac{1+\sqrt{5}}{2}$
χ_4	4	0	1	-1	-1
χ_5	5	1	-1	0	0

Let $u = \sum_{g \in G} u_g g$ be a normalized torsion unit of $\mathbb{Z}A_5$ of order $k > 1$. And write

$$v_i = \varepsilon_{C_i}(u) \text{ for } i \in \{1, 2, \cdots 5\}$$

for the partial augmentations of u. By Theorem 4.2 it follows that $v_1 = 0$. Since u has augmentation one:

$$v_2 + v_3 + v_4 + v_5 = 1.$$

And for all ordinary characters χ of A_5 we have that

$$\chi(u) = v_2\chi(S) + v_3\chi(T) + v_4\chi(V) + v_5\chi(V^2).$$

By Theorem 4.5 the only possible orders of u are divisors of 60, so it will be sufficient to prove that normalized torsion units of order $2, 3$ and 5 are rationally conjugate to group

elements and that there do not exist normalized torsion units of order $4, 6, 10$ and 15. By Theorem 4.11 we can reduce the possible values of partial augmentations:

order	partial augmentations	
$k=2$	$v_3 = v_4 = v_5 = 0$	$v_2 = 1$
$k=3$	$v_2 = v_4 = v_5 = 0$	$v_3 = 1$
$k=4$	$v_3 = v_4 = v_5 = 0$	$v_2 = 1$
$k=5$	$v_2 = v_3 = 0$	$v_4 + v_5 = 1$
$k=6$	$v_4 = v_5 = 0$	$v_2 + v_3 = 1$
$k=10$	$v_3 = 0$	$v_2 + v_4 + v_5 = 1$
$k=15$	$v_2 = 0$	$v_3 + v_4 + v_5 = 1$

When $k \in \{2, 3\}$ it follows by Corollary 4.11 that $u \underset{\mathbb{Q}G}{\sim} S$ and $u \underset{\mathbb{Q}G}{\sim} T$ respectively.

- $k = 4$: By Theorem 4.12 there does not exist a torsion unit of order 4 in $\mathbb{Z}A_5$.
- $k = 5$: Let $Z = e^{\frac{2\pi i}{5}}$, we will need some trace values, which we state in the following table:

$Z = e^{\frac{2\pi i}{5}}$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left((1+\sqrt{5})Z^{-l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left((1-\sqrt{5})Z^{-l}\right)$
$l=1$	4	-6
$l=2$	-6	4

We know that $\chi_3(u) = v_4\chi_3(V) + v_5\chi_3(V^2)$, by Lemma 4.13 we have that

$$\begin{aligned}
\mu_l(u;\chi_3) =& \frac{1}{5}[3 + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_3(u)Z^{-l}\right)] \\
=& \frac{1}{5}[3 + v_4\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_3(V)Z^{-l}\right) + v_5\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_3(V^2)Z^{-l}\right)] \\
=& \frac{1}{5}[3 + \frac{v_4}{2}\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left((1+\sqrt{5})Z^{-l}\right) + \frac{v_5}{2}\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left((1-\sqrt{5})Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{5}[3 + 2v_4 - 3v_5] = \frac{1}{5}[3v_4 + 2v_4] = v_4 \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{5}[3 - 3v_4 + 2v_5] = \frac{1}{5}[3v_5 + 2v_5] = v_5 \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

As $v_4 + v_5 = 1$ and both v_4 and v_5 are non-negative integers, the only possibilities are

$$v_4 = 1, v_5 = 0 \quad \text{or} \quad v_4 = 0, v_5 = 1.$$

So u is rationally conjugate either to V or V^2.

- $k=6$: Let $Z=-e^{\frac{2\pi i}{3}}$, we will need the following trace values:

$Z=-e^{\frac{2\pi i}{3}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l=0$	1	2	2
$l=1$	-1	-1	1
$l=2$	1	-1	-1

We know that $\chi_3(u)=v_2\chi_3(S)+v_3\chi_3(T)$ and combining Lemma 4.13 with the foregoing results yields

$$\begin{aligned}
\mu_l(u;\chi_3)=&\frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_3(u^3)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_3(u^2)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_3(u)Z^{-l}\right)]\\
=&\frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_3(S)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_3(T)Z^{-2l}\right)\\
&+(v_2\chi_3(S)+v_3\chi_3(T))\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)]\\
=&\frac{1}{6}[3-\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)-v_2\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)]\\
=&\begin{cases}\frac{1}{6}[3-1-2v_2]=\frac{1}{6}[2-2v_2]\in\mathbb{Z}_{\geq 0} & \text{if } l=0\\ \frac{1}{6}[3+1-v_2]=\frac{1}{6}[4-v_2]\in\mathbb{Z}_{\geq 0} & \text{if } l=1\\ \frac{1}{6}[3-1+v_2]=\frac{1}{6}[2+v_2]\in\mathbb{Z}_{\geq 0} & \text{if } l=2.\end{cases}
\end{aligned}$$

Let $x=4-v_2$, then x, $6-x=2+v_2$ and $2x-6=2-2v_2$ have to be non-negative multiples of 6. This is only possible if $v_2=-2$, which implies that $v_3=3$. But if we now calculate the value of $\mu_0(u;\chi_5)$ we see that

$$\chi_5(u)=v_2\chi_5(S)+v_3\chi_5(T)=-2\chi_5(S)+3\chi_5(T)=-2-3=-5$$

and

$$\begin{aligned}
\mu_0(u;\chi_5)=&\frac{1}{6}[5+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_5(u^3)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(u^2)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_5(u)Z^{-l}\right)]\\
=&\frac{1}{6}[5+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_5(S)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(T)Z^{-2l}\right)-5\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)]\\
=&\frac{1}{6}[5+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)-\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right)-5\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)]\\
=&\frac{1}{6}[5+1-2-5*2]=\frac{1}{6}[-6]=-1\in\mathbb{Z}_{\geq 0}\quad\text{if } l=0,
\end{aligned}$$

which is a contradiction. Thus there does not exist a torsion unit of order six in $\mathbb{Z}A_5$.

- $k = 10$: Then u^2 is is rationally conjugate to either V or V^2. Replacing u by u^3 if necessary we may assume that u^2 is rationally conjugate to V. Let $Z = -e^{\frac{2\pi i}{5}}$. We will use the following trace values:

$Z = -e^{\frac{2\pi i}{5}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l = 0$	1	4
$l = 1$	-1	1
$l = 5$	-1	-4

Combining Lemma 4.13 with the fact that

$$\chi_5(u) = v_2\chi(S) + v_4\chi_5(V) + v_5\chi_5(V^2) = v_2,$$

we find that

$$\begin{aligned}
\mu_l(u;\chi_5) =& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_5(u)Z^{-l}\right)] \\
=& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(S)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(V \text{ or } V^2)Z^{-2l}\right) + v_2\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) + +v_2\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{10}[5+1+4v_2] = \frac{1}{10}[6+4v_2] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{10}[5-1+v_2] = \frac{1}{10}[4+v_2] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{10}[5-1-4v_2] = \frac{1}{10}[4-4v_2] \in \mathbb{Z}_{\geq 0} & \text{if } l = 5. \end{cases}
\end{aligned}$$

Let $q = 4 + v_2$, then $q, 4q - 10 = 6 + 4v_2$ and $-4q + 20 = 4 - 4v_2$ have to be non-negative multiples of 10 which yields a contradiction. So there does not exist a torsion unit of order ten in $\mathbb{Z}A_5$.

- $k = 15$: In the same spirit as in the case $k = 10$ (replacing u by u^2 if necessary) we can assume that u^3 is rationally conjugate to V. Let $Z = e^{\frac{2\pi i}{5}}e^{\frac{2\pi i}{3}}$. We will need the following trace values:

$Z = e^{\frac{2\pi i}{5}}e^{\frac{2\pi i}{3}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l = 0$	2	8
$l = 3$	2	-2

Remark that $\chi_5(u) = v_3\chi_5(T) + v_4\chi_5(V) + v_5\chi_5(V^2) = -v_3$. By Lemma 4.13 we find that

$$\begin{aligned}
\mu_l(u;\chi_5) =& \frac{1}{15}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_5(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_5(u)Z^{-l}\right)] \\
=& \frac{1}{15}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(T)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_5(V \text{ or } V^2)Z^{-3l}\right) - v_3\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \frac{1}{15}[5 - \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) - v_3\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{15}[5-2-8v_3] = \frac{1}{15}[3-8v_3] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{15}[5-2+2v_3] = \frac{1}{15}[3+2v_3] \in \mathbb{Z}_{\geq 0} & \text{if } l = 3. \end{cases}
\end{aligned}$$

Let $x = 3 + 2v_3$, then x and $15 - 4x = 3 - 8v_3$ have to be non-negative multiples of 15, so $x = 0$. But then $2v_3 = -3$ which is impossible.

■

5.2 Hertweck-Luthar-Passi and S5

In this section we will prove that (ZC) holds for S_5, this result was first proved by Luthar and Trama in [51] without the use of Brauer characters but with a lot of computer computations regarding representations of S_5. We choose to provide the reader with another and much easier proof where Brauer characters are used as well. This is one of my original contributions to this thesis.

Theorem 5.2. *For every torsion unit $u \in V(\mathbb{Z}S_5)$ there exists a $g \in S_5$ such that $u \underset{\mathbb{Q}G}{\sim} g$, i.e. (ZC) holds for S_5.*

Proof. Let $\alpha = (1,2), \alpha' = (1,2)(3,4), \beta = (1,2,3), \gamma = (1,2,3,4), \sigma = (1,2,3,4,5)$ and $\tau = (1,2)(3,4,5)$. The (ordinary) character table of S_5 is given by

		α	α'	β	γ	σ	τ
	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	1	1	1	1	1	1	1
$\chi_{1'}$	1	−1	1	1	−1	1	−1
χ_4	4	2	0	1	0	−1	−1
$\chi_{4'}$	4	−2	0	1	0	−1	1
χ_5	5	1	1	−1	−1	0	1
$\chi_{5'}$	5	−1	1	−1	1	0	−1
χ_6	6	0	−2	0	0	1	0

while the character table in characteristic 5 reads as follows:

		α	α'	β	γ	τ
	C_1	C_2	C_3	C_4	C_5	C_7
ϕ_1	1	1	1	1	1	1
$\phi_{1'}$	1	−1	1	1	−1	−1
ϕ_3	3	1	−1	0	−1	−2
$\phi_{3'}$	3	−1	−1	0	1	2
ϕ_5	5	1	1	−1	−1	1
$\phi_{5'}$	5	−1	1	−1	1	−1

Let $u = \sum_{g \in G} u_g g$ be a normalized torsion unit of $\mathbb{Z}S_5$ of order $k > 1$. And write

$$v_i = \varepsilon_{C_i}(u) \text{ for } i \in \{1, 2, \cdots 7\}$$

for the partial augmentations of u. By Theorem 4.2 it follows that $v_1 = 0$. Since u has augmentation one:

$$v_2 + v_3 + v_4 + v_5 + v_6 + v_7 = 1.$$

And for all ordinary characters χ of S_5 we have that

$$\chi(u) = v_2\chi(\alpha) + v_3\chi(\alpha^{'}) + v_4\chi(\beta) + v_5\chi(\gamma) + v_6\chi(\sigma) + v_7\chi(\tau).$$

While we have for every Brauer character ϕ of S_5 that

$$\phi(u) = v_2\phi(\alpha) + v_3\phi(\alpha^{'}) + v_4\phi(\beta) + v_5\phi(\gamma) + v_7\phi(\tau).$$

By Theorem 4.5 the only possible orders of u are divisors of 120, so it will be sufficient to prove that normalized torsion units of order $2, 3, 4, 5$ and 6 are rationally conjugate to group elements and that there do not exist normalized torsion units of order $8, 10, 12$ and 15. By Theorem 4.11 we can reduce the possible values of partial augmentations:

order	partial augmentations	
$k = 2$	$v_4 = v_6 = v_7 = 0$	$v_2 + v_3 + v_5 = 1$
$k = 3$	$v_2 = v_3 = v_5 = v_6 = v_7 = 0$	$v_4 = 1$
$k = 4$	$v_4 = v_6 = v_7 = 0$	$v_2 + v_3 + v_5 = 1$
$k = 5$	$v_2 = v_3 = v_4 = v_5 = v_7 = 0$	$v_6 = 1$
$k = 6$	$v_6 = 0$	$v_2 + v_3 + v_4 + v_5 + v_7 = 1$
$k = 8$	$v_4 = v_6 = 0$	$v_2 + v_3 + v_5 + v_7 = 1$
$k = 10$	$v_4 = v_7 = 0$	$v_2 + v_3 + v_5 + v_6 = 1$
$k = 12$	$v_6 = 0$	$v_2 + v_3 + v_4 + v_5 + v_7 = 1$
$k = 15$	$v_2 = v_3 = v_5 = v_7 = 0$	$v_4 + v_6 = 1$

When $k \in \{3, 5\}$ it follows by Corollary 4.11 that $u \underset{\mathbb{Q}G}{\sim} \beta$ and $u \underset{\mathbb{Q}G}{\sim} \sigma$ respectively.

- $k = 2$: By Theorem 4.17 we have that $v_5 = 0$, so $v_2 + v_3 = 1$. Let $Z = -1$, we will need the following table:

$Z = -1$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l = 0$	1
$l = 1$	-1

If we combine the fact that $\chi_1^{'}(u) = -v_2 + v_3$ with Lemma 4.13 we have that

$$\begin{aligned}
\mu_l(u; \chi_1^{'}) =& \frac{1}{2}[1 + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_1^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{2}[1 + (-v_2 + v_3)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{2}[1 + (-v_2 + v_3) * 1] = \frac{1}{2}[1 - v_2 + v_3] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{2}[1 + (-v_2 + v_3) * (-1)] = \frac{1}{2}[1 + v_2 - v_3] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1. \end{cases}
\end{aligned}$$

Let $x = 1 - v_2 + v_3$, then x and $2 - x = 1 + v_2 - v_3$ have to be non-negative multiples of 2, so $x = 0$ or $x = 2$. If $x = 0$ we see that $v_2 - v_3 = 1 = v_2 + v_3$ so $v_3 = 0$ and $v_2 = 1$. If $x = 2$ then $-v_2 + v_3 = 1 = v_2 + v_3$ so $v_2 = 0$ and $v_3 = 1$. So either $u \underset{\mathbb{Q}G}{\sim} \alpha$ or $u \underset{\mathbb{Q}G}{\sim} \alpha^{'}$.

- $k = 4$: Let $Z = i$. There are now two possibilities, either $u^2 \underset{\mathbb{Q}G}{\sim} \alpha$ or $u^2 \underset{\mathbb{Q}G}{\sim} \alpha^{'}$. We will use the following field trace values:

$Z = i$	$\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l = 0$	1	2
$l = 1$	-1	0
$l = 2$	1	-2

1. $u^2 \underset{\mathbb{Q}G}{\sim} \alpha$: By Lemma 4.13 and the fact that $\phi_3(u) = v_2 - v_3 - v_5$ we have that

$$\begin{aligned}
\mu_l(u;\phi_3) =& \frac{1}{4}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi(u)Z^{-l}\right)] \\
=& \frac{1}{4}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(\alpha)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi(u)Z^{-l}\right)] \\
=& \frac{1}{4}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (v_2 - v_3 - v_5)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \frac{1}{4}[3 - 1] = \frac{1}{2} \in \mathbb{Z}_{\geq 0} \quad \text{if } l = 1.
\end{aligned}$$

   A contradiction.

2. $u^2 \underset{\mathbb{Q}G}{\sim} \alpha^{'}$: Remark that $\phi_1^{'}(u) = -v_2 + v_3 - v_5$ using Lemma 4.13 we have that

$$\begin{aligned}
\mu_l(u;\phi_1^{'}) =& \frac{1}{4}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_1^{'}(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi(u)Z^{-l}\right)] \\
=& \frac{1}{4}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_1^{'}(\alpha^{'})Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi(u)Z^{-l}\right)] \\
=& \frac{1}{4}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (-v_2 + v_3 - v_5)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{4}[2 + (-v_2 + v_3 - v_5) * 2] = \frac{1}{4}[2 - 2v_2 + 2v_3 - 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{4}[2 + (-v_2 + v_3 - v_5) * (-2)] = \frac{1}{4}[2 + 2v_2 - 2v_3 + 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

   Let $x = 2 - 2v_2 + 2v_3 - 2v_5$, then x and $4 - x = 2 + 2v_2 - 2v_3 + 2v_5$ have to be non-negative multiples of 4, so $x = 0$ or $x = 4$, which implies that

$$v_2 - v_3 - v_5 = \pm 1. \tag{3}$$

Again, by Lemma 4.13 and the fact that $\phi_3(u) = v_2 - v_3 - v_5$ we see that

$$\begin{aligned}
\mu_l(u;\phi_3) =& \frac{1}{4}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3(u)Z^{-l}\right)] \\
=& \frac{1}{4}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(\alpha')Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3(u)Z^{-l}\right)] \\
=& \frac{1}{4}[3 - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (v_2 - v_3 - v_5)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{4}[2 + (v_2 - v_3 - v_5) * 2] = \frac{1}{4}[2 + 2v_2 - 2v_3 - 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{4}[2 + (v_2 - v_3 - v_5) * (-2)] = \frac{1}{4}[2 - 2v_2 + 2v_3 + 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $y = 2+2v_2-2v_3-2v_5$, then y and $4-y = 2-2v_2+2v_3+2v_5$ have to be non-negative multiples of 4, so $y = 0$ or $y = 4$, which implies that

$$-v_2 + v_3 - v_5 = \pm 1. \tag{4}$$

Once again using Lemma 4.13 and this time remarking that $\phi_3'(u) = -v_2 - v_3 + v_5$ we find that

$$\begin{aligned}
\mu_l(u;\phi_3') =& \frac{1}{4}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3'(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3'(u)Z^{-l}\right)] \\
=& \frac{1}{4}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3'(\alpha)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3'(u)Z^{-l}\right)] \\
=& \frac{1}{4}[3 - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (-v_2 - v_3 + v_5)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{4}[2 + (-v_2 - v_3 + v_5) * 2] = \frac{1}{4}[2 - 2v_2 - 2v_3 + 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{4}[2 + (-v_2 - v_3 + v_5) * (-2)] = \frac{1}{4}[2 + 2v_2 + 2v_3 - 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $z = 2-2v_2-2v_3+2v_5$, then z and $4-z = 2+2v_2+2v_3-2v_5$ have to be non-negative multiples of 4, so $z = 0$ or $z = 4$, which implies that

$$-v_2 - v_3 + v_5 = \pm 1. \tag{5}$$

Now, combining the fact that $v_2 + v_3 + v_5 = 1$ and (1) yields the following system of equations.

$$\begin{cases} v_2 + v_3 + v_5 = 1 \\ v_2 - v_3 - v_5 = \pm 1 \end{cases}$$

Which implies that $v_2 \in \{0, 1\}$.
If (3) equals -1, combining this with $v_2 + v_3 + v_5 = 1$ gives rise to the following system of linear equations.

$$\begin{cases} v_2 + v_3 + v_5 = 1 \\ -v_2 - v_3 + v_5 = -1 \end{cases}$$

But this would imply that $v_5 = 0$ which is impossible because of Theorem 4.12. So we conclude that $-v_2 - v_3 + v_5 = 1$, and combining this once more with $v_2 + v_3 + v_5 = 1$

brings us

$$\begin{cases} v_2 + v_3 + v_5 = 1 \\ -v_2 - v_3 + v_5 = 1 \end{cases}$$

which yields that $v_5 = 1$ and $v_2 + v_3 = 0$.
So if $v_2 = 1$ then $v_3 = -1$, but if we look at (1) this would show that $3 = \pm 1$, so we can conclude that $v_2 = v_3 = 0$ and $v_5 = 1$ and we are done.

- $k = 6$: There are now two possibilities, either $u^3 \underset{\mathbb{Q}G}{\sim} \alpha$ or $u^3 \underset{\mathbb{Q}G}{\sim} \alpha^{'}$. Let $Z = -e^{\frac{2\pi i}{3}}$, we will need the following trace values:

$Z = -e^{\frac{2\pi i}{3}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l = 0$	1	2	2
$l = 1$	-1	-1	1
$l = 2$	1	-1	-1

1. $u^3 \underset{\mathbb{Q}G}{\sim} \alpha$: By Lemma 4.13, using that $\phi_1^{'}(u) = -v_2 + v_3 + v_4 - v_5 - v_7$ we see that

$$\begin{aligned}
\mu_l(u;\phi_1^{'}) =& \frac{1}{6}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_1^{'}(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_1^{'}(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_1^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{6}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_1^{'}(\alpha)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_1^{'}(\beta)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_1^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{6}[1 - \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) \\
& + (-v_2 + v_3 + v_4 - v_5 - v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{6}[1 - v_2 + v_3 + v_4 - v_5 - v_7] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{6}[-1 + v_2 - v_3 - v_4 + v_5 + v_7] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

So both $q = 1 - v_2 + v_3 + v_4 - v_5 - v_7$ and $-q = -1 + v_2 - v_3 - v_4 + v_5 + v_7$ have to be non-negative which implies that $q = 0$, or in other words:

$$-v_2 + v_3 + v_4 - v_5 - v_7 = -1. \tag{1}$$

Again by Lemma 4.13 and the fact that $\phi_5(u) = v_2 + v_3 - v_4 - v_5 + v_7$ we see that

$$\begin{aligned}
\mu_l(u;\phi_5) =& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5(u)Z^{-l}\right)] \\
=& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5(\alpha)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5(\beta)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5(u)Z^{-l}\right)] \\
=& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right) - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) \\
& + (v_2 + v_3 - v_4 - v_5 + v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{6}[4 + 2v_2 + 2v_3 - 2v_4 - 2v_5 + 2v_7] = \frac{1}{6}[6 - 4v_4 - 4v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{6}[5 + v_2 + v_3 - v_4 - v_5 + v_7] = \frac{1}{6}[6 - 2v_4 - 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{6}[7 - v_2 - v_3 + v_4 + v_5 - v_7] = \frac{1}{6}[6 + 2v_4 + 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Where we used that $v_2+v_3+v_4+v_5+v_7=1$ to simplify our equations. Let $x=6-2v_4-2v_5$, then x, $12-x=6+2v_4+2v_5$ and $2x-6=6-4v_4-4v_5$ have to be non-negative multiples of 6 which implies that $x\in\{6,12\}$, which yields

$$v_4+v_5\in\{0,-3\}. \tag{2}$$

Once more using Lemma 4.13 and the fact that $\phi_5^{'}(u)=-v_2+v_3-v_4+v_5-v_7$ it follows that

$$\begin{aligned}
\mu_l(u;\phi_5^{'}) =&\frac{1}{6}[5+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5^{'}(u^3)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5^{'}(u^2)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5^{'}(u)Z^{-l}\right)]\\
=&\frac{1}{6}[5+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5^{'}(\alpha)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5^{'}(\beta)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5^{'}(u)Z^{-l}\right)]\\
=&\frac{1}{6}[5-\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)-\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right)\\
&+(-v_2+v_3-v_4+v_5-v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)]\\
=&\begin{cases}\frac{1}{6}[2-2v_2+2v_3-2v_4+2v_5-2v_7]=\frac{1}{6}[4v_3+4v_5]\in\mathbb{Z}_{\geq 0} & \text{if } l=0\\ \frac{1}{6}[7-v_2+v_3-v_4+v_5-v_7]=\frac{1}{6}[6+2v_3+2v_5]\in\mathbb{Z}_{\geq 0} & \text{if } l=1\\ \frac{1}{6}[5+v_2-v_3+v_4-v_5+v_7]=\frac{1}{6}[6-2v_3-2v_5]\in\mathbb{Z}_{\geq 0} & \text{if } l=2.\end{cases}
\end{aligned}$$

Where we used again that $v_2+v_3+v_4+v_5+v_7=1$ to simplify our equations. Setting $y=6+2v_3+2v_5$, then y, $12-y=6-2v_3-2v_5$ and $2y-6=4v_3+4v_5$ have to be non-negative multiples of 6 which implies that $y\in\{6,12\}$, which yields

$$v_3+v_5\in\{0,3\}. \tag{3}$$

If we now use Lemma 4.13 to look at $\phi_{3'}$ and use that $\phi_3^{'}(u)=-v_2-v_3+v_5+2v_7$ we see that

$$\begin{aligned}
\mu_l(u;\phi_3^{'}) =&\frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3^{'}(u^3)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3^{'}(u^2)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3^{'}(u)Z^{-l}\right)]\\
=&\frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3^{'}(\alpha)Z^{-3l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3^{'}(\beta)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3^{'}(u)Z^{-l}\right)]\\
=&\frac{1}{6}[3-\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)+(-v_2-v_3+v_5+2v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)]\\
=&\begin{cases}\frac{1}{6}[2-2v_2-2v_3+2v_5+4v_7]\in\mathbb{Z}_{\geq 0} & \text{if } l=0\\ \frac{1}{6}[4-v_2-v_3+v_5+2v_7]\in\mathbb{Z}_{\geq 0} & \text{if } l=1\\ \frac{1}{6}[2+v_2+v_3-v_5-2v_7]\in\mathbb{Z}_{\geq 0} & \text{if } l=2.\end{cases}
\end{aligned}$$

Let $z=2+v_2+v_3-v_5-2v_7$, then z, $6-2z=2-2v_2-2v_3+2v_5+4v_7$ and $6-z=4-v_2-v_3+v_5+2v_7$ have to be non-negative multiples of 6 which implies that $z=0$, which yields

$$v_2+v_3-v_5-2v_7=-2. \tag{4}$$

Combining $(1)-(4)$ with the fact that $v_2+v_3+v_4+v_5+v_7=1$ we obtain the following system of linear equations:

$$\begin{cases} v_2+v_3+v_4+v_5+v_7=1 \\ -v_2+v_3+v_4-v_5-v_7=-1 \\ v_4+v_5\in\{0,-3\} \\ v_3+v_5\in\{0,3\} \\ v_2+v_3-v_5-2v_7=-2 \end{cases}$$

There are only two integer solutions which fulfill this system of equations, namely

$$(v_2,v_3,v_4,v_5,v_7)=(0,0,0,0,1) \text{ and } (-1,3,-3,0,2).$$

But filling in those solutions into $\mu_1(u;\phi_3)$, using that $\phi_3(u)=v_2-v_3-v_5-2v_7$ yields

$$\begin{aligned} \mu_1(u;\phi_3) =& \frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3(u^3)Z^{-3}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(u^2)Z^{-2}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3(u)Z^{-1}\right)] \\ =& \frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3(\alpha)Z^{-3}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(\beta)Z^{-2}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3(u)Z^{-1}\right)] \\ =& \frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3}\right)+(v_2-v_3-v_5-2v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-1}\right)] \\ =& \frac{1}{6}[2+v_2-v_3-v_5-2v_7]=\frac{1}{6}[2-1-3-4]=-1\in\mathbb{Z}_{\geq 0} \quad \text{if } l=1, \end{aligned}$$

a contradiction.

2. $u^3 \underset{\mathbb{Q}G}{\sim} \alpha'$: Recall that $\phi_1'(u) = -v_2 + v_3 + v_4 - v_5 - v_7$, by Lemma 4.13 we have that

$$\begin{aligned}
\mu_l(u;\phi_1') =& \frac{1}{6}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_1'(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_1'(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_1'(u)Z^{-l}\right)] \\
=& \frac{1}{6}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_1'(\alpha')Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_1'(\beta)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_1'(u)Z^{-l}\right)] \\
=& \frac{1}{6}[1 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) \\
&+(-v_2 + v_3 + v_4 - v_5 - v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{6}[-1 - v_2 + v_3 + v_4 - v_5 - v_7] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{6}[1 + v_2 - v_3 - v_4 + v_5 + v_7] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

So both $q = -1 - v_2 + v_3 + v_4 - v_5 - v_7$ and $-q = 1 + v_2 - v_3 - v_4 + v_5 + v_7$ have to be non-negative which implies that $q = 0$, in other words:

$$-v_2 + v_3 + v_4 - v_5 - v_7 = 1. \tag{1}$$

Remember that $\phi_5(u) = v_2 + v_3 - v_4 - v_5 + v_7$, again by Lemma 4.13 we see that

$$\begin{aligned}
\mu_l(u;\phi_5) =& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5(u)Z^{-l}\right)] \\
=& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5(\alpha')Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5(\beta)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5(u)Z^{-l}\right)] \\
=& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right) - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) \\
&+(v_2 + v_3 - v_4 - v_5 + v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{6}[4 + 2v_2 + 2v_3 - 2v_4 - 2v_5 + 2v_7] = \frac{1}{6}[6 - 4v_4 - 4v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{6}[5 + v_2 + v_3 - v_4 - v_5 + v_7] = \frac{1}{6}[6 - 2v_4 - 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{6}[7 - v_2 - v_3 + v_4 + v_5 - v_7] = \frac{1}{6}[6 + 2v_4 + 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Where we used that $v_2+v_3+v_4+v_5+v_7 = 1$ to simplify our equations. Let $x = 6-2v_4-2v_5$, then x, $12 - x = 6 + 2v_4 + 2v_5$ and $2x - 6 = 6 - 4v_4 - 4v_5$ have to be non-negative multiples of 6 which implies that $x \in \{6, 12\}$, which yields

$$v_4 + v_5 \in \{0, -3\}. \tag{2}$$

We already know that $\phi_5^{'}(u) = -v_2 + v_3 - v_4 + v_5 - v_7$ and by using once more Lemma 4.13 it follows that

$$\begin{aligned}
\mu_l(u;\phi_5^{'}) =& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5^{'}(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5^{'}(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{6}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_5^{'}(\alpha^{'})Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_5^{'}(\beta)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_5^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{6}[5 + 4\,\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right) - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) \\
&+ (-v_2 + v_3 - v_4 + v_5 - v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{6}[4 - 2v_2 + 2v_3 - 2v_4 + 2v_5 - 2v_7] = \frac{1}{6}[2 + 4v_3 + 4v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{6}[5 - v_2 + v_3 - v_4 + v_5 - v_7] = \frac{1}{6}[4 + 2v_3 + 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{6}[7 + v_2 - v_3 + v_4 - v_5 + v_7] = \frac{1}{6}[8 - 2v_3 - 2v_5] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Where we used again that $v_2 + v_3 + v_4 + v_5 + v_7 = 1$ to simplify our equations. Setting $y = 4 + 2v_3 + 2v_5$, then y, $12 - y = 8 - 2v_3 - 2v_5$ and $2y - 6 = 2 + 4v_3 + 4v_5$ have to be non-negative multiples of 6 which implies that $y \in \{6, 12\}$, which yields

$$v_3 + v_5 \in \{1, 4\}. \tag{3}$$

Finally, because $\phi_3^{'}(u) = -v_2 - v_3 + v_5 + 2v_7$ and Lemma 4.13 we see that

$$\begin{aligned}
\mu_l(u;\phi_3^{'}) =& \frac{1}{6}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3^{'}(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3^{'}(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{6}[3 + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3^{'}(\alpha^{'})Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3^{'}(\beta)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{6}[3 - \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right) + (-v_2 - v_3 + v_5 + 2v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{6}[2 - 2v_2 - 2v_3 + 2v_5 + 4v_7] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{6}[4 - v_2 - v_3 + v_5 + 2v_7] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{6}[2 + v_2 + v_3 - v_5 - 2v_7] \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $z = 2 + v_2 + v_3 - v_5 - 2v_7$, then z, $6 - 2z = 2 - 2v_2 - 2v_3 + 2v_5 + 4v_7$ and $6 - z = 4 - v_2 - v_3 + v_5 + 2v_7$ have to be non-negative multiples of 6 which implies that $z = 0$, which yields

$$v_2 + v_3 - v_5 - 2v_7 = -2. \tag{4}$$

Combining $(1)-(4)$ with the fact that $v_2+v_3+v_4+v_5+v_7=1$ we find the following system of linear equations:

$$\begin{cases} v_2+v_3+v_4+v_5+v_7=1 \\ -v_2+v_3+v_4-v_5-v_7=1 \\ v_4+v_5\in\{0,-3\} \\ v_3+v_5\in\{1,4\} \\ v_2+v_3-v_5-2v_7=-2 \end{cases}$$

there remain only two possibilities:

$$(v_2,v_3,v_4,v_5,v_7)=(-1,1,0,0,1) \text{ or } (-2,4,-3,0,2).$$

But if we now take a look at $\mu_0(u;\phi_3)$, using that $\phi_3(u)=v_2-v_3-v_5-2v_7$, we see that

$$\begin{aligned} \mu_0(u;\phi_3) =& \frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3(u^3)Z^{-3}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(u^2)Z^{-2}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3(u)Z^{-1}\right)] \\ =& \frac{1}{6}[3+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\phi_3(\alpha')Z^{-3}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\phi_3(\beta)Z^{-2}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\phi_3(u)Z^{-1}\right)] \\ =& \frac{1}{6}[3-\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(1\right)+(v_2-v_3-v_5-2v_7)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(1\right)] \\ =& \frac{1}{6}[2+2v_2-2v_3-2v_5-4v_7]\in\mathbb{Z}_{\geq 0} \quad \text{if } l=0. \end{aligned}$$

Plugging in $(v_2,v_3,v_4,v_5,v_7)=(-1,1,0,0,1)$ yields $\frac{1}{6}[2-2-2-4]=-1\in\mathbb{Z}_{\geq 0}$, a contradiction. If we substitute $(v_2,v_3,v_4,v_5,v_7)=(-2,4,-3,0,2)$ into this equation, we find that $\frac{1}{6}[2-4-8-8]=-3\in\mathbb{Z}_{\geq 0}$ a contradiction. So there does not exist a torsion unit of order 6 in $\mathbb{Z}S_5$.

- $k=8$: As there does not exist an element of order 8 in S_5, Theorem 4.12 yields that there does not exist a unit of order 8 in $\mathbb{Z}S_5$.
- $k=10$: Let $Z=-e^{\frac{2\pi i}{5}}$. There are now two possibilities, either $u^5 \underset{\mathbb{Q}G}{\sim} \alpha$ or $u^5 \underset{\mathbb{Q}G}{\sim} \alpha'$. We will need the following table

$Z=-e^{\frac{2\pi i}{5}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l=0$	1	4	4
$l=1$	-1	-1	1
$l=2$	1	-1	-1

1. $u^5 \underset{\mathbb{Q}G}{\sim} \alpha$: By Lemma 4.13 and the fact that $\chi_4(u) = 2v_2 - v_6$ we have that

$$\begin{aligned}
\mu_l(u;\chi_4) =& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4(u)Z^{-l}\right)] \\
=& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4(\alpha)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4(\sigma)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4(u)Z^{-l}\right)] \\
=& \frac{1}{10}[4 + 2\,\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (2v_2 - v_6)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{10}[2 + 8v_2 - 4v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{10}[3 + 2v_2 - v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{10}[7 - 2v_2 + v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $x = 3 + 2v_2 - v_6$, then x, $10 - x = 7 - 2v_2 + v_6$ and $4x - 10 = 2 + 8v_2 - 4v_6$ have to be non-negative multiples of 10 which implies that $x = 10$, hence

$$2v_2 - v_6 = 7. \tag{1}$$

Combining $\chi_4^{'}(u) = -2v_2 - v_6$ with Lemma 4.13 we find that

$$\begin{aligned}
\mu_l(u;\chi_4^{'}) =& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4^{'}(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4^{'}(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4^{'}(\alpha)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4^{'}(\sigma)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{10}[4 - 2\,\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (-2v_2 - v_6)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{10}[-2 - 8v_2 - 4v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{10}[7 - 2v_2 - v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{10}[3 + 2v_2 + v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $y = 7 - 2v_2 - v_6$, then y, $10 - y = 3 + 2v_2 + v_6$ and $4y - 30 = -2 - 8v_2 - 4v_6$ have to be non-negative multiples of 10 which implies that $y = 10$, hence

$$-2v_2 - v_6 = 3. \tag{2}$$

Again by Lemma 4.13 and because $\chi_5(u) = v_2 + v_3 - v_5$ we have that

$$\begin{aligned}
\mu_l(u;\chi_5) =& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_5(u)Z^{-l}\right)] \\
=& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(\alpha)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(\sigma)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_5(u)Z^{-l}\right)] \\
=& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) + (v_2 + v_3 - v_5)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{10}[6 + 4v_2 + 4v_3 - 4v_5] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{10}[4 + v_2 + v_3 - v_5] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{10}[6 - v_2 - v_3 + v_5] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $z = 4 + v_2 + v_3 - v_5$, then z, $10 - z = 6 - v_2 - v_3 + v_5$ and $4z - 10 = 6 + 4v_2 + 4v_3 - 4v_5$ have to be non-negative multiples of 10 which implies that $z = 10$, hence

$$v_2 + v_3 - v_5 = 6. \tag{3}$$

Combining $(1) - (3)$ with the fact that $v_2 + v_3 + v_5 + v_6 = 1$ we arrive at the following system of linear equations

$$\begin{cases} v_2 + v_3 + v_5 + v_6 = 1 \\ 2v_2 - v_6 = 7 \\ -2v_2 - v_6 = 3 \\ v_2 + v_3 - v_5 = 6 \end{cases}$$

The solution is given by

$$(v_2, v_3, v_5, v_6) = (1, 5, 0, -5).$$

But filling in the last one into $\mu_0(u; \chi_6)$, taking into account that $\chi_6(u) = -2v_3 + v_6$ yields

$$\begin{aligned} \mu_0(u;\chi_6) =& \frac{1}{10}[6 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_6(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-l}\right)] \\ =& \frac{1}{10}[6 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_6(\alpha)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(\sigma)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-l}\right)] \\ =& \frac{1}{10}[6 + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(1\right) + (-2v_3 + v_6)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(1\right)] \\ =& \frac{1}{10}[10 - 8v_3 + 4v_6] = \frac{1}{10}[10 - 8 * 5 + 4 * (-5)] = -5 \in \mathbb{Z}_{\geq 0} \quad \text{if } l = 0 \end{aligned}$$

a contradiction.

2. $u^5 \underset{\mathbb{Q}G}{\sim} \alpha'$: Recall that $\chi_4(u) = 2v_2 - v_6$, by Lemma 4.13 we have that

$$\begin{aligned} \mu_l(u;\chi_4) =& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4(u)Z^{-l}\right)] \\ =& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4(\alpha')Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4(\sigma)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4(u)Z^{-l}\right)] \\ =& \frac{1}{10}[4 - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (2v_2 - v_6)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\ =& \begin{cases} \frac{1}{10}[8v_2 - 4v_6] = \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{10}[5 + 2v_2 - v_6] = \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{10}[5 - 2v_2 + v_6] = \in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases} \end{aligned}$$

Let $x = 5 + 2v_2 - v_6$, then x, $10 - x = 5 - 2v_2 + v_6$ and $4x - 20 = 8v_2 - 4v_6$ have to be non-negative multiples of 10 which implies that $x = 10$, hence

$$2v_2 - v_6 = 5. \tag{1}$$

Remember that $\chi_4^{'}(u) = -2v_2 - v_6$, using Lemma 4.13 we obtain

$$\begin{aligned}
\mu_l(u;\chi_4^{'}) =& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4^{'}(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4^{'}(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{10}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4^{'}(\alpha^{'})Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_4^{'}(\sigma)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4^{'}(u)Z^{-l}\right)] \\
=& \frac{1}{10}[4 - \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right) + (-2v_2 - v_6)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{10}[-8v_2 - 4v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{10}[5 - 2v_2 - v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{10}[5 + 2v_2 + v_6] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $y = 5 - 2v_2 - v_6$, then y, $10 - y = 5 + 2v_2 + v_6$ and $4y - 20 = -8v_2 - 4v_6$ have to be non-negative multiples of 10 which implies that $y = 10$, hence

$$-2v_2 - v_6 = 5. \tag{2}$$

Applying Lemma 4.13 once more and taking into account that $\chi_5(u) = v_2 + v_3 - v_5$ we have that

$$\begin{aligned}
\mu_l(u;\chi_5) =& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(u^2)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_5(u)Z^{-l}\right)] \\
=& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_5(\alpha^{'})Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_5(\sigma)Z^{-2l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_5(u)Z^{-l}\right)] \\
=& \frac{1}{10}[5 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) + (v_2 + v_3 - v_5)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{10}[6 + 4v_2 + 4v_3 - 4v_5] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{10}[4 + v_2 + v_3 - v_5] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{10}[6 - v_2 - v_3 + v_5] =\in \mathbb{Z}_{\geq 0} & \text{if } l = 2. \end{cases}
\end{aligned}$$

Let $z = 4 + v_2 + v_3 - v_5$, then z, $10 - z = 6 - v_2 - v_3 + v_5$ and $4z - 10 = 6 + 4v_2 + 4v_3 - 4v_5$ have to be non-negative multiples of 10 which implies that $z = 10$, hence

$$v_2 + v_3 - v_5 = 6. \tag{3}$$

Combining $(1) - (3)$ with the fact that $v_2 + v_3 + v_5 + v_6 = 1$ we find the following system of linear equations

$$\begin{cases} v_2 + v_3 + v_5 + v_6 = 1 \\ 2v_2 - v_6 = 5 \\ -2v_2 - v_6 = 5 \\ v_2 + v_3 - v_5 = 6 \end{cases}$$

the solution is given by:

$$(v_2, v_3, v_5, v_6) = (0, 6, 0, -5).$$

But substituting this into $\mu_0(u;\chi_6)$, using that, $\chi_6(u) = -2v_3 + v_6$, yields

$$\begin{aligned}\mu_0(u;\chi_6) =&\frac{1}{10}[6+\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_6(u^5)Z^{-5l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(u^2)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-l}\right)]\\ =&\frac{1}{10}[6+\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_6(\alpha')Z^{-5l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(\sigma)Z^{-2l}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-l}\right)]\\ =&\frac{1}{10}[6-2\,\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(1\right)+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(1\right)+(-2v_3+v_6)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(1\right)]\\ =&\frac{1}{10}[8-8v_3+4v_6]=\frac{1}{10}[8-8*6+4*(-5)]=-6\in\mathbb{Z}_{\geq 0}\quad\text{if } l=0,\end{aligned}$$

a contradiction. We conclude that there does not exist a torsion unit of order 10 in $\mathbb{Z}S_5$.

- $k=12$: There are now two cases, either $u^6 \underset{\mathbb{Q}G}{\sim} \alpha$ or $u^6 \underset{\mathbb{Q}G}{\sim} \alpha'$. We will need the following trace field values:

$Z=e^{\frac{\pi i}{6}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^6)/\mathbb{Q}}\left(Z^{-6l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z^4)/\mathbb{Q}}\left(Z^{-4l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)$
$l=1$	-1	-1	0

$Z=e^{\frac{\pi i}{6}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(Z^{-2l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l=1$	1	0

1. $u^6 \underset{\mathbb{Q}G}{\sim} \alpha'$: Remark that $\chi_6(u)=-2v_3$. By Lemma 4.13 we have that

$$\begin{aligned}\mu_1(u;\chi_6) =&\frac{1}{12}[6+\mathrm{Tr}_{\mathbb{Q}(Z^6)/\mathbb{Q}}\left(\chi_6(u^6)Z^{-6}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^4)/\mathbb{Q}}\left(\chi_6(u^4)Z^{-4}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_6(u^3)Z^{-3}\right)\\ &+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(u^2)Z^{-2}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-1}\right)]\\ =&\frac{1}{12}[6+\mathrm{Tr}_{\mathbb{Q}(Z^6)/\mathbb{Q}}\left(\chi_6(\alpha')Z^{-6}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^4)/\mathbb{Q}}\left(\chi_6(\beta)Z^{-4}\right)+\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_6(\gamma)Z^{-3}\right)\\ &+\mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(\sigma)Z^{-2}\right)+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-1}\right)]\\ =&\frac{1}{12}[6-2\,\mathrm{Tr}_{\mathbb{Q}(Z^6)/\mathbb{Q}}\left(Z^{-6}\right)-2v_3\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-1}\right)]\\ =&\frac{1}{12}[6+2-2v_3*0]=\frac{1}{12}[8]\in\mathbb{Z}_{\geq 0}\quad\text{if } l=1,\end{aligned}$$

which is a contradiction.

2. $u^6 \underset{\mathbb{Q}G}{\sim} \alpha$: Again using that $\chi_6(u) = -2v_3$ and Lemma 4.13 we have that

$$\begin{aligned}
\mu_1(u;\chi_6) =& \frac{1}{12}[6 + \mathrm{Tr}_{\mathbb{Q}(Z^6)/\mathbb{Q}}\left(\chi_6(u^6)Z^{-6}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^4)/\mathbb{Q}}\left(\chi_6(u^4)Z^{-4}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_6(u^3)Z^{-3}\right) \\
&+ \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(u^2)Z^{-2}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-1}\right)] \\
=& \frac{1}{12}[6 + \mathrm{Tr}_{\mathbb{Q}(Z^6)/\mathbb{Q}}\left(\chi_6(\alpha)Z^{-6}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^4)/\mathbb{Q}}\left(\chi_6(\beta)Z^{-4}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_6(\gamma)Z^{-3}\right) \\
&+ \mathrm{Tr}_{\mathbb{Q}(Z^2)/\mathbb{Q}}\left(\chi_6(\sigma)Z^{-2}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_6(u)Z^{-1}\right)] \\
=& \frac{1}{12}[6 - 2v_3\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-1}\right)] \\
=& \frac{1}{12}[6 - 2v_3 * 0] = \frac{1}{12}[6] \in \mathbb{Z}_{\geq 0} \quad \text{if } l = 1,
\end{aligned}$$

which is a contradiction, so we can conclude that there does not exist a torsion unit of order 12 in $\mathbb{Z}S_5$.

- $k = 15$: Let $Z = e^{\frac{2\pi i}{5}}e^{\frac{2\pi i}{3}}$. We will need the following field trace values:

$Z = -e^{\frac{2\pi i}{5}}e^{\frac{2\pi i}{3}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l = 0$	2	4	8
$l = 1$	-1	-1	1
$l = 3$	2	-1	-2

Because of $\chi_4(u) = v_4 - v_6$ and by Lemma 4.13 we see that

$$\begin{aligned}
\mu_l(u;\chi_4) =& \frac{1}{15}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_4(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4(u)Z^{-l}\right)] \\
=& \frac{1}{15}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi_4(\beta)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi_4(\sigma)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi_4(u)Z^{-l}\right)] \\
=& \frac{1}{15}[4 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) - \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(Z^{-3l}\right) + (v_4 - v_6)\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)] \\
=& \begin{cases} \frac{1}{15}[2 + 8v_4 - 8v_6] \in \mathbb{Z}_{\geq 0} & \text{if } l = 0 \\ \frac{1}{15}[4 + v_4 - v_6] \in \mathbb{Z}_{\geq 0} & \text{if } l = 1 \\ \frac{1}{15}[7 - 2v_4 + 2v_6] \in \mathbb{Z}_{\geq 0} & \text{if } l = 3. \end{cases}
\end{aligned}$$

Let $x = 4 + v_4 - v_6$, then x, $8x - 30 = 2 + 8v_4 - 8v_6$ and $15 - 2x = 7 - 2v_4 + 2v_6$ have to be non-negative multiples of 15 which is impossible. So there does not exist a torsion unit of order 15 in $\mathbb{Z}S_5$, which finishes the proof.

■

6 The Prime Graph Question

6.1 A reduction theorem

> "If a 'religion' is defined to be a system of ideas that contains unprovable statements, then Gödel taught us that mathematics is not only a religion, it is the only religion that can prove itself to be one."
>
> — J.D. Barrow, The Artful Universe (1995)

After the recent counterexample of Leo Margolis and Florian Eisele to the Zassenhaus Conjecture, see [31], interest shifted towards related problems. One of these questions is the so-called prime graph question, which was first introduced by Kimmerle in [47]. We will first recall the definition of the prime graph of a group en then restate the Prime Graph Question itself.

Let G be a group, recall that its prime graph $\prod(G)$ is defined as follows. The vertices of $\prod(G)$ are the primes p for which $p \mid |G|$ and two different vertices p and q are joined by an edge if and only if G contains an element of order $p \cdot q$.

> **The Prime Graph Question (PQ)**: Given a finite group G, do G and $V(\mathbb{Z}G)$ have the same prime graph?

In contrast to the Zassenhaus Conjecture, the Prime Graph Questions admits a powerful reduction theorem, which will be treated in the first section. This reduction theorem will shift our attention to almost simple groups. After giving a brief introduction to the HeLP package in GAP [6], this package will be used in order to provide proofs of the Prime Graph Question for some sporadic simple groups. The prime graph of the unit group of a group ring is sometimes called differently:

Definition 6.0.1. The **Gruenberg-Kegel graph** $\Gamma(R)$ of a group ring R is the prime graph of its normalized unit group $V(R)$, i.e. $\prod(V(R))$.

We will now prove a lemma due to Kimmerle and Konovalov [48] which is vital in proving the reduction theorem.

Lemma 6.1. *Let N be a normal subgroup of G and let $E = G/N$. Assume that $\prod(V(\mathbb{Z}E)) = \prod(E)$. Let $q \in \pi(G)\backslash\pi(N)$ and let $p \in \pi(G)$ with $p \neq q$. Then $V(\mathbb{Z}G)$ has elements of order $p \cdot q$ if and only if G has elements of order $p \cdot q$, i.e. (PQ) holds for G.*

Proof. That $V(\mathbb{Z}G)$ contains elements of order $p \cdot q$ if G contains such elements is obvious as $G \subset V(\mathbb{Z}G)$.

For the other direction, assume that $\mathbb{Z}G$ has a unit u of order $p \cdot q$ but that G does not have an element of this order. Let $f : G \to E : g \to \overline{g}$ be the quotient map and let ϕ be the extension of f to the normalized unit groups, i.e.

$$\phi : V(\mathbb{Z}G) \to V(\mathbb{Z}E) : \sum_{g \in G} u_g g \to \sum_{g \in G} u_g \overline{g}.$$

Now $o(\phi(u)) \in \{1, p, q, pq\}$, but $o(\phi(u)) = 1$ is impossible as $q \in \pi(G)\backslash\pi(N)$. Suppose that $o(\phi(u)) = p \cdot q$, as $\prod(V(\mathbb{Z}E)) = \prod(E)$ this would imply that E and thus G would contain an element of order $p \cdot q$ which is impossible by assumption So either

$$\phi(u^p) = 1 \text{ or } \phi(u^q) = 1.$$

In each case we have an element $x \in Ker(\phi)$ with $o(x) = k$, where k is a prime number. But this implies, by Theorem 4.12, that there exists a $g \in G$ of order k such that $\varepsilon_g(x) \neq 0$. Which in turn implies that $\overline{g} \in N$, so $o(x) \mid |N|$. As $q \notin \pi(N)$ it follows that $\phi(u^q) = 1$. For a group H and a prime r, denote by $H(r)$ the set of all elements of H which order is a strict positive power of r. Now we can write u in the following way:

$$u = \sum_{g \in G(p)} a_g g + \sum_{g \in G(q)} b_g g + \sum_{g \in R} c_g g.$$

Where $R = G\backslash(G(p) \cup G(q))$. By Theorem 4.2 we know that $c_1 = 0$ and because of Theorem 4.11 it follows that

$$\varepsilon_g(u) = 0 \quad \text{for all } g \in R \implies \sum_{g \in R} c_g = 0. \tag{1}$$

On the other hand we can also write that

$$\phi(u) = \sum_{g \in G(p)} a_g \phi(g) + \sum_{g \in G(q)} b_g \phi(g) + \sum_{\substack{g \in R \\ \phi(g) \in E(p)}} c_g \phi(g) + \sum_{\substack{g \in R \\ \phi(g) \in E(q)}} c_g \phi(g) + \sum_{\substack{g \in R \\ \phi(g) \in R'}} c_g \phi(g).$$

Where $R' = E\backslash(E(p) \cup E(q))$. We also know that $\phi(u)$ has order q, so again using Theorem 4.11 we see that

$$\sum_{g \in G(p)} a_g + \sum_{\substack{g \in R \\ \phi(g) \in E(p)}} c_g + \sum_{\substack{g \in R \\ \phi(g) \in R'}} c_g = 0. \tag{2}$$

But if we now combine this with (1) and use the fact that the augmentation of $\phi(u) = 1$ we can conclude that

$$\sum_{g \in G(p)} a_g = 0 \text{ and } \sum_{g \in G(q)} b_g = 1.$$

If we now take a look at

$$u^q = \sum_{g \in G(p)} \alpha_g g + \sum_{g \in G(q)} \beta_g g + \sum_{g \in R} \gamma_g g.$$

Because u^q has order p it follows by Theorem 4.11 that

$$\sum_{g \in G(q)} \beta_g = 0. \tag{3}$$

Moreover, by Theorem 4.2 we see that $\gamma_1 = 0$. Because q is a prime, we can invoke Theorem 4.6 to obtain

$$u^q \equiv \sum_{g \in G(p)} a_g g^q + \sum_{g \in G(q)} b_g g^q + \sum_{g \in R} c_g g^q \mod [\mathbb{Z}G, \mathbb{Z}G] + q\mathbb{Z}G.$$

Remark that if $o(g) \neq q$ we have that g^q is a q-element if and only if g is a q-element. As $u^q \neq 1$ it follows by Theorem 4.2 that the coefficient of u at 1 is zero, i.e.

$$c_1 + \sum_{\substack{g \in G(q) \\ o(g)=q}} b_g \equiv 0 \mod q \text{ but as } c_1 = 0 \quad \text{this implies that} \sum_{\substack{g \in G(q) \\ o(g)=q}} b_g \equiv 0 \mod q.$$

So we see that

$$\sum_{g \in G(q)} b_g \equiv \sum_{\substack{g \in G(q) \\ o(g) \neq q}} b_g \equiv \sum_{g \in G(q)} \beta_g \equiv 0 \mod q$$

which is a contradiction to (3). So (PQ) holds for G. ■

Corollary 6.1.1. *Let G be a group and let N be a normal subgroup of G. If $\prod(N)$ is a complete graph and $\prod(V(\mathbb{Z}G/N)) = \prod(G/N)$ then $\prod(V(\mathbb{Z}G)) = \prod(G)$.*

Proof. By Lemma 6.1 it suffices to show the statement for different primes who divide $|N|$. But these primes are connected in $\prod(G)$ by assumption and thus also in $\prod(V(\mathbb{Z}G))$. ■

Theorem 6.2. *(Reduction theorem to almost simple groups) Let G be a finite group. If (PQ) holds for every almost simple group X which occurs as image of G then (PQ) holds for G itself.*

Proof. We will prove this by induction on the length of a chief series of G. The theorem holds by assumption if G itself is an almost simple group.

Suppose that G has a minimal normal subgroup N which is abelian. In particular N is nilpotent and we can write N as the direct product of its Sylow subgroups.

$$N = \prod_{P_i \text{ Sylow subgroup}} P_i.$$

But as $N \lhd G$ this implies that

$$N^g = \prod_{P_i \text{ Sylow subgroup}} P_i^g = \prod_{P_i \text{ Sylow subgroup}} P_i$$

which implies that $P_i \lhd G$, by minimality it follows that $N = P_i$, an abelian p-group. It follows by the induction hypothesis that (PQ) holds for $G \backslash N$. If we take $q \neq p$ in $\pi(G)$ it follows that $q \in \pi(G) \backslash \pi(N)$ because N is a p-group. So we can apply Lemma 6.1, which shows that (PQ) holds for G in this case.

If G does not have an abelian minimal normal subgroup then all minimal normal subgroups have to be perfect. Suppose not, so let N be a minimal normal subgroup such that $N^{'} \neq 1$, as $N^{'}$ is a characteristic subgroup of N it follows that $N^{'} \lhd G$. But N is minimal so $N = N^{'}$. Again by Lemma 6.1 it follows that G has an element of order $p \cdot q$ if there exists a subgroup $N \lhd G$ with $p \in \pi(G) \backslash \pi(N)$ or $q \in \pi(G) \backslash \pi(N)$. So assume that p and q divide the order of each minimal normal subgroup of G, we distinguish between three cases:

a If G contains at least two different minimal subgroups N_1 and N_2 then $N_1 \times N_2 \subset G$. But $N_1 \times N_2$ has an element of order $p \cdot q$ and so does G.

b Suppose N is unique and non-simple minimal normal subgroup. By Theorem 2.4, N is the direct product of at least two conjugate simple subgroups $S_i \lhd N$. So there exists S_1 and S_2 simple such that

$$S_1 \times S_2 \subset N \subset G.$$

As $S_1 \times S_2$ contains an element of order $p \times q$ we are done.

c If G has a unique non-abelian minimal normal subgroup S which is simple, then G itself is an almost simple group so this case holds by assumption.

■

In the same spirit one can prove that:

Corollary 6.2.1. *Let G be a finite group with minimal normal subgroup N. If N is either abelian or not simple perfect with $\prod(\mathbb{Z}G/N) = \prod(G/N)$ then*

$$\prod(\mathbb{Z}G) = \prod(G).$$

Corollary 6.2.2. *The prime graph question holds for solvable groups.*

Proof. For an abelian simple group S the only torsion elements of $V(\mathbb{Z}G)$ are given by the elements of G itself. Because a solvable group of length l can be obtained as an extension of a solvable group of length $l-1$ by an abelian group A, induction yields the desired result. ■

6.2 The HeLP Package

The HeLP package was developed in 2015 by Andreas Bächle and Leo Margolis in an effort to create a GAP Package which provides functionalities to work with torsion units in integral group rings. For more information about the package, please see [6]. We will give a brief introduction on how we used this package to obtain our results.

First of all, it can be very helpful to use a high info level in order to get an idea about how fast computations are progressing. We used SetInfoLevel(HeLP_Info,5); .

There are functions available in this package like HeLP_ZC(OrdinaryCharacterTable | Group); or HeLP_PQ(OrdinaryCharacterTable | Group); which checks the Zassenhaus conjecture respectively the Prime Graph Question. But those functions are not really suited for our purposes as those functions use a rather brute force approach which can take ages if the order of the group becomes bigger. Instead we started by using the function HeLP_WithGivenOrder(CharacterTable, Order); which returns, given an ordinary table or a Brauer table and a specified order, a list of all possible partial augmentations of the given order and saves them in the variable HeLP_sol. But it does more than that. Given a unit $u \in V(\mathbb{Z}G)$ of order k it calculates all possible partial augmentations for u and the partial augmentations for all powers u^d of u for which $d \mid k$. For instance for $k = 6$ an entry of HeLP_sol[6] might be of the following form:

$$[[0, 1], [-3, 2, 0], [4, -6, 2, -8, 9]].$$

The first nested tuple $[0, 1]$ corresponds to the partial augmentations of u^3. So u^3 has partial augmentation 0 at the first conjugacy class of order 2 and partial augmentation 1 at the second conjugacy class of order 2, where the ordering on the conjugacy classes is taken from the GAP Character Table Library [19]. The second nested tuple $[-3, 2, 0]$ corresponds to the partial augmentations of u^2 and where -3 is the partial augmentation of u^2 at the first conjugacy class of order 3 and so on. It follows by Theorem 4.11 that all the information regarding the partial augmentations of u and its powers is contained in this list.

After using this function for the prime divisors of a group G, it is a good idea to try to reduce the number of admissible partial augmentations for units of order $p \cdot q$ by using p-constant characters. This can be done using the function
HeLP_WithGivenOrderSConstant(CharacterTable,p,q);. In many cases choosing a good table is sufficient to let this function return that there are 0 solutions left for units of order $p \cdot q$, i.e. to disprove their existence. It is important to remark that this function is equal to the function HeLP_WithGivenOrder(CharacterTable, q); in the case where there is a unique conjugacy class of order p. Moreover, this function only saves the output in HeLP_sol if it arrives at the conclusion that there are zero remaining possibilities for units of order $p \cdot q$.

Otherwise it returns a tuple, consisting of two nested tuples, where the first tuple corresponds to the partial augmentations of u^p, and the second tuple corresponds to the partial augmentations of u, but the first entry of this second tuple is nothing more than the sum of all the partial augmentations on conjugacy classes of order p.

After doing this for all possible combinations of p and q, the function
HeLP_AllOrdersPQ(CharacterTable) can be used. This function uses all previously calculated results and tries to discard the remaining solutions, if any, by using the Wagner Test, which is a consequence of Theorem 4.6:

Theorem 6.3. *(Wagner Test) Let u be a torsion unit in $V(\mathbb{Z}G)$, let g be a group element, and let $p, j \in \mathbb{Z}_{\geq 1}$ with p prime, then:*

$$\sum_{x^{p^j} \sim g} \varepsilon_x(u) \equiv \varepsilon_s(u) \mod p.$$

Proof. See Proposition 3.1 in [6]. ■

Theorem 6.4. *Let G be a finite group whose order is divisible by at most three primes, then*

$$\prod(\mathbb{Z}G) = \prod(G).$$

Proof. There is nothing to prove for the trivial group or for p-groups. If $|G| = p^a q^b$ for primes p and q then Theorem 2.3 implies that G is solvable so the prime graph question holds for G by Corollary 6.2.2 in this case. If $|G|$ is divisible by exactly three primes we can use Theorem 6.2, so it suffices to consider the almost simple groups whose order is divisible by exactly three primes. By the Classification of Finite Simple Groups [3], the finite simple groups of order divisible by exactly 3 primes and their automorphism groups are given in the tables below, together with the characters needed to exclude the existence of units of order $p \cdot q$ in their integral group ring. All of the simple groups listed below have $|Out(G)| = 2$ except $PSL(2,8)$ for which $|Out(PSL(2,8))| = 3$ and A_6 for which $|Out(A_6)| = 4$. So in all cases except for A_6 it suffices to show that the Prime Graph Question holds for the simple groups and their automorphism groups. For A_6 one should also check the Prime Graph Question for the groups M_{10} and $PGL(2,9)$ but this can not be done by the HeLP-method alone. For a proof of these cases we refer the reader to [5].

We used the HeLP Package for this calculations. The number of admissible partial augmentations for units of prime order where calculated by using the function HeLP_WithGivenOrder(CharacterTable, Order);. The characters in the third and fourth column below are those that were used in the function HeLP_WithGivenOrderSConstant(Characters, p,q);. We added a $*$ after the orders in the first column for which there is a unique conjugacy class of order p, in which case the use of HeLP_WithGivenOrderSConstant(Characters, p,q); is essentially the same as the use of HeLP_WithGivenOrder(Characters, p,q);. We will continue with this notation throughout this thesis.

Data for units in $V(\mathbb{Z}S)$ where S is a simple group with exactly three prime divisors .

Orders $p \cdot q$	q	p	r	Number of partial augmentations for units of order p and q	Ordinary characters used as p-constant characters	Brauer characters mod r used as p-constant characters	Number of remaining partial augmentations
A_5							
6∗	2	3	/	$1,1$	$2,5$	/	0
10	2	5	/	$1,2$	4	/	0
15	3	5	/	$1,2$	4	/	0
$PSL(2,7)$							
6∗	2	3	7	$1,1$	/	$2,3$	0
14	2	7	/	$1,2$	4	/	0
21	3	7	/	$1,2$	4	/	0
$PSL(2,8)$							
6∗	2	3	/	$1,1$	$2,3$	/	0
14	2	7	/	$1,15$	2	/	0
21	3	7	/	$1,15$	2	/	0
A_6							
6	2	3	/	$1,2$	$4,7$	/	1
10	2	5	/	$1,4$	2	/	0
15	3	5	/	$2,4$	2	/	0
$PSL(2,17)$							
6∗	2	3	17	$1,1$	/	$2,3$	0
34	2	17	/	$1,2$	4	/	0
51	3	17	/	$1,2$	4	/	0
$PSL(3,3)$							
26	2	13	/	$1,56$	2	/	0
39	3	13	/	$6,56$	2	/	0
$U(4,2)$							
10∗	2	5	/	$3,1$	$2,8$	/	0
15∗	3	5	/	$9,1$	$2,8$	/	0
$U(3,3)$							
14	2	7	/	$1,10$	2	/	0
21	3	7	/	$3,10$	2	/	0

The remaining partial augmentation in the case of A_6 corresponds to the tuple

$$[[2],[3,-2]].$$

Which was excluded by Hertweck in [40].

Data for units in $V(\mathbb{Z}Aut(S))$ where S is a simple group with exactly three prime divisors .

Orders $p \cdot q$	q	p	Number of partial augmentations for units of order p and q	Ordinary characters used as p-constant characters	Number of remaining partial augmentations
$Aut(A_5)$					
$10*$	2	5	$2,1$	$2,3$	0
$15*$	3	5	$1,1$	3	0
$Aut(PSL(2,7))$					
$14*$	2	7	$2,1$	3	0
$21*$	3	7	$1,1$	3	0
$Aut(PSL(2,8))$					
$14*$	2	7	$1,1$	4	0
$21*$	3	7	$3,1$	4	0
$Aut(A_6)$					
$15*$	3	5	$3,1$	5	0
$Aut(PSL(2,17))$					
$34*$	2	17	$2,1$	3	0
$51*$	3	17	$1,1$	3	0
$Aut(PSL(3,3))$					
26	2	13	$2,6$	3	0
39	3	13	$6,6$	3	0
$Aut(U(4,2))$					
$15*$	3	5	$7,1$	$3,4,7$	0
$Aut(U(3,3))$					
$14*$	2	7	$2,1$	3	0
$21*$	3	7	$3,1$	3	0

All of these characters are chosen minimal in the sense that a proper subset of the characters used in the table will not be sufficient to obtain the same results. ■

6.3 Sporadic Simple Groups

Theorem 6.5. *The prime graph question holds for the following sporadic simple groups:*

$$M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, J_1, J_2, J_3, HS, McL, He, Ru, Suz, Co_1, Co_2, Co_3.$$

Proof. Data can be found in the tables below. We were able to check that all the used characters, except for order 119 in He, are minimal in the sense that a proper subset of the used characters is not sufficient to exclude the existence of the units in question.

Data for units in $V(\mathbb{Z}M_{ij})$, where M_{ij} is a Mathieu group.

Orders $p \cdot q$	q	p	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters
	M_{11}				
10∗	2	5	1	1	2, 6
15∗	3	5	1	1	2
22	2	11	1	4	2
33	3	11	1	4	2
55	5	11	1	4	2
	M_{12}				
15∗	3	5	5	1	2, 4
22	2	11	6	4	2, 6
33	3	11	5	4	2
55	5	11	1	4	2
	M_{22}				
10∗	2	5	1	1	2, 3
14	2	7	1	14	2
15∗	3	5	1	1	2
21	3	7	1	14	5
22	2	11	1	52	2
33	3	11	1	52	2
35	5	7	1	14	2
55	5	11	1	52	2
77	7	11	14	52	2
	M_{23}				
10∗	2	5	1	1	2, 3
21	3	7	1	14	2, 10
22	2	11	1	164	2
33	3	11	1	164	2
35	5	7	1	14	2
46	2	23	2	68	2
55	5	11	1	164	2
69	3	23	1	68	2
77	7	11	14	164	2
115	5	23	1	68	2
161	7	23	14	68	2
253	11	23	164	68	2
	M_{24}				
22∗	2	11	6	1	2, 5
33∗	3	11	6	1	3
35	5	7	1	14	2
46	2	23	6	68	3
55∗	5	11	5	1	2
69	3	23	6	68	3
77∗	7	11	14	1	2
115	5	23	1	68	2
161	7	23	14	68	2
253	11	23	1	68	2

Data for units in $V(\mathbb{Z}J_i)$ where J_i is a Janko group.

Orders $p \cdot q$	q	p	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters
	J_1				
14∗	2	7	1	1	4
21∗	3	7	1	1	2
22∗	2	11	1	1	2
33∗	3	11	1	1	2
35∗	5	7	12	1	2
38	2	19	1	210	2
55∗	5	11	12	1	4
57	3	19	1	210	4
77∗	7	11	1	1	2
95	5	19	12	210	2
133	7	19	1	210	2
209	11	19	1	210	2
	J_2				
14∗	2	7	6	1	$4, 7$
21∗	3	7	4	1	$2, 4$
35∗	5	7	54	1	2
	J_3				
34	2	17	1	144	2
38∗	19	2	10	1	2
51	3	17	15	144	$2, 4$
57	3	19	15	10	9
85	5	17	130	144	6
95	5	19	130	10	6
323	19	17	10	144	2

Data for units in $V(\mathbb{Z}HS), V(\mathbb{Z}McL), V(\mathbb{Z}He), V(\mathbb{Z}Ru)$ and $V(\mathbb{Z}Suz)$.

Orders $p \cdot q$	q	p	r	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters	Brauer characters mod r used as p-constant characters
	HS						
14∗	2	7	/	6	1	2, 3	/
21∗	3	7	/	1	1	3	/
22	2	11	/	6	164	2, 3	/
33	3	11	/	1	164	2	/
35∗	5	7	/	63	1	2, 3, 4	/
55	5	11	/	63	164	2, 3	/
77	7	11	/	1	164	2	/
	McL						
21	3	7	/	4	2292	3, 5	/
22	2	11	/	1	164	2	/
33	3	11	/	4	164	2, 3	/
35	5	7	3	6	2292	/	2, 3
55	5	11	/	6	164	2, 3	/
77	11	7	/	164	2292	2	/
	He						
34	2	17	/	13	122	2, 4	/
35	5	7	2	1	8102	/	2, 4, 6
51	3	17	/	13	122	4, 6	/
85	5	17	/	1	122	2	/
119	7	17	2	8102	30	/	2, 3, 4, 5
	Ru						
21∗	3	7	2	1	1	/	2
35∗	5	7	2	8	1	/	3, 4
39∗	3	13	2	1	1	/	2
58	2	29	/	22	7628	2, 4	/
65∗	5	13	/	102	1	2	/
87	3	29	/	1	7628	2	/
91∗	7	13	/	1	1	2	/
145	5	29	/	102	7628	2	/
203	7	29	/	1	7628	2	/
377	13	29	/	1	7628	4	/
	Suz						
22∗	2	11	/	17	1	2, 4	/
26	2	13	/	17	14394	2, 3, 4	/
33∗	3	11	/	165	1	2, 3, 4	/
35∗	5	7	3	17	1	/	3
39	3	13	/	165	14394	2, 3, 4	/
55∗	5	11	/	27	1	2, 3	/
65	5	13	/	27	14394	2, 3	/
77∗	7	11	/	1	1	2	/
91	7	13	/	1	14394	2	/
143	11	13	/	1	14394	3	/

Data for units in $V(\mathbb{Z}Co_i)$ where Co_i is a Conway group.

Orders $p \cdot q$	q	p	r	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters	Brauer characters mod r used as p-constant characters	Number of remaining partial augmentations
	Co_1							
46	2	23	/	219	58588	$2,3,4$	/	0
55∗	5	11	/	1041	1	$2,3,4$	/	120
65	5	13	/	1041	1	$2,3,4,5$	/	14
69	3	23	/	15239	58588	$2,3,4,5$	/	0
77∗	7	11	/	47	1	3	/	0
91	7	13	/	47	1	3	/	0
115	5	23	/	1041	58588	$2,3,4$	/	0
143	11	13	/	1	1	2	/	0
161	7	23	/	47	58588	3	/	0
253	11	23	/	1	58588	2	/	0
299	13	23	/	1	58588	2	/	0
	Co_2							
21∗	3	7	/	4	1	$3,5$	/	0
22∗	2	11	/	48	1	$2,4,5$	/	0
33∗	3	11	/	4	1	3	/	0
35∗	5	7	/	6	1	$2,3$	/	2
46	2	23	/	48	838	$2,4,5$	/	0
55∗	5	11	/	6	1	3	/	0
69	3	23	/	4	838	3	/	0
77∗	7	11	/	1	1	2	/	0
115	5	23	/	6	838	3	/	0
161	7	23	/	1	838	2	/	0
253	11	23	/	1	838	2	/	0
	Co_3							
33	3	11	2	155	164	/	$2,3,6$	2
35∗	5	7	/	6	1	$2,3$	/	2
46	2	23	/	6	838	$2,3$	/	0
55	5	11	/	6	164	3	/	0
69	3	23	/	171	838	$2,3,6$	/	0
77	7	11	/	1	164	2	/	0
115	5	23	/	6	838	3	/	0
161	7	23	/	1	838	2	/	0
253	11	23	/	164	838	2	/	0

The two remaining possibilities for partial augmentations of order 33 in $V(\mathbb{Z}Co_3)$ were eliminated using the Wagner test. The other cases were handled by Margolis in [53]. We will use the same method as Margolis used here in the proof of the Prime Graph Question for $Aut(Fi22)$ later on. ■

Theorem 6.6. *Let G be one of the following sporadic simple groups:*

$$M_{12}, M_{22}, J_2, J_3, HS, McL, He, Suz.$$

Then the prime graph question holds for $Aut(G)$, in other words:

$$\prod(\mathbb{Z}Aut(G)) = \prod(Aut(G)).$$

Proof. Data can be found in the tables below. We were able to check that all the used characters are minimal in the sense that a proper subset of the used characters is not sufficient to exclude the existence of the units in question.

Data for units in $V(\mathbb{Z}Aut(M_{ij}))$ and $V(\mathbb{Z}Aut(J_k))$,

where M_{ij} is a Mathieu group and where J_k is a Janko group.

Orders $p \cdot q$	q	p	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters
$Aut(M_{12})$					
15∗	3	5	5	1	3, 4
22∗	2	11	11	1	2, 3, 4
33∗	3	11	5	1	3
55∗	5	11	1	1	3
$Aut(M_{22})$					
15∗	3	5	1	1	3
21	3	7	1	14	9
22∗	2	11	10	1	2, 3, 9
33∗	3	11	1	1	3
35	5	7	1	14	3
55∗	5	11	1	1	3
77∗	7	11	1	1	3
$Aut(J_2)$					
21∗	3	7	4	1	3, 4
35∗	5	7	5	1	3
$Aut(J_3)$					
38∗	2	19	2	1	2, 3
51	3	17	15	144	3, 4
57∗	3	19	15	1	3, 4
85	5	17	1	144	5
95∗	5	19	1	1	3
323∗	17	19	144	1	2, 3

Data for units in $V(\mathbb{Z}Aut(HS)), V(\mathbb{Z}Aut(McL)), V(\mathbb{Z}Aut(He))$ and $V(\mathbb{Z}Aut(Suz))$.

Orders $p \cdot q$	q	p	r	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters	Brauer characters mod r used as p-constant characters
$Aut(HS)$							
21∗	3	7	/	1	1	5	/
22∗	2	11	/	31	1	$2, 3, 4, 5$	/
33∗	3	11	/	1	1	3	/
35∗	5	7	/	63	1	$3, 6, 7$	/
55∗	5	11	/	63	1	$3, 5$	/
77∗	7	11	/	1	1	3	/
$Aut(McL)$							
21∗	3	7	/	4	1	$5, 9$	/
33	3	11	/	4	164	$3, 5$	/
35∗	5	7	3	6	1	/	$3, 5$
55	5	11	/	6	164	$3, 5$	/
77	7	11	/	1	164	3	/
$Aut(He)$							
34∗	2	17	/	26	1	$2, 3, 7$	/
35∗	7	5	/	612	1	$3, 4, 6, 11$	/
51∗	3	17	/	13	1	$4, 6$	/
85∗	5	17	/	1	1	3	/
119∗	7	17	/	612	1	$3, 4, 5$	/
$Aut(Suz)$							
26∗	2	13	3	103	1	/	$2, 3, 4, 5$
33∗	3	11	/	165	1	$3, 5, 7$	/
35∗	5	7	3	17	1	/	$3, 5$
39∗	3	13	/	165	1	$3, 5, 7$	/
55∗	5	11	/	27	1	$3, 5$	/
65∗	5	13	/	27	1	$3, 5$	/
77∗	7	11	/	1	1	3	/
91∗	7	13	/	1	1	3	/
143∗	11	13	/	1	1	5	/

■

Theorem 6.7. *The prime graph question holds for the Tits group T and its automorphism group.*

Proof. Data can be found in the table below. All the used characters are minimal.

Data for units in $V(\mathbb{Z}T)$, and $V(\mathbb{Z}Aut(T))$.

Orders $p \cdot q$	q	p	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters
T					
$15*$	3	5	1	1	2
26	2	13	7	316	$2, 7$
39	3	13	1	316	2
65	5	13	1	316	2
$Aut(T)$					
$15*$	3	5	1	1	3
$26*$	2	13	7	1	$3, 10$
$39*$	3	13	1	1	3
$65*$	5	13	1	1	3

■

According to [25], the following sporadic simple groups have trivial $Out(G)$:

$$M_{11}, M_{23}, M_{24}, J_1, Ru, Co_1, Co_2, Co_3.$$

We obtain:

Theorem 6.8. *The prime graph question holds for a finite group provided its composition factors S are isomorphic to one of the following sporadic simple groups:*

$$M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, J_1, J_2, J_3, Co_1, Co_2, Co_3, HS, McL, He, Ru, Suz, T, Fi_{22}$$

Proof. By Theorem 6.2, it suffices to look at all almost simple quotients of G. Let X be such a quotient, by Schreiers conjecture it follows that X is an extension of a solvable group A by a simple group S. As the set of composition factors of a group H is the union of the composition factors of a normal subgroup N and those of the quotient group H/N it follows that S is a composition factor of G. So if the prime graph question holds for all composition factors of G and their automorphism groups, then the prime graph question holds for all almost simple quotients of G, which settles the proof. The fact that the Prime Graph Question holds fo Fi_{22} and its automorphism group will be proven in the next section. ■

6.3.1 Some results on Fi_{22}

Using the HeLP package we obtained the following results on the sporadic simple group Fi_{22}:

Orders $p \cdot q$	q	p	Number of partial augmentations for units of order q	Number of partial augmentations for units of order p	Ordinary characters used as p-constant characters	Number of remaining partial augmentations
Fi_{22}						
26	2	13	68	21324	$2, 3, 4$	0
33	3	11	5320	105892	$2, 3, 4, 5$	4
35∗	5	7	1	1	2	0
39	3	13	5320	21324	$2, 4, 5$	0
55	5	11	1	105892	2	0
65	5	13	1	21324	2	0
77	7	11	1	105892	3	0
91	7	13	1	21324	2	0
143	11	13	105892	21324	2	0
$Aut(Fi_{22})$						
26∗	2	13	6259	1	$2, ..., 8$	0
33∗	3	11	5320	1	$3, 5, 7, 9$	4
35∗	5	7	1	1	3	0
39∗	3	13	5320	1	$3, 7, 9$	0
55∗	5	11	1	1	3	0
65∗	5	13	1	1	3	0
77∗	7	11	1	1	5	0
91∗	7	13	1	1	3	0
143∗	11	13	1	1	3	0

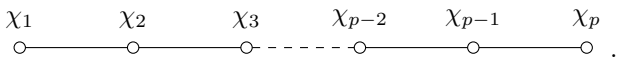
We checked minimality for all used characters, except for units of order 26 in $Aut(Fi_{22})$. The 4 possibilities which are left for units of order 33 correspond in both cases to the tuples:

$$[[1, -1, 3, -2], [-11, 1, -1, 3, 9]], [[1, 0, 3, -3], [-11, 1, 0, 3, 8]],$$

$$[[1, 9, 3, -12], [-11, 1, -2, 3, 10]], [[1, 10, 3, -13], [-11, 1, -1, 3, 9]].$$

But it should be remarked that there are two conjugacy classes of order 11 in Fi_{22}, so in this case we still do not know the partial augmentations of u^3, moreover the first entry of the second nested tuple, the -11 is nothing more than the sum of the two partial augmentations of u at the conjugacy classes of order 11. The Prime Graph Question for Fi_{22} and $Aut(Fi_{22})$ were open problems at the start of this thesis, but we will now prove that there are no units of order 33 in $V(\mathbb{Z}Aut(Fi_{22}))$ which settles the Prime Graph Question for both Fi_{22} and $Aut(Fi_{22})$. Remark that we can identify $V(\mathbb{Z}Fi_{22})$ with part of $V(\mathbb{Z}Aut(Fi_{22}))$ so that it is indeed the case that it suffices to exclude units of order 33 in the integral group ring of the automorphism group. For this we will use a method which is due to Margolis [53]. We will state the main theorem and lemma but we will not delve deeper into the proofs.

Theorem 6.9. *(Margolis) Let G be a finite group, let p and n be positive integers where $p \nmid n$. Assume that some p-block of G is a Brauer Graph Algebra consisting of the ordinary characters $\chi_1, \cdots, \chi_p$ where the Brauer Graph Algebra has the following form:*



Moreover assume that the prime p is unramified in the character fields of the χ_i. Let u be a unit of order n in $V(\mathbb{Z}G)$ and let ξ be an nth root of unity. Then the following inequality holds:

$$\mu(\xi \cdot \zeta_p, u, \chi_{p-1}) - \mu(\xi \cdot \zeta_p, u, \chi_p) \leq \mu(\xi, u, \chi_1) - \sum_{i=1}^{p-2} (-1)^i \mu(\xi \cdot \zeta_p, u, \chi_i)$$

where $\mu(\alpha, u, \chi)$ denotes the multiplicity of α as an eigenvalue of the representation associated with χ.

Proof. See Theorem 1.2 in [53]. ■

Definition 6.9.1. An algebraic field field extension F of $\mathbb{Q}$ is called a **number field** if and only if it is an extension of finite degree.

Definition 6.9.2. A prime p is called a **ramified prime** in a number field K if and only if the prime ideal factorization of (p) does not has a component with multiplicity greater than one.

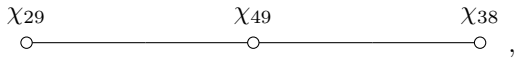
We will not have to deal with checking ramification as all the characters we will use have only integral values and there do not exist any ramified primes in $\mathbb{Q}$. In this case we can also use the following lemma to calculate the multiplicities in Theorem 6.9 :

Lemma 6.10. *(Margolis) Let G be a finite group and let u be a unit of order pq in $V(\mathbb{Z}G)$ where p and q are different primes. Let χ be a character of G which takes only integral values. Set $\chi(1) = d$, $\chi(u^p) = x$, $\chi(u^q) = y$ and $\chi(u) = z$. Then the following formulas hold.*

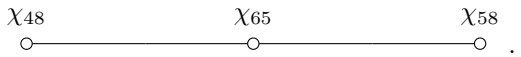
$$\begin{aligned} pq \cdot \mu(1, u, \chi) &= d + (q-1)x + (p-1)y + (p-1)(q-1)z, \\ pq \cdot \mu(\zeta_p, u, \chi) &= d + (q-1)x - y - (q-1)z, \\ pq \cdot \mu(\zeta_q, u, \chi) &= d - x + (p-1)y - (p-1)z, \\ pq \cdot \mu(\zeta_{pq}, u, \chi) &= d - x - y + z. \end{aligned}$$

Proof. See Lemma 2.3 in [53]. ■

In order to try this method on Fi_{22} and its automorphism group, we calculated the Brauer trees which fulfill the requirements of Theorem 6.9, we used the same numbering as in [19]. For Fi_{22} we found the following admissible trees:



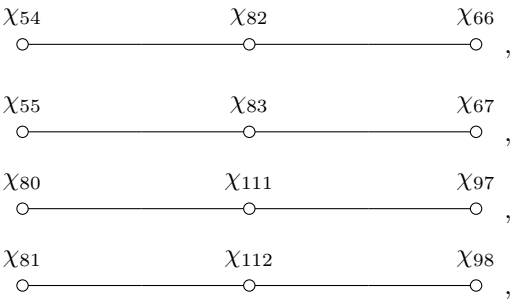
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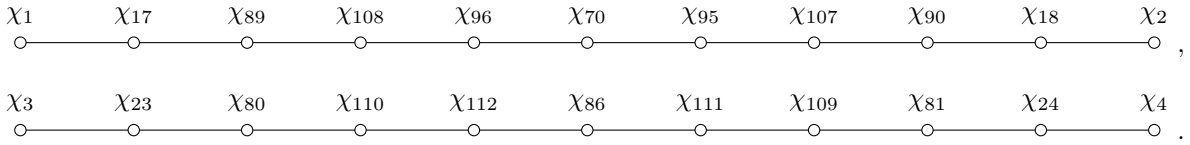


.

But after applying Theorem 6.9 we were not able to achieve a contradiction.

For $Aut(Fi_{22})$ we found the following admissible Brauer trees:





The trees of length 3 were not sufficient to achieve a contradiction after applying Theorem 6.9. But let us now take a look at the second Brauer tree of length 11, we have chosen this tree as all the characters in this tree have only integral values, so Lemma 6.10 is applicable in this case. We obtained the multiplicities below by using the function HeLP_MultiplicitiesOfEigenvalues(Character, Order, Partial Augmentations); which returns a tuple where the ith spot corresponds to the multiplicity of $E(k)^{i-1}$, and where E(k) denotes a primitive k-th root of unity. Please beware of the fact that the partial augmentations one wishes to use should be given in the same order as in [19]. We obtained the following multiplicities:

	$[[1,-1,3,-2],[1],[1,-1,3,9,-11]]$		$[[1,9,3,-12],[1],[1,-2,3,10,-11]]$	
	$[[1,0,3,-3],[1],[1,0,3,8,-11]]$		$[[1,10,3,-13],[1],[1,-1,3,9,-11]]$	
χ	$\mu(\zeta_3,u,\chi)$	$\mu(\zeta_3\zeta_{11},u,\chi)$	$\mu(\zeta_3,u,\chi)$	$\mu(\zeta_{11}\zeta_3,u,\chi)$
χ_3	2	1	2	1
χ_{23}		1322		1340
χ_{80}		25846		25846
χ_{110}		77444		77372
χ_{112}		82708		82708
χ_{86}		59582		59690
χ_{111}		82708		82708
χ_{109}		77444		77372
χ_{81}		25846		25846
χ_{24}		1322		1340
χ_4		1		1

So using Theorem 6.9 yields two times $1321 \leq 1319$ and two times $1339 \leq 1337$, which are contradictions. Thus we may conclude that there are no units of order 33 in $Aut(Fi_{22})$ which settles the Prime Graph Question for both Fi_{22} and $Aut(Fi_{22})$.

6.4 PQ for almost simple groups with alternating socle

The last section in this thesis is devoted to the study of the Prime Graph Question for almost simple groups with alternating socle. As the Reduction Theorem 6.2 shows, almost simple groups are of particular interest regarding the Prime Graph Question. The main goal of this section is providing a proof that the Prime Graph Question holds for almost simple groups with socle A_n for $n \leq 17$. We will follow [22].

Lemma 6.11. *Let P be the representation which corresponds to the character π.*

1. *Let r be a prime and let $v \in V(\mathbb{Z}S_n)$ be a unit of order k. Then the multiplicity of a primitive k-th root of unity as an eigenvalue of $P(v)$ is given by*

$$\mu_1(v, \pi) = \sum_{j=1}^{\lfloor \frac{n}{k} \rfloor} j\varepsilon_{k.j}(v).$$

2. *Let p and q be two different primes such that $p+q > n$. For an element $u \in V(\mathbb{Z}S_n)$ of order pq, the multiplicity of a primitive pq-th root of unity as an eigenvalue of $P(u)$ is given by*

$$\mu_1(u, \pi) = \frac{1}{pq}\Big[q\sum_{j=1}^{\lfloor \frac{n}{q} \rfloor} j(\varepsilon_{q.j}(u^p) - \varepsilon_{q.j}(u)) + p\sum_{i=1}^{\lfloor \frac{n}{p} \rfloor} i(\varepsilon_{p.i}(u^q) - \varepsilon_{p.i}(u))\Big].$$

Proof.

1. We know by Theorem 4.17 that the element v can only have nonzero partial augmentations at conjugacy classes of elements of order k. Using the HeLP-formula 4.16 yields

$$\mu_1(v, \pi) = \frac{1}{k}(\pi(1) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}(\pi(v), \eta)).$$

   Where $Z = \mathbb{Q}(\eta)$ and η is a primitive k-th root of unity. But using Theorem 4.11 we obtain that

$$\pi(v) = \sum_{j=1}^{\lfloor \frac{n}{k} \rfloor} \varepsilon_{\mathsf{k.j}}(v)\pi(\mathsf{k.j}) = \sum_{j=1}^{\lfloor \frac{n}{k} \rfloor} \varepsilon_{\mathsf{k.j}}(v)(\pi(1) - kj).$$

   If we now combine this with the fact that $\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}(\eta) = 1$, we find, using that v has augmentation 1:

$$\mu_1(v, \pi) = \frac{1}{k}(\pi(1) + \sum_{j=1}^{\lfloor \frac{n}{k} \rfloor} \varepsilon_{\mathsf{k.j}}(v)(\pi(1) - kj)) = \sum_{j=1}^{\lfloor \frac{n}{k} \rfloor} j\varepsilon_{\mathsf{k.j}}(v).$$

2. Let u be an element of $V(\mathbb{Z}S_n)$ of order pq, as S_n has no elements of order pq we find, using the HeLP-formula that

$$\mu_1(u,\pi) = \frac{1}{pq}(\pi(1)+\mathrm{Tr}_{\mathbb{Q}(Z^q)/\mathbb{Q}}(\pi(u^q)\eta^{-q})+\mathrm{Tr}_{\mathbb{Q}(Z^p)/\mathbb{Q}}(\pi(u^p)\eta^{-p})+\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}(\pi(u)\eta^{-1}).$$

Where η is a primitive pq-root of unity. Using that $\mathrm{Tr}_{\mathbb{Q}(Z^q)/\mathbb{Q}}(\eta^{-q}) = \mathrm{Tr}_{\mathbb{Q}(Z^p)/\mathbb{Q}}(\eta^{-p}) = -1$ and $\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}(\eta^{-1}) = 1$ we see that

$$\mu_1(u,\pi) = \frac{1}{pq}(\pi(1) - \pi(u^q) - \pi(u^p) + \pi(u)). \tag{1}$$

Analogous to 1. we can rewrite the $\pi(x)$, for example

$$\pi(u^q) = \sum_{j=1}^{\lfloor \frac{n}{q} \rfloor} \varepsilon_{\mathsf{p.j}}\,(v)\pi(\mathsf{p.j}) = \pi(1) - p\sum_{j=1}^{\lfloor \frac{n}{q} \rfloor} j\varepsilon_{\mathsf{p.j}}\,(u^q).$$

Doing the same for $\pi(u^p)$ and $\pi(u)$ and substituting this into (1) yields:

$$\begin{aligned}
\mu_1(u,\pi) =& \frac{1}{pq}(\pi(1) - (\pi(1) - p\sum_{j=1}^{\lfloor \frac{n}{p} \rfloor} j\varepsilon_{\mathsf{p.j}}\,(u^q)) - (\pi(1) - q\sum_{i=1}^{\lfloor \frac{n}{q} \rfloor} i\varepsilon_{\mathsf{q.i}}\,(u^p))) \\
&+(\pi(1) - p\sum_{j=1}^{\lfloor \frac{n}{p} \rfloor} j\varepsilon_{\mathsf{p.j}}\,(u)) - q\sum_{i=1}^{\lfloor \frac{n}{q} \rfloor} i\varepsilon_{\mathsf{q.i}}\,(u))) \\
=& \frac{1}{pq}(q(\sum_{i=1}^{\lfloor \frac{n}{q} \rfloor} i\varepsilon_{\mathsf{q.i}}\,(u^p) - \sum_{i=1}^{\lfloor \frac{n}{q} \rfloor} i\varepsilon_{\mathsf{q.i}}\,(u)) + p(\sum_{j=1}^{\lfloor \frac{n}{p} \rfloor} j\varepsilon_{\mathsf{p.j}}\,(u^q) - \sum_{j=1}^{\lfloor \frac{n}{p} \rfloor} j\varepsilon_{\mathsf{q.j}}\,(u))) \\
=& \frac{1}{pq}\Big[q\sum_{j=1}^{\lfloor \frac{n}{q} \rfloor} j(\varepsilon_{\mathsf{q.j}}\,(u^p) - \varepsilon_{\mathsf{q.j}}\,(u)) + p\sum_{i=1}^{\lfloor \frac{n}{p} \rfloor} i(\varepsilon_{\mathsf{p.i}}\,(u^q) - \varepsilon_{\mathsf{p.i}}\,(u))\Big].
\end{aligned}$$

■

Lemma 6.12. *Let n be an integer, not smaller then 7, and let p and q be primes with $p > \frac{n}{2}$ and $n \leq q \leq 3$. Assume that $u \in V(\mathbb{Z}S_n)$ has order pq. Then it follows that*

$$\mu_1(u,\pi) = 0 \quad \textit{and} \quad \mu_q(u,\pi) = 1.$$

If, additionally, $p + q \geq n$, we have that

$$\mu_p(u,\pi) = \mu_1(u^p,\pi) \in \{0,1\}.$$

Where 1 is only possible if $p + q \in \{n, n+1\}$.

Proof. Let us extend our representation P to a representation of $V(\mathbb{Z}S_n)$. As u has finite order, we can diagonalize $P(u)$ over $\mathbb{C}$. As $P(u)$ is a rational matrix , we can write

$$P(u) \sim diag(m_1 \times \boxed{1}, m_q \times \boxed{\eta_q, \eta_q^2 \cdots, \eta_q^{q-1}}, m_p \times \boxed{\eta_p, \eta_p^2 \cdots, \eta_p^{p-1}}, m_{pq} \times \boxed{\eta_{pq}, \eta_{pq}^2 \cdots, \eta_{pq}^{(p-1)(q-1)}}).$$

Where η_k is a primitive k-th root of unity, and where

- $m_1 = \mu_0(u, \pi)$ is the multiplicity of 1 as an eigenvalue of $P(u)$.
- $m_q = \mu_p(u, \pi)$ is the number of $(q-1) \times (q-1)$ blocks with primitive qth roots of unity.
- $m_p = \mu_q(u, \pi)$ is the number of $(p-1) \times (p-1)$ blocks with primitive pth roots of unity.
- $m_{pq} = \mu_1(u, \pi)$ is the number of $(q-1)(p-1) \times (q-1)(p-1)$ blocks with primitive pqth roots of unity.

But then it follows that

$$P(u^q) \sim diag((m_1 + (q-1)m_q) \times \boxed{1}, (m_p + (q-1)m_{pq}) \times \boxed{\eta_p, \eta_p^2 \cdots, \eta_p^{p-1}}).$$

Further, because $p > \frac{n}{2}$, there is a unique conjugacy class with elements of order p in S_n, namely p-1, it follows that u^q is rationally conjugate to an element in p-1 and hence there is exactly one block of primitive pth roots of unity in the diagonalization of $P(u^q)$. So we obtain that

$$m_q + (q-1)m_{pq} = 1.$$

As $q \geq 3$ this implies that $\mu_1(u, \pi) = m_{pq} = 0$ and $\mu_q(u, \pi) = m_p = 1$.
If $p + q \geq n$, then looking at $P(u^{pq})$ and using the above results we see that

$$n - 1 = \pi(1) = m_1 + m_q(q-1) + (p-1).$$

Thus $m_q \in \{0, 1\}$. As $m_{pq} = 0$ we get that $\mu_1(u^p, \pi) = m_q$. ■

6.4.1 Units of Odd Order

In this section we will treat units of order pq where p and q are odd primes. We will start with an example, which illustrates the use of the so-called p-constant characters.

Example 6.13. *We will prove that there do not exist units of order* 15 *in* $V(\mathbb{Z}S_7)$. *Assume by means of contradiction that* u *is such a unit. By theorem 4.11 it follows that* u *can only posses nonzero partial augmentations at conjugacy classes of order* 3 *and* 5, *i.e. at the classes 3.1,3.2 and 5.1. Consider the character* χ *belonging to the partition* $(4, 1^3)$ *of* 7. *By the Frobenius Formula 3.3 we obtain the following values:*

	1	3.1	3.2	5.1
χ	20	2	2	0

As the character is constant on the classes which contains elements of order 3, *the multiplicities of the eigenvalues will not depend on the prartial augmentations of* u^5 *as such. Using the HeLP-formula we obtain*

$$\begin{aligned}\mu_l(u;\chi) =& \frac{1}{15}[20 + \mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(\chi(u^5)Z^{-5l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z^3)/\mathbb{Q}}\left(\chi(u^3)Z^{-3l}\right) + \mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(\chi(u)Z^{-l}\right)]\\ =& \frac{1}{15}[20 + 2\,\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right) + (1 - \varepsilon_{5.1}(u))\,\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)]\\ =& \begin{cases} \frac{1}{15}[24 + 16 - 16\varepsilon_{5.1}(u)] = \frac{8}{3} - \frac{16}{15}\varepsilon_{5.1}(u) \in \mathbb{Z}_{\geq 0} & \text{if } l = 0\\ \frac{1}{15}[18 - 8 - 8\varepsilon_{5.1}(u)] = \frac{2}{3} + \frac{8}{15}\varepsilon_{5.1}(u) \in \mathbb{Z}_{\geq 0} & \text{if } l = 5. \end{cases}\end{aligned}$$

Which is a contradiction. Here we used that

$Z = -e^{\frac{2\pi i}{5}}e^{\frac{2\pi i}{3}}$	$\mathrm{Tr}_{\mathbb{Q}(Z^5)/\mathbb{Q}}\left(Z^{-5l}\right)$	$\mathrm{Tr}_{\mathbb{Q}(Z)/\mathbb{Q}}\left(Z^{-l}\right)$
$l = 0$	2	8
$l = 5$	-1	-4

Theorem 6.14. *Let* $n \in \mathbb{Z}_{\geq 1}$ *and let* p *and* q *be odd primes with* $p, q > \frac{n}{3}$. *Then there exist an element of order* pq *in* S_n *if and only if there is an element of order* pq *in* $V(\mathbb{Z}S_n)$.

Proof. As necessity is rather obvious it suffices to prove sufficiency. Assume that $u \in V(\mathbb{Z}S_n)$ is a unit of order pq with $p+q > n$ (so that there does not exist an element of order pq in S_n.). By [24], Corollary 4.1. There exists an element of order p^2 in S_n if and only if there exists an element of order pq in $V(\mathbb{Z}S_n)$. Thus we can assume that $p > q$. We also know, by [48], that the prime graph question holds for S_n if $n \leq 6$. Combining this with the example above we may assume that $p \geq 7$ and $q \geq 3$. We can also assume that $p > \frac{n}{2}$, for otherwise there is an element of order pq in S_n. Thus there exists a unique conjugacy class p.1, of elements of order p in S_n and u^q is rationally conjugate to an element in that class. We will distinguish between two cases: $q > \frac{n}{2}$ or $\frac{n}{2} \geq q > \frac{n}{3}$ which is equivalent to $\lfloor \frac{n}{q} \rfloor = 1$ or $\lfloor \frac{n}{q} \rfloor = 2$.

If $\lfloor \frac{n}{q} \rfloor = 1$, we know that the element u^p, which has order q, is rationally conjugate to a group element which lies in the conjugacy class $\mathsf{q.1}$. If we now use Lemma 6.11 to calculate $\mu_1(u, \pi)$ we find that

$$\mu_1(u,\pi) = \frac{1}{pq}[q(1-\varepsilon_{\mathsf{q.1}}(u)) + p(1-\varepsilon_{\mathsf{p.1}}(u))] = \frac{1}{pq}[\varepsilon_{\mathsf{p.1}}(u)(q-p)+p].$$

If we combine this with the fact that the augmentation of u has to be $1 = \varepsilon_{\mathsf{p.1}}(u) + \varepsilon_{\mathsf{q.1}}(u)$ and with Lemma 6.12, which implies that $\mu_1(u, \pi) = 0$ we get that $\varepsilon_{\mathsf{p.1}}(u) = \frac{p}{p-q} \in \mathbb{Z}$. So $(p, q) = (3, 2)$ and $n = 3$, a case for which the result is known.

On the other hand if $\lfloor \frac{n}{q} \rfloor = 2$, we can once more use Lemma 4.11 to obtain that $\mu_1(u^p, \pi) = \varepsilon_{\mathsf{q.1}}(u^p) + 2\varepsilon_{\mathsf{q.2}}(u^p)$ and if we use 6.12 it follows that $\mu_1(u^p, \pi) \in \{0, 1\}$ where 1 is only possible if $n = p + q - 1$. If we combine this with the fact that $\varepsilon_{\mathsf{q.1}}(u^p) + \varepsilon_{\mathsf{q.2}}(u^p) = 1$ we find the following system of linear equations:

$$\begin{cases} \varepsilon_{\mathsf{q.1}}(u^p) + 2\varepsilon_{\mathsf{q.2}}(u^p) \in \{0,1\} & (i) \\ \varepsilon_{\mathsf{q.1}}(u^p) + \varepsilon_{\mathsf{q.2}}(u^p) = 1 & (ii) \end{cases}$$

which yields the following solutions:

$$(\varepsilon_{\mathsf{q.1}}(u^p), \varepsilon_{\mathsf{q.2}}(u^p)) = (2, -1)$$

$$(\varepsilon_{\mathsf{q.1}}(u^p), \varepsilon_{\mathsf{q.2}}(u^p)) = (1, 0) \quad \text{and} \quad n = p + q - 1.$$

We will now use another character, $\rho = \chi_{(n-2,1,1)}$, we can compute its values using the Frobenius formula, Theorem 3.3, to obtain the following table:

	1	$\mathsf{q.1}$	$\mathsf{q.2}$	$\mathsf{p.1}$
π	$n-1$	$\pi(1)-q$	$\pi(1)-2q$	$\pi(1)-p$
ρ	$\frac{1}{2}(n-1)(n-2)$	$\rho(1)-\frac{1}{2}q(2n-q-3)$	$\rho(1)-q(2n-2q-3)$	$\rho(1)-\frac{1}{2}p(2n-p-3)$

.

(i) : Assume that $(\varepsilon_{\mathsf{q.1}}(u^p), \varepsilon_{\mathsf{q.2}}(u^p)) = (2, -1)$. If we use the HeLP-formula we see that

$$\mu_1(u^p,\rho) = \frac{1}{q}(\rho(1) - 2(\rho(1) - q\frac{2n-q-3}{2}) - (\rho(1) - q(2n-2q-3))) = q.$$

But arguing as in the proof of Lemma 6.12 we may conclude that this has to be equal to $\mu_p(u, \rho) + (p-1)\mu_1(u, \rho)$ which implies that $\mu_1(u, \rho) = 0$ and $\mu_p(u, \rho) = q$.

(ii) : Assume that $(\varepsilon_{\mathsf{q.1}}(u^p), \varepsilon_{\mathsf{q.2}}(u^p)) = (1,0)$ and that $n = p+q-1$. Using the HeLP-formula yields, as $q \geq 3$:

$$\mu_1(u^p, \rho) = \frac{1}{q}(\rho(1) - (\rho(1) - (\rho(1) - q\frac{2n-2q-3}{2})) = \frac{2n-q-3}{2} \leq n-3.$$

But again, arguing as in the proof of Lemma 6.12, this equals $\mu_p(u,\rho) + (p-1)\mu_1(u,\rho)$ which forces that $\mu_1(u,\rho) \in \{0,1\}$, as $p > \frac{n}{2}$.

Thus we can conclude that

$$\begin{cases} \mu_1(u,\rho) = 0 \\ \mu_1(u,\rho) = 1 \quad \text{with } n = p+q-1. \end{cases}$$

Combining the HeLP-formula with Lemma 6.11 for the character ρ we get that:

$$\begin{aligned} \mu_1(u,\pi) =& \frac{1}{pq}[-q\varepsilon_{\mathsf{q.1}}(u) - 2q\varepsilon_{\mathsf{q.2}}(u) - p\varepsilon_{\mathsf{p.1}}(u) + q(\varepsilon_{\mathsf{q.1}}(u^p) + 2\varepsilon_{\mathsf{q.2}}(u^p)) + p], \\ \mu_1(u,\rho) =& \frac{1}{2pq}[-(2n-q-3)q\varepsilon_{\mathsf{q.1}}(u) - 2(2n-2q-3)q\varepsilon_{\mathsf{q.2}}(u) - (2n-p-3)p\varepsilon_{\mathsf{p.1}}(u) \\ & +q((2n-q-3)\varepsilon_{\mathsf{q.1}}(u^p) + 2(2n-2q-3)\varepsilon_{\mathsf{q.2}}(u^p)) + p(2n-p-3)]. \end{aligned}$$

If we now take a look at the following system of linear equations

$$\begin{cases} \mu_1(u,\pi) = 0 \\ \mu_1(u,\rho) = t \in \{0,1\} \\ \varepsilon(u) = 1 \end{cases} \tag{1}$$

This system has a unique solution over $\mathbb{Q}$ and if we solve it for $\varepsilon_{\mathsf{p.1}}$ we find that

$$\varepsilon_{\mathsf{p.1}}(u) = \frac{p(p-q(3-2t))}{(p-q)(p-2q)}.$$

Which equals respectively in the cases $t=0$ and $t=1$:

$$\frac{p(p-3q)}{(p-q)(p-2q)} \quad \text{and} \quad \frac{p}{p-2q}.$$

The first value is an integer if and only if $(p,q) \in \{(3,2),(5,3)\}$, cases we excluded at the start of the proof. The second possibility can only occur if $n = p+q-1$. If we require the second possibility to be an integer we obtain that $p-2q = \pm 1$. But if $p-2q = 1$ then $n = 3q$ which is a contradiction. So $p-2q = -1$ and the unique integral solution of (1) is given by

$$(\varepsilon_{\mathsf{q.1}}(u), \varepsilon_{\mathsf{q.2}}(u), \varepsilon_{\mathsf{p.1}}(u)) = (1, p, -p).$$

In this case we will need another character. Set $\tau = \chi_{(n-3,2,1)}$ and by using the Frobenius formula, Theorem 3.3, we find that

$$\tau(1) = \frac{1}{3}n(n-2)(n-4),$$

$$\tau(\mathsf{r.j}) = \frac{(n-rj)}{3}((n-rj-1)(n-rj-5)+3) \quad \text{with} \quad \mathsf{r.j} \in \{\mathsf{q.1}, \mathsf{q.2}, \mathsf{p.1}\}.$$

Once more using the HeLP-formula provides us with

$$\begin{aligned}\mu_p(u,\tau) =& \frac{1}{pq}(\tau(1) - \tau(u^p) + (p-1)\tau(u^q) - (p-1)\tau(u)) \\ =& \frac{1}{(2q-1)q}(\tau(1) - (2q-1)\tau(\mathsf{q.1}) - (2q-1)(2q-2)\tau(\mathsf{q.2}) \\ &+2q(2q-2)\tau(\mathsf{p.1}) = -\frac{1}{3}q(q-4),\end{aligned}$$

which can only be non-negative if $q \leq 4$ and hence $p = 7$ and $q = 3$, but this is in contradiction with the assumption that $p = 2q - 1$. ■

We know by [59] that the prime graph question holds for A_n if $n \leq 10$, as we can embed $\mathbb{Z}A_n$ into $\mathbb{Z}S_n$ and there is an element of order pq in A_n if and only if there is an element of order pq in S_n, we immediately obtain the following corollary, which will be of vital importance in the proof of our main theorem later on:

Corollary 6.14.1. *Let $n \in \mathbb{Z}_{\geq 1}$ and let p and q be two primes with $p, q > \frac{n}{3}$. Then there is an element of order pq in A_n if and only if there is an element of order pq in $V(\mathbb{Z}A_n)$.*

6.4.2 Units of Even Order

It this section, p will be an odd prime. This section is devoted to units in $V(\mathbb{Z}S_n)$ of order $2p$. In the prime graph of S_n the prime 2 is connected to any other vertex except in the cases when $n \in \{p, p+1\}$. Remark that the conjugacy classes of elements of order 2, denoted by 2.j, in S_n are contained in A_n if and only if j is even.

Lemma 6.15. *Let $n \in \mathbb{Z}_{\geq 1}$, p an odd prime and $u \in V(\mathbb{Z}S_n)$ of order $2p$. Then $\varepsilon_{A_n}(u) = \varepsilon_{A_n}(u^p) \in \{0, 1\}$.*

Proof. Remark that $S_n \backslash A_n = \langle t \rangle \cong C_2$ and consider the natural ring epimorphism

$$f : \mathbb{Z}S_n \to \mathbb{Z}S_n \backslash A_n \cong \mathbb{Z}C_2.$$

Now an element $x \in \mathbb{Z}S_n$ is mapped to $f(v) = \varepsilon_{A_n}(v) + \varepsilon_{S_n \backslash A_n} t$. But the ring $\mathbb{Z}C_2$ has only trivial torsion units so one of the values $\varepsilon_{A_n}(v)$ and $\varepsilon_{S_n \backslash A_n}$ equals 1 and the other is 0. Furthermore for every odd $k \in \mathbb{Z}_{\geq 1}$:

$$\begin{aligned}f(u^k) =& \varepsilon_{A_n}(u^k) + \varepsilon_{S_n \backslash A_n}(u^k)t = f(u)^k \\ =& \sum_{j=0}^{k} \binom{k}{j} \varepsilon_{A_n}(u)^j \varepsilon_{S_n \backslash A_n}(u)^{k-j} t^{k-j} = \varepsilon_{A_n}(u)^k + \varepsilon_{S_n \backslash A_n}(u)^k t\end{aligned}$$

from which the claim follows. ■

Lemma 6.16. *Let p be an odd prime and let $u \in V(\mathbb{Z}S_p)$ be an element of order $2p$. Then we have $\mu_1(u,\pi) = 0$ and $\sum_{j=1}^{\lfloor \frac{p}{2} \rfloor} j\varepsilon_{2.j}(u^p) = 0$, more precisely:*

$$\sum_{\substack{j=1 \\ 2\nmid j}}^{\lfloor \frac{p}{2} \rfloor} j\varepsilon_{2.j}(u^p) = \sum_{\substack{j=1 \\ 2 | j}}^{\lfloor \frac{p}{2} \rfloor} j\varepsilon_{2.j}(u^p) = 0.$$

Proof. We will need the following character values, which can be obtained using Theorem 3.3.

$j \in \{1, 2, \cdots, \lfloor \frac{n}{2} \rfloor\}$	1	2.j	p.1
$\pi = \chi_{(n-1,1)}$	$n-1$	$\pi(1) - 2j$	$\pi(1) - p$
$\pi \otimes sgn = \chi_{(2,1^{n-2})}$	$n-1$	$(-1)^j(\pi(1) - 2j)$	$\pi(1) - p$

Because there is a unique conjugacy class of elements of order p in S_p, namely p.1, we know that u^2 is rationally conjugate to an element in that class. Thus

$$P(u^2) \sim diag(1 \times \boxed{\eta_p, \eta_p^2, \cdots, \eta_p^{p-1}}).$$

Where η_p is a primitive p-th root of unity. But as $\pi(u) \in \mathbb{Z}$ this implies that

$$P(u) \sim diag(1 \times \boxed{\eta_p, \eta_p^2, \cdots, \eta_p^{p-1}}) \quad \text{or} \quad P(u) \sim diag(1 \times \boxed{-\eta_p, -\eta_p^2, \cdots, -\eta_p^{p-1}}).$$

Looking at the decomposition matrix for the prime p, see [44], we can see that the restriction of π to the p-regular conjugacy classes of S_p has a trivial constituent, using Theorem 3.9 we obtain $\pi(u^p) \geq -(p-3)$ which makes the second option impossible. So $\mu_1(u,\pi) = 0$ and $u^p \in Ker(P)$. We can calculate the multiplicity of -1 as an eigenvalue of $P(u^p)$ by Lemma 6.11 to see that

$$\mu_1(u^p, \pi) = \sum_{j=1}^{\lfloor \frac{n}{2} \rfloor} j\varepsilon_{2.j}(u^p).$$

Using that $P(u^p)$ is the identity, we find that this sum equals zero. Now consider a representation L associated with the character $\pi \otimes sgn$. With the same reasoning as before we find that

$$L(u) \sim diag(1 \times \boxed{\eta_p, \eta_p^2, \cdots, \eta_p^{p-1}}) \quad \text{or} \quad L(u) \sim diag(1 \times \boxed{-\eta_p, -\eta_p^2, \cdots, -\eta_p^{p-1}}).$$

From the decomposition matrix for the prime p one can also conclude that the restriction of $\pi \otimes sgn$ to the p-regular conjugacy classes of S_p contains sgn as a constituent. Because $sgn(u^p)$ only depends on the value $\varepsilon_{A_p}(u) = \varepsilon_{A_p}(u^p) \in \{0, 1\}$. We obtain that $L(u)$ has the first form if and only if $\varepsilon_{A_p}(u^p) = 1$ and it has the second form if and only if $\varepsilon_{A_p}(u^p) = 0$. So $\mu_1(u^p, \pi \otimes sgn)$ is zero if $\varepsilon_{A_p}(u^p) = 1$ and is $p-1$ if $\varepsilon_{A_p}(u^p) = 0$.

If we combine the HeLP-formula with the fact that $\sum_{j=1}^{\lfloor \frac{p}{2} \rfloor} j\varepsilon_{2.\mathrm{j}}(u^p) = 0$ to find the multiplicity of -1 as an eigenvalue of $R(u^p)$ we see that:

$$\begin{aligned}\mu_1(u^p, \pi \otimes sgn) =& \frac{1}{2}[\pi(1) - \sum_{j=1}^{\lfloor \frac{p}{2} \rfloor}(-1)^j(\pi(1) - 2j)\varepsilon_{2.\mathrm{j}}(u^p)] \\ =& \frac{1}{2}[\sum_{j=1}^{\lfloor \frac{p}{2} \rfloor}(\pi(1) - 2j)\varepsilon_{2.\mathrm{j}}(u^p) - \sum_{j=1}^{\lfloor \frac{p}{2} \rfloor}(-1)^j(\pi(1) - 2j)\varepsilon_{2.\mathrm{j}}(u^p)] \\ =& \sum_{\substack{j=1 \\ 2|j}}^{\lfloor \frac{p}{2} \rfloor}(\pi(1) - 2j)\varepsilon_{2.\mathrm{j}}(u^p) = \pi(1)\varepsilon_{S_p \setminus A_p}(u^p) - 2\sum_{\substack{j=1 \\ 2\nmid j}}^{\lfloor \frac{p}{2} \rfloor} j\varepsilon_{2.\mathrm{j}}(u^p).\end{aligned}$$

Plugging in the possibilities for $\varepsilon_{S_p \setminus A_p}(u^p)$ yields the demanded results. ■

6.4.3 Applications

We will use our previous results to verify the prime graph question for certain almost simple groups. Additional calculations were performed in GAP, by using the HeLP package.

Theorem 6.17. *The prime question is true for all almost simple groups with socle an alternating group of degree at most* 17*.*

Proof. By a result of Salim [59] we know that the prime graph question has a positive answer for alternating groups of degree up to 10. Besides this the Prime Graph Question holds for all almost simple groups with socle isomorphic to A_n for $n \leq 6$ [46]. It suffices to exclude the existence of a normalized unit of order pq in the integral group ring of the largest symmetric (or alternating) group which does not contain an element of order pq. After using Theorem 6.14 and Corollary 6.14.1, we are left with the triples (n, p, q), corresponding to a unit of order pq in $V(\mathbb{Z}S_n)$ or $V(\mathbb{Z}A_n)$ given in the tables below. Thanks to the fact that the HeLP Package performs much better these days , it is possible achieve the same results as in [22] with less calculations. We used the function HeLP_WithGivenOrder(Characters, Order): to obtain the following results:

Data for units in $V(\mathbb{Z}A_n)$.

n	q	p	r	Number of partial augmentations for units of order q	Number after using the characters	Ordinary characters used	Brauer characters mod r used
14	2	11	/	48	0	$3, 4, 5$	/
16	2	13	/	813	0	$2, 3, 5, 9$	/

Data for units in $V(\mathbb{Z}S_n)$.

n	q	p	r	Number of partial augmentations for units of order q	Number after using the characters	Ordinary characters used	Brauer characters mod r used
7	3	5	/	4	0	$2, 4$	/
8	2	7	/	15	0	$4, 5, 6$	/
9	3	7	/	21	0	$3, 4, 5$	/
12	2	11	3	343	0	/	$2, .., 6, 9$
13	3	11	/	556	0	$4, 5, 13, 16$	/
14	2	13	7	3341	0	/	$2, .., 4, 8, ..11$
15	3	13	2	749	0	/	$2, .., 6$
15	5	11	/	19	0	$3, 5, 9$	/
17	2	17	7	111917	2	/	$2, ..., 10$
17	3	17	2	1016	0	/	$2, .., 5, 7$
17	5	17	/	27	0	$6, 9$	/
17	5	13	/	27	0	$4, 8, 10$	/

All these characters, except the ones used to exclude units of order $17 \cdot 2$ are minimal in the sense that a proper subset of the used characters will not be sufficient to exclude units of order pq in the corresponding integral group ring. The two cases which are left as possible partial augmentations for units of order 34 in $V(\mathbb{Z}S_{17})$ correspond to the tuples:

$$[[3, -3, 0, 1, 0, 0, 0, 0], [1, -1, 3, 1, -3, 0, 0, 0, 0]] \quad [[4, -6, -1, 4, 0, 0, 0, 0], [1, 0, 0, 0, 0, 0, 0, 0, 0]].$$

Both of these tuples can be eliminated using Lemma 6.16.

The main problem in extending this list is the fact the Brauer tables of S_n for $n > 17$ are not available in GAP. Especially the Brauer tables mod 3 would be very helpful to reduce the rapidly rising number of possible partial augmentations for units of order $2 \cdot p$ in S_p. ■

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